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**Services Stage 3**
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# Foreword

This Technical Specification has been produced by the 3rd Generation Partnership Project (3GPP).

The contents of the present document are subject to continuing work within the TSG and may change following formal TSG approval. Should the TSG modify the contents of the present document, it will be re-released by the TSG with an identifying change of release date and an increase in version number as follows:

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z the third digit is incremented when editorial only changes have been incorporated in the document.

In the present document, modal verbs have the following meanings:

**shall** indicates a mandatory requirement to do something

**shall not** indicates an interdiction (prohibition) to do something

The constructions "shall" and "shall not" are confined to the context of normative provisions, and do not appear in Technical Reports.

The constructions "must" and "must not" are not used as substitutes for "shall" and "shall not". Their use is avoided insofar as possible, and they are not used in a normative context except in a direct citation from an external, referenced, non-3GPP document, or so as to maintain continuity of style when extending or modifying the provisions of such a referenced document.

**should** indicates a recommendation to do something

**should not** indicates a recommendation not to do something

**may** indicates permission to do something

**need not** indicates permission not to do something

The construction "may not" is ambiguous and is not used in normative elements. The unambiguous constructions "might not" or "shall not" are used instead, depending upon the meaning intended.

**can** indicates that something is possible

**cannot** indicates that something is impossible

The constructions "can" and "cannot" are not substitutes for "may" and "need not".

**will** indicates that something is certain or expected to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document

**will not** indicates that something is certain or expected not to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document

**might** indicates a likelihood that something will happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

**might not** indicates a likelihood that something will not happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

In addition:

**is** (or any other verb in the indicative mood) indicates a statement of fact

**is not** (or any other negative verb in the indicative mood) indicates a statement of fact

The constructions "is" and "is not" do not indicate requirements.

# 1 Scope

The present document specifies the stage 3 protocol and data model for the Nimsas Service Based Interface. It provides stage 3 protocol definitions and message flows, and specifies the API for each service offered by the IMS AS.

The 5G System stage 2 architecture and procedures are specified in TS 23.501 [2] and TS 23.502 [3].

The IP Multimedia Subsystem (IMS) supporting DC architecture and procedures are specified in TS 23.228 [14].

The Technical Realization of the Service Based Architecture and the Principles and Guidelines for Services Definition are specified in TS 29.500 [4] and TS 29.501 [5].

# 2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
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[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[2] 3GPP TS 23.501: "System Architecture for the 5G System; Stage 2".

[3] 3GPP TS 23.502: "Procedures for the 5G System; Stage 2".

[4] 3GPP TS 29.500: "5G System; Technical Realization of Service Based Architecture; Stage 3".

[5] 3GPP TS 29.501: "5G System; Principles and Guidelines for Services Definition; Stage 3".

[6] OpenAPI: "OpenAPI Specification Version 3.0.0", https://spec.openapis.org/oas/v3.0.0.

[7] 3GPP TR 21.900: "Technical Specification Group working methods".

[8] 3GPP TS 33.501: "Security architecture and procedures for 5G system".

[9] IETF RFC 6749: "The OAuth 2.0 Authorization Framework".

[10] 3GPP TS 29.510: "5G System; Network Function Repository Services; Stage 3".

[11] IETF RFC 7540: "Hypertext Transfer Protocol Version 2 (HTTP/2)".

[12] IETF RFC 8259: "The JavaScript Object Notation (JSON) Data Interchange Format".

[13] IETF RFC 7807: "Problem Details for HTTP APIs".

[14] 3GPP TS 23.228: "IP Multimedia Subsystem (IMS); Stage 2".

[15] 3GPP TS 29.562: "5G System; Home Subscriber Server (HSS) Services; Stage 3".

[16] 3GPP TS 29.571: "5G System; Common Data Types for Service Based Interfaces; Stage 3".

[17] 3GPP TS 29.176: "IP Multimedia Subsystems (IMS); Media Function (MF) Services; Stage 3".

[18] 3GPP TS 24.610:" Communication HOLD (HOLD) using IP Multimedia (IM) Core Network (CN) subsystem; Protocol specification".

[19] 3GPP TS 24.186:" IMS Data Channel applications; Protocol specification".

[20] IETF RFC 8259: "The JavaScript Object Notation (JSON) Data Interchange Format".

[21] 3GPP TS 24.604: "Communication Diversion (CDIV) using IP Multimedia (IM); Core Network (CN) subsystem; Protocol specification".

[22] 3GPP TS 29.503: "5G System; Unified Data Management Services; Stage 3".

[23] 3GPP TS 24.229: " IP multimedia call control protocol based on Session Initiation Protocol (SIP) and Session Description Protocol (SDP); Stage 3".

# 3 Definitions, symbols and abbreviations

## 3.1 Definitions

For the purposes of the present document, the terms and definitions given in TR 21.905 [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in TR 21.905 [1].

For the purposes of the present document, the following terms and definitions given in TS 23.228 [14] apply:

- **Bootstrap data channel**
- **DC Application Server**
- **Rich Call Data (RCD) server**
- **Rich Call Data (RCD) server address**
- **Rich Call Data (RCD) information**
- **Rich Call Data (RCD) URL**
- **Rich Call Data (RCD) properties**

## 3.2 Symbols

For the purposes of the present document, the following symbols given in TS 23.228 [14] apply:

DC1 Reference Point between an SBI capable IMS AS and DCSF.
DC2 Reference Point between an SBI capable IMS AS and MF.

## 3.3 Abbreviations

For the purposes of the present document, the abbreviations given in TR 21.905 [1], TS 23.228 [14] and the following apply. An abbreviation defined in the present document or TS 23.228 [14] takes precedence over the definition of the same abbreviation, if any, in TR 21.905 [1].

AR Augmented Reality
DTLS Datagram Transport Layer Security
IMS AS IP Multimedia Subsystem Application Server
MF Media Function
NEF Network Exposure Function
SCP Service Communication Proxy

# 4 Overview

The IP Multimedia Subsystem (IMS) architecture supporting the Data Channel services is defined in TS 23.228 [14].

The IP Multimedia Subsystem (IMS) Application Server (AS) is enhanced to offer services to e.g. DCSF, HSS, NEF or trusted AF via the Nimsas service-based interface (see TS 23.228 [14]) and support the following functionalities:

- The IMS AS interacts with the DCSF via DC1 for event notifications;
- The IMS AS receives the data channel control instructions from the DCSF and accordingly interacts with the MF via DC2 for data channel media resource management;
- The IMS AS receives the data channel management requirement from NEF or trusted AF and interacts with the DCSF and MF for data channel management.
- The IMS AS interacts with the HSS/NEF for subscriber related events subscription, and notifies the NEF with the subscriber related events which are subscribed, and interacts with the NEF for non-subscriber related events subscription and notification, e.g. subscriptions to any UE;
- The IMS AS receives the provisioning information (e.g. RCD properties) from NEF or trusted AF and stores in HSS repository data to support the 3rd party user identity.

Figure 4-1 provides the reference model (in service-based interface representation and in reference point representation), with focus on the IMS AS.

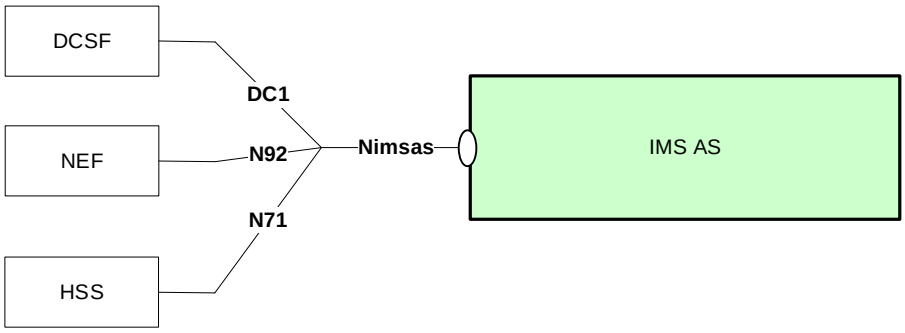



**Figure 4-1: Reference model – IMS AS**

DC1 is the reference point between an SBI capable IMS AS and DCSF.

N92 is the reference point between an SBI capable IMS AS and NEF.

N71 is the reference point between an SBI capable IMS AS and HSS.

# 5 Services offered by the IMS AS

## 5.1 Introduction

The IMS AS offers to other NFs the following services:

- Nimsas_SessionEventControl. This service enables the consumer to be notified about session events when served IMS subscribers take part in IMS sessions.
- Nimsas_MediaControl. This service enables the consumer to control IMS AS handling of media flow within an IMS session.
- Nimsas_ImsSessionManagement. This service enables the consumer to request IMS AS to create or release an IMS session or modify the media in a specific IMS session.
- Nimsas_EventExposure. This service enables the service consumer to subscribe to subscriber-related events (e.g. HSS) and non-subscriber related events (e.g. NEF).
- Nimsas_ImsParameterProvision. This service enables the consumer to provision the IMS user related information (e.g. RCD URL, RCD server address or RCD information) for the target IMPU.

Table 5.1-1 summarizes the SBI services produced by the IMS AS and the corresponding APIs defined for this specification.

**Table 5.1-1: API Descriptions**

| Service Name | Clause | Description | OpenAPI Specification File | apiName | Annex |
|---|---|---|---|---|---|
| Nimsas_SessionEventControl | 6.1 | IMS AS SessionEventControl Service | TS29175_Nimsas_SessionEventControl.yaml | nimsas-sec | A.2 |
| Nimsas_MediaControl | 6.2 | IMS AS MediaControl Service | TS29175_Nimsas_MediaControl.yaml | nimsas-mc | A.3 |
| Nimsas_ImsSessionManagement | 6.3 | IMS AS ImsSessionManagement Service | TS29175_Nimsas_ImsSessionManagement.yaml | nimsas-ism | A.4 |
| Nimsas_EventExposure | 6.4 | IMS AS EventExposure Service | TS29175_Nimsas_ImsEE.yaml | nimsas-ee | A.5 |
| Nimsas_ImsParameterProvision | 6.5 | IMS AS ImsParameterProvision Service | TS29175_Nimsas_ImsParameterProvision.yaml | nimsas-ipp | A.6 |

# 5.2 Nimsas_SessionEventControl Service

## 5.2.1 Service Description

The Nimsas_SessionEventControl service enables the consumer to be notified about session events when served IMS subscribers take part in IMS sessions.

## 5.2.2 Service Operations

### 5.2.2.1 Introduction

The Service operations defined for the Nimsas_SessionEventControl service are as follows:

- Notify: It allows the IMS AS to notify the IMS session control events of a specific IMS subscriber to NFs. This service operation reports the event to the consumer that has subscribed implicitly.

NOTE: Explicit subscription to receive session events is another service operation defined in TS 23.228 [14] which has not specified in this Release. In this Release an implicit subscription is assumed where the IMS AS notifies a configured or discovered the NF consumer of a call event through a Notify.

**Table 5.2.2.1-1: Service operations supported by the Nimsas_SessionEventControl service**

| Service Operations | Description | Operation Semantics | Example Consumer(s) |
|---|---|---|---|
| Notify | Notification about IMS session control events of a specific IMS subscriber. | Subscribe/Notify | DCSF |

### 5.2.2.1A Subscribe

This is a pseudo operation, the IMS AS does not actually provide Subscribe service operation through Nimsas_SessionEventControl service in this Release. The actual subscription is implicitly subscribed with the SessionEventNotificationUri.

#### 5.2.2.2 Notify

##### 5.2.2.2.1 General

This service operation is invoked by IMS AS and used to enables the IMS AS to notify the NF service consumers (e.g. DCSF) of IMS session events related to a specific served IMS subscriber requesting use of IMS media (e.g. data channel media).

##### 5.2.2.2.2 Notification for Session Event

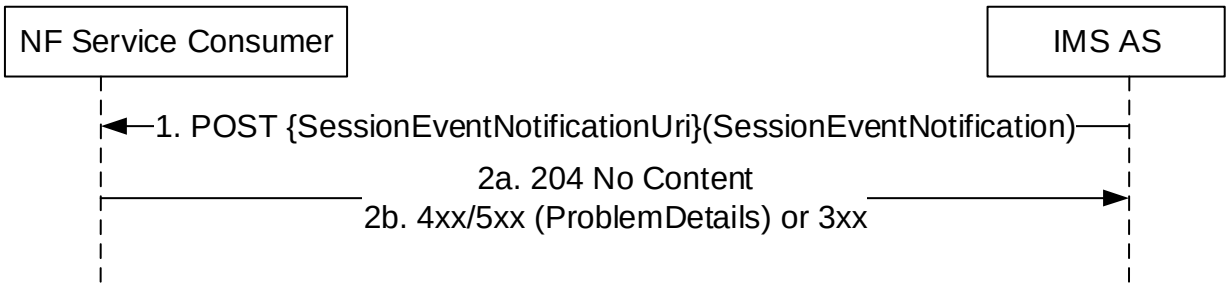



**Figure 5.2.2.2.2-1: Notification for Session Event**

1. If the IMS AS observes the IMS session control events of a specific IMS subscriber that requesting use of IMS data channel media, and it determines based on the IMS subscriber's service profile that the events need to be notified to the DCSF, the IMS AS shall send a POST request to the SessionEventNotificationUri as specified in clause 6.1.5.2.2.

   The IMS AS discovers the DCSF and its session event notification URI based on local configuration or via NRF, as specified in clause AC.7.1 of TS 23.228 [14].

   Based on the operators's policy, if the terminating IMS AS observes the IMS session control events based on the terminating IMS subscriber's DC subscription and DC capability that the events need to be notified to the DCSF, the IMS AS shall send a POST request to the SessionEventNotificationUri as specified in clause 6.1.5.2.2.

2a. Upon success, the NF Service Consumer responds with "204 No Content". The NF service consumer may invoke the Nimsas_MediaControl service before sending the session event notification response to IMS AS if the NF service consumer determines that media resources for data channel are needed.

2b. On failure or redirection:

- If the NF Service Consumer does not consider the "SessionEventNotificationUri" as a valid notification URI, the NF Service Consumer shall return "404 Not Found" status code with the ProblemDetails IE providing details of the error.
- In the case of redirection, the NF service consumer shall return 3xx status code, which shall contain a Location header with an URI pointing to the endpoint of another NF service consumer endpoint. A RedirectResponse IE may be included in the content of POST response.

## 5.3 Nimsas_MediaControl Service

### 5.3.1 Service Description

The Nimsas_MediaControl service enables the consumer to control IMS AS handling of media flow within an IMS session. The service can be used by the consumer before responding to a Nimsas_SessionEventControl_Notity request.

If interworking between DC capable UE and non-DC capable UE is supported, the Nimsas_MediaControl shall support to indicate transforming data channel media to video media.

## 5.3.2 Service Operations

### 5.3.2.1 Introduction

The Service operations defined for the Nimsas_MediaControl service are as follows:

- MediaInstruction: It provides instructions to the IMS AS for all media flows a consumer wants to control based on its policies for the received IMS session event, and that may require media resource reservation on MF.

NOTE: For the case when a specific media flow needs to be terminated in MF media resource (i.e. termination of a Data Channel media descriptor offered by one of the IMS subscribers) or originated by the MF (i.e. origination of a data channel media descriptor to be offered towards one of the IMS subscribers), the consumer must provide a complete MF media specification including information required by MF to know how to terminate or originate the media flow.

**Table 5.3.2.1-1: Service operations supported by the Nimsas_MediaControl service**

| Service Operations | Description | Operation Semantics | Example Consumer(s) |
|---|---|---|---|
| MediaInstruction | Instructions to the IMS AS for all media flows a consumer wants to control based on its policies for the received IMS session event | Request/Response | DCSF |

### 5.3.2.2 MediaInstruction

#### 5.3.2.2.1 General

This service operation provides instructions to the IMS AS for all media flows a consumer wants to control based on its policies for the received IMS session event, and that may require resource reservation in media resource

The MediaInstruction service operation defined for Nimsas_MediaControl service is used:

- to enable the NF consumer to instruct the IMS AS how to manage the MF media resource. E.g., setting up the bootstrap data channel with MF both for originating and terminating side, terminating the media flow of the originating UE to MF and updating the MF resource with MDC2 media endpoint information of DC Application Server.

#### 5.3.2.2.2 Media Instruction

Figure 5.3.2.2.2-1 shows a scenario where the NF service consumer (e.g. DCSF) sends a request to the IMS AS to instruct the IMS AS how to manage the MF media resource. The request contains the IMS session identity (/{sessionId}) which is assigned by the IMS AS during the IMS session setup and will be notified to consumer via Nimsas_SessionEventControl notify service operation. After that, the consumer shall reuse the session ID it received from the IMS AS for referencing the same session.

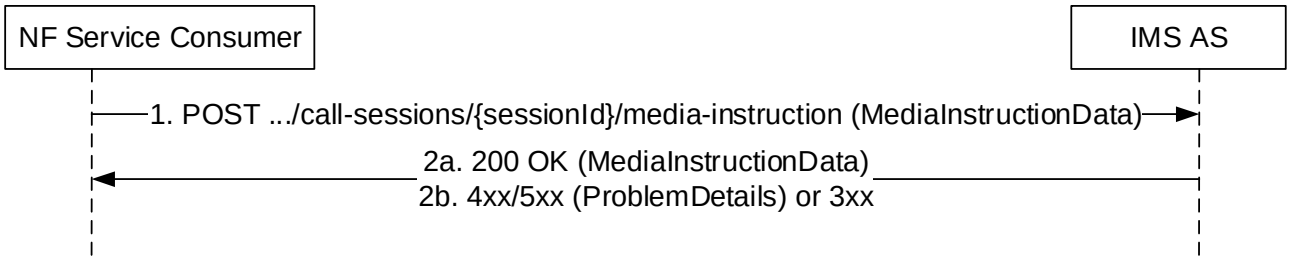



**Figure 5.3.2.2.2-1: Media instruction from NF service consumer**

1. The NF service consumer shall send a POST request to the resource representing the individual session resource in the IMS AS. The payload body of the POST request shall contain the media instructions for all the media flows that the NF Service Consumer want to control based on its policies for the received IMS session event via Nimsas_SessionEventControl notification.

For the scenario of interworking via MF, the NF service consumer should instruct the data channel media with "TERMINATE_MEDIA" in data type MediaInstructions and indicate to negotiate the video media with "audioVideoReNegotiationInd" in data type MediaInstructions.

2a. Upon success, the IMS AS responds with "200 OK" with the message body containing a representation of the created media instruction data.

2b. On failure or redirection:

In the case of failure, one of the HTTP status code listed in Table 6.2.3.2.4.2.2-2 shall be returned. For a 4xx/5xx response, the message body shall contain a ProblemDetails structure with the "cause" attribute set to one of the application errors listed in Table 6.2.3.2.4.2.2-2.

In the case of redirection, the NF service producer shall return 3xx status code, which shall contain a Location header with an URI pointing to the endpoint of another NF service producer endpoint.

# 5.4 Nimsas_ImsSessionManagement Service

## 5.4.1 Service Description

The Nimsas_ImsSessionManagement service enables the NF service consumer to request IMS AS to create or release an IMS session or modify the media in a specific IMS session, or enables the IMS AS to notify the IMS session events to the interested NF service consumer.

## 5.4.2 Service Operations

### 5.4.2.1 Introduction

The service operations defined for the Nimsas_ImsSessionManagement service are as follows:

- Create: It allows the NF consumer to create an IMS session with standalone data channel media.
- Update: It allows the the NF consumer to modify the media in a specific IMS session, such as adding, updating or removing the corresponding data channel media.
- Delete: It allows the NF consumer to release the specific IMS session with standalone data channel media.
- Notify: It allows the IMS AS to notify the subscribed IMS session events to the NF consumer.

**Table 5.4.2.1-1: Service operations supported by the Nimsas_ImsSessionManagement service**

| Service Operations | Description | Operation Semantics | Example Consumer(s) |
|---|---|---|---|
| Create | Create an IMS session with standalone data channel media. | Request/Response | NEF, trusted AF |
| Update | Modify the media in a specific IMS session. | Request/Response | NEF, trusted AF |
| Delete | Release a specific IMS session with standalone data channel media. | Request/Response | NEF, trusted AF |
| Notify | Notify the IMS session events. | Subscribe/Notify | NEF, trusted AF |

### 5.4.2.2 Create Service Operation

#### 5.4.2.2.1 General

This service operation is used by the service consumer (e.g AF, NEF) to create an IMS session with standalone data channel media.

#### 5.4.2.2.2 Creation of a new IMS session

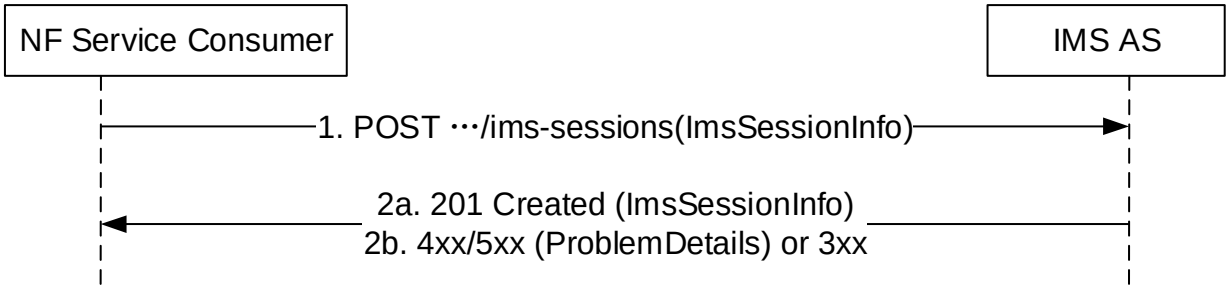



**Figure 5.4.2.2.2-1: Creation of a new IMS session**

1. The NF Service Consumer shall send a POST request to create an "Individual IMS Session" resource in the IMS AS. The payload body of the POST request shall contain a representation of the individual IMS session resource (ImsSessionInfo) to be created.

   If the "notifUri" and "notifCorrelationId" attributes are included within the "ImsSessionInfo" data type, this request creates an implicit subscription to IMS session event notification related to this session.

2a. Upon the reception of the HTTP POST request, if the request is accepted and no error occur, the IMS AS shall create a new session ID for the session and the mediaCorrelationId for each media used in the IMS session.

   The IMS AS shall include a HTTP Location header to provide the location of a newly created resource (ImsSessionInfo) together with the status code 201 Created indicating the requested resource is created in the response message.

2b. On failure, one of the HTTP status code listed in table 6.3.3.2.3.1-3 shall be returned. For a 4xx/5xx response, the message body shall contain a ProblemDetails structure with the "cause" attribute set to one of the application errors listed in table 6.3.3.2.3.1-3.

   On redirection, the IMS AS shall include 3xx status code, which shall contain a Location header with an URI pointing to the endpoint to another IMS AS (service) instance.

### 5.4.2.3 Update Service Operation

#### 5.4.2.3.1 General

This service operation is used by the service consumer (e.g. AF, NEF) to modify the media in a specific IMS session, such as creating or closing application data channel(s).

#### 5.4.2.3.2 Partial updating an existing IMS session

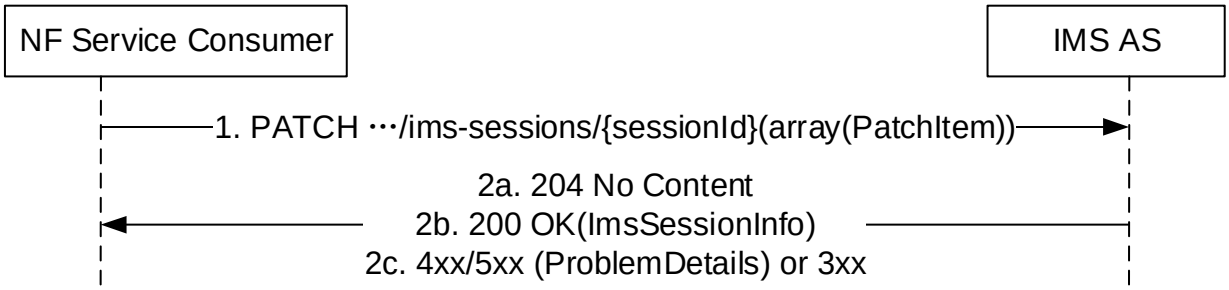



**Figure 5.4.2.3.2-1: Update an IMS session**

1. The NF Service Consumer shall send a PATCH request to modify an "Individual IMS Session" resource in the IMS AS. The modification may be for adding, updating or deleting a media in the existing IMS session.

   For adding, removing or updating a media in an existing ImsSessionInfo, the payload body of the PATCH request shall contain an "add", "remove" or "replace" PATCH operation respectively, with one item of the attribute "mediaInfoSet" of the ImsSessionInfo. The IMS AS shall use the mediaCorrelationId to check whether the operated media exists in the IMS session.

If the payload of the PATCH request includes the "notifUri" and "notifCorrelationId" within the "ImsSessionInfo" data type, this request creates an implicit subscription to IMS session event notification related to this session.

2a. On success, if the change is to delete the existing media and IMS AS accepts the request change, the IMS AS shall return the status code 204 No Content.

2b. On success, if the change is to add a new media or to update the existing media within an IMS session and the IMS AS accepts the request change, the IMS AS shall return the status code 200 OK. The response shall contain the new resource representation of the resource Individual IMS Session, which includes the added and modified resource or its sub-resource.

2c On failure, one of the HTTP status code listed in table 6.3.3.3.3.1-3 indicating the error shall be returned. For a 4xx/5xx response, the message body shall contain a ProblemDetails structure with the "cause" attribute set to one of the application errors listed in table 6.3.3.3.3.1-3.

On redirection, the IMS AS shall return 3xx status code, which shall contain a Location header with an URI pointing to the endpoint to another IMS AS (service) instance.

#### 5.4.2.3.3 Updating an existing IMS session

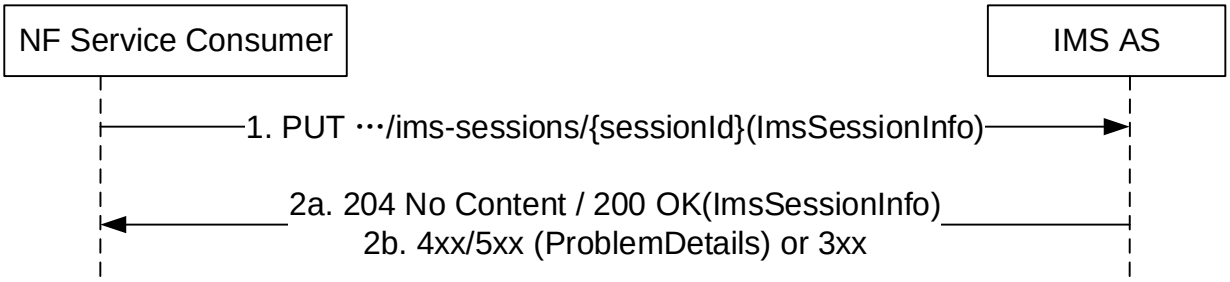



**Figure 5.4.2.3.3-1: Updating an existing IMS session**

1. The NF Service Consumer shall send a PUT request with the Resource URI "{apiRoot}/nimsas-ism/<apiVersion>/ims-sessions/{sessionId} " to update an "existing IMS session" resource in the IMS AS identified by the {sessionId}.

2a. On success, the IMS AS shall return the status code "204 No Content" or "200 OK" with the resource representation of the updated resource Individual Ims Event Subscription.

2b On failure, one of the HTTP status code listed in table 6.4.3.3.3.2-3 indicating the error shall be returned. For a 4xx/5xx response, the message body shall contain a ProblemDetails structure with the "cause" attribute set to one of the application errors listed in table 6.4.3.3.3.2-3.

On redirection, the IMS AS shall return 3xx status code, which shall contain a Location header with an URI pointing to the endpoint to another IMS AS (service) instance.

### 5.4.2.4 Delete Service Operation

#### 5.4.2.4.1 General

This service operation is used by the service consumer (e.g. AF, NEF) to release the specific IMS session.

#### 5.4.2.4.2 Releasing an existing IMS Session

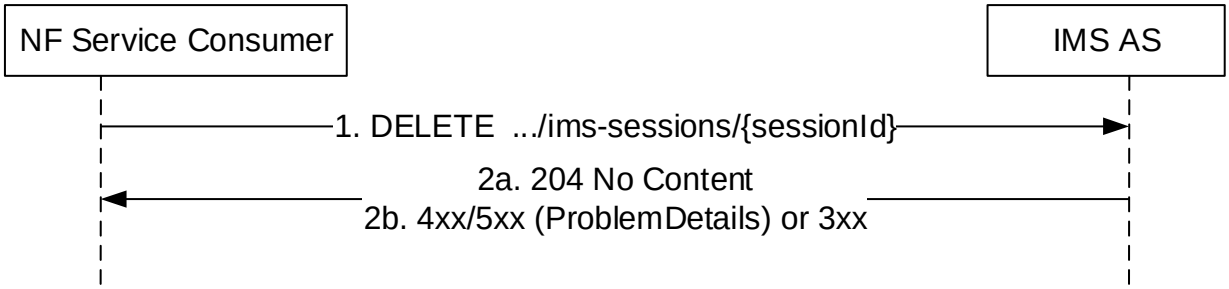



**Figure 5.4.2.4.2-1: Releasing an existing IMS Session**

1. The NF service consumer shall send a DELETE request to delete an existing Individual IMS Session resource in the IMS AS.

2a. On success, the request is accepted, the IMS AS shall reply with the status code "204 No Content" indicating that the resource identified by sessionId is successfully deleted in the response message.

2b. On failure, one of the HTTP status codes listed in table 6.3.3.3.3.2-3 indicating the error shall be returned. For a 4xx/5xx response, the message body shall contain a ProblemDetails structure with the "cause" attribute set to one of the application errors listed in table 6.3.3.3.3.2-3.

On redirection, the IMS AS shall return 3xx status code, which shall contain a Location header with an URI pointing to the endpoint to another IMS AS (service) instance.

### 5.4.2.5 Notify

#### 5.4.2.5.1 General

This service operation is invoked by IMS AS and used to enable the IMS AS to notify the NF service consumers (e.g. NEF, trusted AF) of IMS session events related to a specific IMS subscriber.

#### 5.4.2.5.2 Notification for IMS Session Event

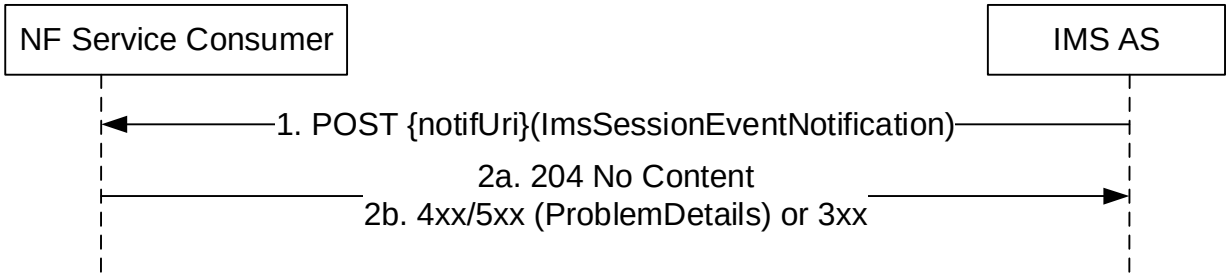



**Figure 5.4.2.5.2-1: Notification for IMS Session Event**

1. If the IMS AS detects the IMS session events of a specific IMS subscriber, and it determines based on the event subscriptions that the IMS session events need to be notified to the NF service consumer, the IMS AS shall send a POST request to the notifUri as specified in clause 6.3.5.2.2.

2a. Upon success, the NF Service Consumer responds with "204 No Content".

2b. On failure or redirection:

- If the NF Service Consumer receives a notification from IMS AS for session event that is not found for the service consumer, the NF Service Consumer shall return "404 Not Found" status code with the ProblemDetails IE providing details of the error.

- In the case of redirection, the NF service consumer shall return 3xx status code, which shall contain a Location header with an URI pointing to the endpoint of another NF service consumer endpoint. A RedirectResponse IE may be included in the content of POST response.

# 5.5 Nimsas_EventExposure Service

## 5.5.1 Service Description

The Nimsas_EventExposure service enables the service consumer (e.g. HSS on behalf of NEF) to subscribe to subscriber related events so that originally subscribing NF (e.g. NEF) is notified about the subscribed events detected by IMS AS (see TS 23.228 [14]).

The Nimsas_EventExposure service enables the service consumer (e.g. NEF) to subscribe to the non-subscriber related events (e.g. any UE) and to be notified about the subscribed events detected by IMS AS (see TS 23.228 [14]).

## 5.5.2 Service Operations

### 5.5.2.1 Introduction

The service operations defined for the Nimsas_EventExposure service are as follows:

- Subscribe: It allows the NF consumer to subscribe for a specific subscriber or non-subscriber related IMS event.
- Unsubscribe: It allows the the NF consumer to delete the subscription for a specific subscriber or non-subscriber related event.
- Notify: It allows the IMS AS to notify the specific subscriber or non-subscriber related event to NFs (e.g. NEF). This service operation reports the information associated with a subscribed event.

**Table 5.5.2.1-1: Service operations supported by the Nimsas_EventExposure service**

| Service Operations | Description | Operation Semantics | Example Consumer(s) |
|---|---|---|---|
| Subscribe | Subscribe to a specific subscriber or non-subscriber related IMS event. | Subscribe/Notify | NEF, HSS, trusted AF |
| Unsubscribe | Delete the subscription to a specific subscriber or non-subscriber related IMS event. | Subscribe/Notify | NEF, HSS, trusted AF |
| Notify | Notification about events detected by IMS AS. | Subscribe/Notify | NEF, trusted AF |

### 5.5.2.2 Subscribe

#### 5.5.2.2.1 General

This service operation is used by the service consumer (e.g HSS, NEF) to subscribe the IMS events on the IMS AS or update an existing subscription on the IMS AS.

The IMS AS may offer this service as a Service Producer to enable an NF to subscribe to event notifications, e.g. on behalf of another NF, and get notified about an event.

The IMS AS shall include the events it supports as part of its NF profile registered in NRF.

NOTE: The events supported by IMS AS in the NF profile can be known or unknown. The NF consumer (e.g. HSS) does not need to know the semantics of an unknown event.

#### 5.5.2.2.2 Subscription for event notifications

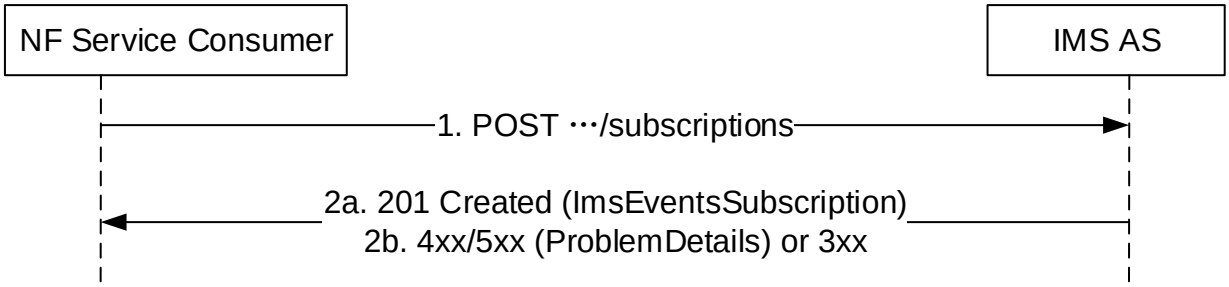



**Figure 5.5.2.2.2-1: Creation of a new subscription**

1. The NF Service Consumer shall send a POST request to create an "Individual IMS Event Subscription" resource in the IMS AS. The payload body of the POST request shall contain a representation of the individual IMS event subscription resource (ImsEventsSubscription) to be created. The request may contain an expiry time representing the time upto which the subscription is to be kept active (i.e. the time after which the subscribed event(s) shall stop generating reports).

2a. Upon the reception of the HTTP POST request, if the request is accepted and no error occur, the IMS AS shall create a new subscription ID for the subscription.

The IMS AS shall include a HTTP Location header to provide the location of a newly created resource (ImsEventsSubscriptionCreated) together with the status code "201 Created" indicating the requested resource is created in the response message. The response may include an expiry time, which may be equal to the requested expiry time or lower.

2b. On failure, one of the HTTP status code listed in table 6.4.3.2.3.1-3 shall be returned. For a 4xx/5xx response, the message body shall contain a ProblemDetails structure with the "cause" attribute set to one of the application errors listed in table 6.4.3.2.3.1-3.

On redirection, the IMS AS shall include 3xx status code, which shall contain a Location header with an URI pointing to the endpoint to another IMS AS (service) instance.

#### 5.5.2.2.3 Updating an exisiting subscription

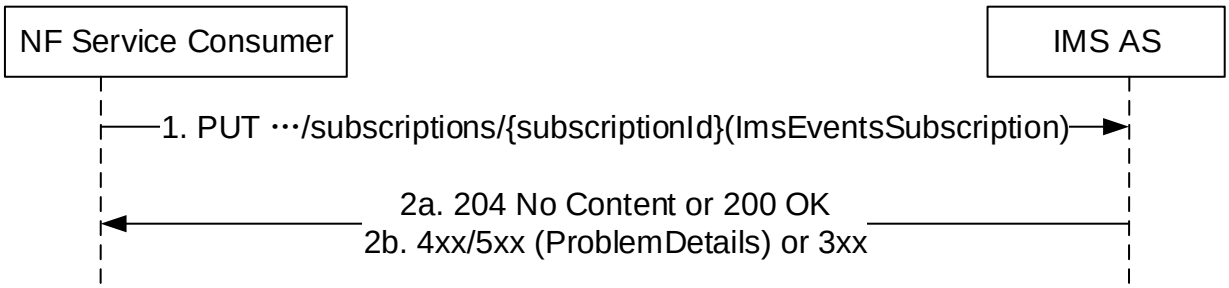



**Figure 5.5.2.2.3-1: Updating an existing subscription**

1. The NF Service Consumer shall send a PUT request with the Resource URI "{apiRoot}/nimsas-ee/<apiVersion>/subscriptions/{subscriptionId}" to update an "Individual IMS Event Subscription" resource in the IMS AS identified by the {subscriptionId}. The ImsEventsSubscription data structure provided in the request body shall include the same contents as described in clause 5.5.2.2.2.

2a. On success, the IMS AS shall return the status code "204 No Content" or "200 OK" with the resource representation of the updated resource Individual Ims Event Subscription.

2c On failure, one of the HTTP status code listed in table 6.4.3.3.3.2-3 indicating the error shall be returned. For a 4xx/5xx response, the message body shall contain a ProblemDetails structure with the "cause" attribute set to one of the application errors listed in table 6.4.3.3.3.2-3.

On redirection, the IMS AS shall return 3xx status code, which shall contain a Location header with an URI pointing to the endpoint to another IMS AS (service) instance.

#### 5.5.2.2.4 Subscription Partial Update

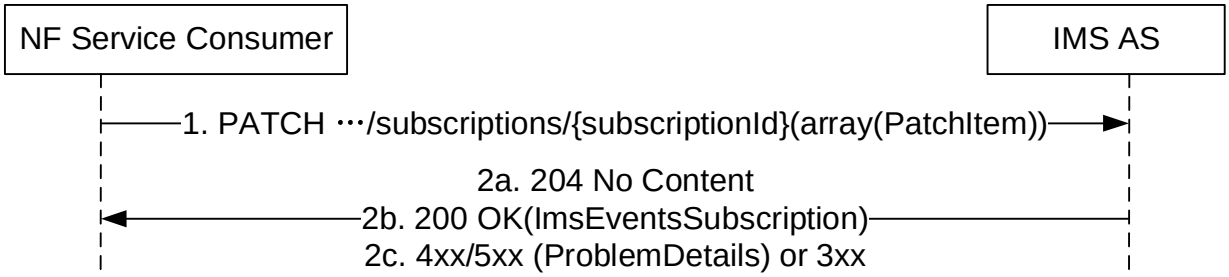



**Figure 5.5.2.2.4-1: Partially update a subscription**

1. The NF Service Consumer shall send a PATCH request to modify an "Individual IMS Event Subscription" resource in the IMS AS identified by the {subscriptionId}. The content of the PATCH request shall contain the list of operations (add/delete/replace) to be applied to the subscription; these operations may be directed to individual parameters of the subscription or to the list of the subscribed events.

2a. On success, if the change is to delete the events to be subscribed and IMS AS accepts the request change, the IMS AS shall return the status code 204 No Content.

2b. On success, if the change is to add new events to be subscribed or to update the existing subscribed event within a subscription and the IMS AS accepts the request change, the IMS AS shall return the status code 200 OK. The response shall contain the new resource representation of the resource Individual IMS Event Subcription.

2c On failure, one of the HTTP status code listed in table 6.4.3.3.3.3-3 indicating the error shall be returned. For a 4xx/5xx response, the message body shall contain a ProblemDetails structure with the "cause" attribute set to one of the application errors listed in table 6.4.3.3.3.3-3.

On redirection, the IMS AS shall return 3xx status code, which shall contain a Location header with an URI pointing to the endpoint to another IMS AS (service) instance.

### 5.5.2.3 Unsubscribe

#### 5.5.2.3.1 General

This service operation is used by the service consumer (e.g. HSS, NEF) to unsubscribe from event notifications.

#### 5.5.2.3.2 Unsubscribe from notifications

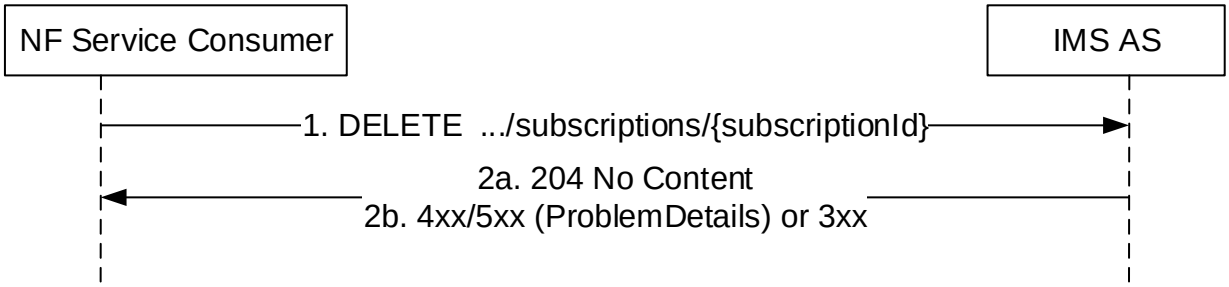



**Figure 5.5.2.3.2-1: Unsubscribe from notifications**

1. The NF service consumer shall send a DELETE request to delete an existing Individual IMS Event Subscription resource in the IMS AS.

2a. On success, the request is accepted, the IMS AS shall reply with the status code "204 No Content" indicating that the resource identified by subscriptionId is successfully deleted in the response message.

2b. On failure, one of the HTTP status code listed in table 6.4.3.3.3.2-3 indicating the error shall be returned. For a 4xx/5xx response, the message body shall contain a ProblemDetails structure with the "cause" attribute set to one of the application errors listed in table 6.4.3.3.3.2-3.

On redirection, the IMS AS shall return 3xx status code, which shall contain a Location header with an URI pointing to the endpoint to another IMS AS (service) instance.

### 5.5.2.4 Notify

#### 5.5.2.4.1 General

This service operation is used by the IMS AS to enables the IMS AS to notify the NF service consumers (e.g. HSS, NEF) of IMS session events related to a specific served IMS subscriber requesting use of IMS media (e.g. data channel media).

#### 5.5.2.4.2 Notification for IMS Event

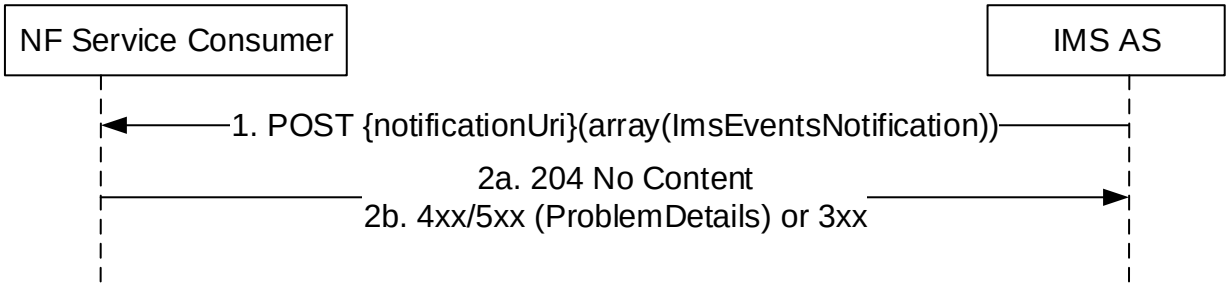



**Figure 5.2.2.2.2-1: Notification for IMS Event**

1. If the IMS AS observes the IMS events of a specific IMS subscriber, the IMS AS shall send a POST request to the request with "{notificationURI}" received in the Nimsas_ImsEE_Subscribe service operation as Resource URI.

2a. Upon success, the NF Service Consumer responds with "204 No Content".

2b. On failure or redirection:

- If the NF Service Consumer receives a notification from IMS AS for IMS event subscription that is not found for the service consumer, the NF Service Consumer shall return "404 Not Found" status code with the ProblemDetails IE providing details of the error.
- In the case of redirection, the NF service consumer shall return 3xx status code, which shall contain a Location header with an URI pointing to the endpoint of another NF service consumer endpoint. A RedirectResponse IE may be included in the content of POST response.

### 5.5.2.5 Subscription Info Retrieval

#### 5.5.2.5.1 General

This service operation allows the retrieval of subscription information.

#### 5.5.2.5.2 Subscription Info Retrieval

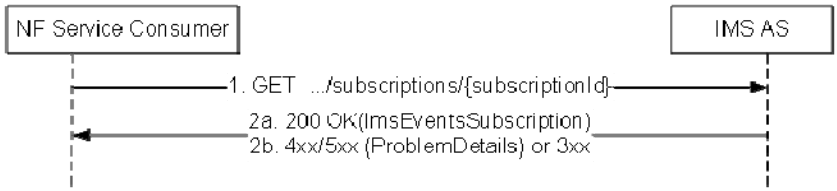



**Figure 5.5.2.4.2-1: Subscription info retrieval**

1. The NF Service Consumer shall send an HTTP GET request to the resource URI "subscription" collection resource.

2a. On success, the IMS AS shall return the status code 200 OK. The response shall contain the resource representation of the resource Individual IMS Event Subscription.

2b. On failure, one of the HTTP status codes listed in table 6.4.3.3.3.1-3 indicating the error shall be returned. For a 4xx/5xx response, the message body shall contain a ProblemDetails structure with the "cause" attribute set to one of the application errors listed in table 6.4.3.3.3.1-3.

On redirection, the IMS AS shall return 3xx status code, which shall contain a Location header with an URI pointing to the endpoint to another IMS AS (service) instance.

# 5.6 Nimsas_ImsParameterProvision Service

## 5.6.1 Service Description

The Nimsas_ImsParameterProvision service enables the service consumer (e.g. the NEF or trusted AF) to provision the IMS user related information for the target IMPU to support the 3rd party user identity information (see TS 23.228 [14]).

## 5.6.2 Service Operations

### 5.6.2.1 Introduction

The service operations defined for the Nimsas_ImsParameterProvision service are as follows:

- Create: It allows the NF consumer to provision the IMS user related information for a target subscriber.
- Update: It allows the NF consumer to update the IMS user related information for a target subscriber.
- Delete: It allows the NF consumer to delete the IMS user related information for a target subscriber.

**Table 5.6.2.1-1: Service operations supported by the Nimsas_ImsParameterProvision service**

| Service Operations | Description | Operation Semantics | Example Consumer(s) |
|---|---|---|---|
| Create | Create the IMS user related information for a target subscriber. | Request/Response | NEF, trusted AF |
| Update | Update the IMS user related information for a target subscriber. | Request/Response | NEF, trusted AF |
| Delete | Delete the IMS user related information for a target subscriber. | Request/Response | NEF, trusted AF |

### 5.6.2.2 Create Service Operation

#### 5.6.2.2.1 General

This service operation is used by the service consumer (e.g. the NEF) to create the IMS Parameter Provisioning Data entry for a certain AF which contains the IMS user related information (e.g., RCD URL, RCD server address or RCD information for the target IMPU).

#### 5.6.2.2.2 Creation of the Parameter Provisioning Data Entry per AF

The request contains the IMS UE's identity (/{imsUeId}) which shall be an IMPU.

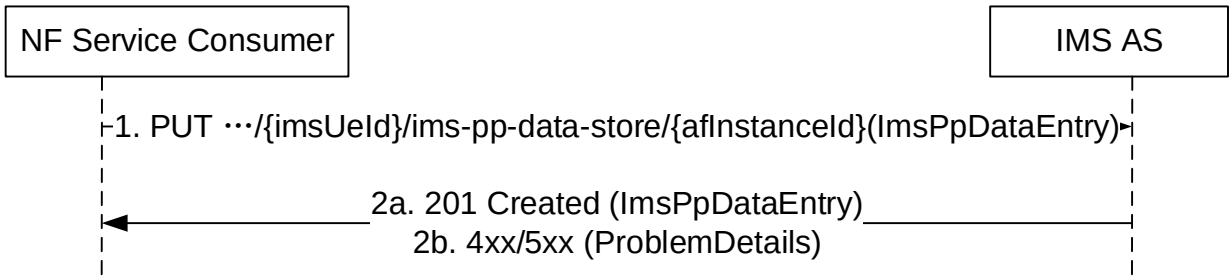



**Figure 5.6.2.2.2-1: Creation of the IMS Parameter Provisioning Data Entry per AF**

1. The NF Service Consumer (e.g. the NEF) sends a PUT request to the resource that represents the new IMS Parameter Provisioning Data entry for the AF identified by the afInstanceId. The request body shall contain an ImsPpDataEntry object representing the value of the new resource.

2a. Upon the reception of the HTTP PUT request, if the request is accepted and no error occurs, the IMS AS shall respond with "201 Created" and the response body shall contain the ImsPpDataEntry object representing the new created resource.

2b. On failure, one of the HTTP status codes listed in table 6.5.3.2.3.1-3 shall be returned. For a 4xx/5xx response, the message body shall contain a ProblemDetails structure with the "cause" attribute set to one of the application errors listed in table 6.5.3.2.3.1-3.

### 5.6.2.3 Update Service Operation

#### 5.6.2.3.1 General

This service operation is used by the service consumer (e.g. the NEF) to modify the IMS Parameter Provisioning Data entry for a certain AF which contains the IMS user related information (e.g., RCD URL, RCD server address or RCD information for the target IMPU).

#### 5.6.2.3.2 Updating the Parameter Provisioning Data Entry per AF

The request contains the IMS UE's identity (/{imsUeId}) which shall be an IMPU.

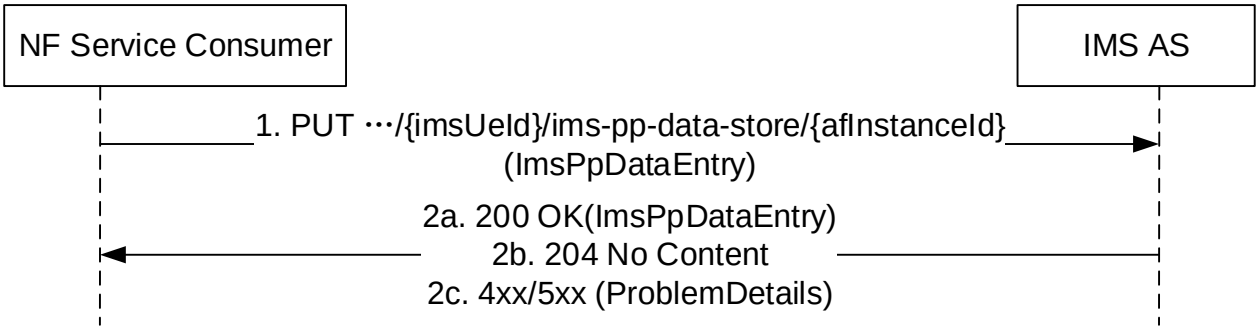



**Figure 5.6.2.3.2-1: Update the IMS Parameter Provisioning Data Entry per AF**

1. The NF Service Consumer shall send a PUT request to modify the resource that represents the IMS Parameter Provisioning Data entry for the AF identified by the afInstanceId. The request body shall contain an ImsPpDataEntry object representing the value of the modified resource.

2a., 2b. On success, the IMS AS shall return the status code "200 OK" with the resource representation of the updated resource ImsPpDataEntry or "204 No Content".

2c. On failure, one of the HTTP status codes listed in table 6.5.3.3.3.1-3 indicating the error shall be returned. For a 4xx/5xx response, the message body shall contain a ProblemDetails structure with the "cause" attribute set to one of the application errors listed in table 6.5.3.3.3.1-3.

#### 5.6.2.4 Delete Service Operation

##### 5.6.2.4.1 General

This service operation is used by the service consumer (e.g. AF, NEF) to delete the IMS user related information for a target subscriber.

##### 5.6.2.4.2 Deleting the Parameter Provisioning Data Entry per AF

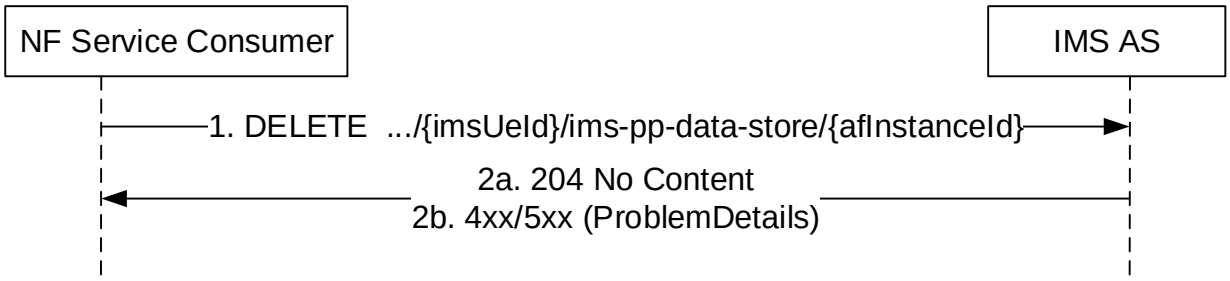



**Figure 5.6.2.4.2-1: Deleting the IMS Parameter Provisioning Data Entry per AF**

1. The NF service consumer shall send a DELETE request to delete the resource that represents the IMS Parameter Provisioning Data entry for the AF identified by the afInstanceId.

2a. On success, the request is accepted, the IMS AS shall reply with the status code "204 No Content" indicating that the resource identified by ImsPpDataEntry is successfully deleted in the response message.

2b. On failure, one of the HTTP status codes listed in table 6.5.3.3.3.2-3 indicating the error shall be returned. For a 4xx/5xx response, the message body shall contain a ProblemDetails structure with the "cause" attribute set to one of the application errors listed in table 6.5.3.3.3.2-3.

# 6 API Definitions

## 6.1 Nimsas_SessionEventControl Service API

### 6.1.1 API URI

The Nimsas_SessionEventControl service shall use the Nimsas_SessionEventControl API.

The API URI of the Nimsas_SessionEventControl API shall be:

**{apiRoot}/<apiName>/<apiVersion>**

The request URIs used in HTTP requests from the NF service consumer towards the NF service producer shall have the Resource URI structure defined in clause 4.4.1 of TS 29.501 [5], i.e.:

**{apiRoot}/<apiName>/<apiVersion>/<apiSpecificResourceUriPart>**

with the following components:

- The {apiRoot} shall be set as described in TS 29.501 [5].
- The <apiName> shall be "nimsas-sec".
- The <apiVersion> shall be "v1".
- The <apiSpecificResourceUriPart> shall be set as described in clause 6.1.3.

## 6.1.2 Usage of HTTP

### 6.1.2.1 General

HTTP/2, IETF RFC 7540 [11], shall be used as specified in clause 5 of TS 29.500 [4].

HTTP/2 shall be transported as specified in clause 5.3 of TS 29.500 [4].

The OpenAPI [6] specification of HTTP messages and content bodies for the Nimsas_SessionEventControl API is contained in Annex A.

### 6.1.2.2 HTTP standard headers

#### 6.1.2.2.1 General

See clause 5.2.2 of TS 29.500 [4] for the usage of HTTP standard headers.

#### 6.1.2.2.2 Content type

JSON, IETF RFC 8259 [12], shall be used as content type of the HTTP bodies specified in the present specification as specified in clause 5.4 of TS 29.500 [4]. The use of the JSON format shall be signalled by the content type "application/json".

"Problem Details" JSON object shall be used to indicate additional details of the error in a HTTP response body and shall be signalled by the content type "application/problem+json", as defined in IETF RFC 7807 [13].

### 6.1.2.3 HTTP custom headers

The mandatory HTTP custom header fields specified in clause 5.2.3.2 of TS 29.500 [4] shall be supported, and the optional HTTP custom header fields specified in clause 5.2.3.3 of TS 29.500 [4] may be supported.

## 6.1.3 Resources

### 6.1.3.1 Overview

This clause describes the structure for the Resource URIs and the resources and methods used for the service.

Figure 6.1.3.1-1 depicts the resource URIs structure for the Nimsas_SessionEventControl service API.

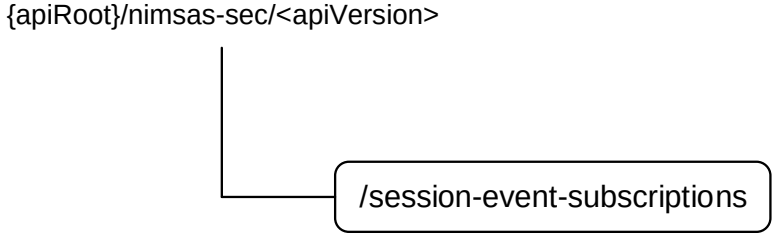



**Figure 6.1.3.1-1: Resource URI structure of the Nimsas_SessionEventControl API**

Table 6.1.3.1-1 provides an overview of the resources and applicable HTTP methods.

**Table 6.1.3.1-1: Resources and methods overview**

| Resource purpose/name | Resource URI (relative path after API URI) | HTTP method or custom operation | Description (service operation) |
|---|---|---|---|
| Session Event Subscriptions (Collection) | /session-event-subscriptions | POST | This is a pseudo resource. |

## 6.1.3.2 Resource: Session Event Subscriptions

### 6.1.3.2.1 Description

### 6.1.3.2.2 Resource Definition

Resource URI: **{apiRoot}/nimsas-sec/<apiVersion>/session-event-subscriptions**

This resource shall support the resource URI variables defined in table 6.1.3.2.2-1.

**Table 6.1.3.2.2-1: Resource URI variables for this resource**

| Name | Data type | Definition |
|---|---|---|
| apiRoot | string | See clause 6.1.1 |

### 6.1.3.2.3 Resource Standard Methods

#### 6.1.3.2.3.1 POST

This method shall support the URI query parameters specified in table 6.1.3.2.3.1-1.

**Table 6.1.3.2.3.1-1: URI query parameters supported by the POST method on this resource**

| Name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| n/a | | | | | |

This method shall support the request data structures specified in table 6.1.3.2.3.1-2 and the response data structures and response codes specified in table 6.1.3.2.3.1-3.

**Table 6.1.3.2.3.1-2: Data structures supported by the POST Request Body on this resource**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| Any | | | |

**Table 6.1.3.2.3.1-3: Data structures supported by the POST Response Body on this resource**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| n/a | | | | |
| NOTE: The mandatory HTTP error status code for the POST method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply. | | | | |

### 6.1.3.2.4 Resource Custom Operations

None.

## 6.1.4 Custom Operations without associated resources

None in this release of the specification.

## 6.1.5 Notifications

### 6.1.5.1 General

Notifications shall comply to clause 6.2 of TS 29.500 [4] and clause 4.6.2.3 of TS 29.501 [5].

**Table 6.1.5.1-1: Notifications overview**

| Notification | Callback URI | HTTP method or custom operation | Description (service operation) |
|---|---|---|---|
| Session Event Notification | {SessionEventNotificationUri} | POST | Report the session event observed. |

### 6.1.5.2 Session Event Notification

#### 6.1.5.2.1 Description

The Session Event Notification is used by the IMS AS to report the observed Session Event to notification endpoints.

#### 6.1.5.2.2 Target URI

The Callback URI "{SessionEventNotificationUri}" shall be used with the callback URI variables defined in table 6.1.5.2.2-1.

**Table 6.1.5.2.2-1: Callback URI variables**

| Name | Definition |
|---|---|
| SessionEventNotificationUri | The notification URI of the Service Consumer (i.e., DCSF) to receive the session events.<br><br>In this release, the SessionEventNotificationUri of the Service Consumer is locally configured in the IMS AS or discovered by the IMS AS via NRF (in the default notification subscription with the "DC_SESSION_EVENT_NOTIFICATION" notification type). |

#### 6.1.5.2.3 Standard Methods

##### 6.1.5.2.3.1 POST

This method shall support the request data structures specified in table 6.1.5.2.3.1-1 and the response data structures and response codes specified in table 6.1.5.2.3.1-1.

**Table 6.1.5.2.3.1-1: Data structures supported by the POST Request Body**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| SessionEventNotification | M | 1 | The session event notification to notification endpoints. |

**Table 6.1.5.2.3.1-2: Data structures supported by the POST Response Body**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| n/a | | | 204 No Content | This case represents a successful notification of the event. |
| ProblemDetails | O | 0..1 | 404 Not Found | Indicates the session event notification has failed due to application error.<br><br>The "cause" attribute may be used to indicate one of the following application errors:<br>- USER_NOT_FOUND, e.g. if the NF Service Consumer (e.g. DCSF) does not serve this service user;<br>- NOTIFICATION_URI_NOT_FOUND, if the NF Service Consumer (e.g. DCSF) considers the "SessionEventNotificationUri" is not recognized. |
| RedirectResponse | O | 0..1 | 307 Temporary Redirect | Temporary redirection.<br>(NOTE 2) |
| RedirectResponse | O | 0..1 | 308 Permanent Redirect | Permanent redirection.<br>(NOTE 2) |
| NOTE 1: The mandatory HTTP error status codes for the POST method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply.<br>NOTE 2: RedirectResponse may be inserted by an SCP, see clause 6.10.9.1 of TS 29.500 [4]. | | | | |

**Table 6.1.5.2.3-3: Headers supported by the 307 Response Code on this endpoint**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | A URI pointing to the endpoint of the NF service consumer instance to which the request should be sent. For the case, when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF instance ID towards which the notification is redirected |

**Table 6.1.5.2.3-4: Headers supported by the 308 Response Code on this endpoint**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | A URI pointing to the endpoint of the NF service consumer instance to which the request should be sent. For the case, when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF instance ID towards which the notification is redirected |

## 6.1.6 Data Model

### 6.1.6.1 General

This clause specifies the application data model supported by the API.

Table 6.1.6.1-1 specifies the data types defined for the Nimsas_SessionEventControl service based interface protocol.

**Table 6.1.6.1-1: Nimsas_SessionEventControl specific Data Types**

| Data type | Clause defined | Description |
|---|---|---|
| SessionEventNotification | 6.1.6.2.2 | The session event notification item. |
| NotificationEvent | 6.1.6.2.3 | The notification event information. |
| SessionInfo | 6.1.6.2.4 | The session information related to the notification event. |
| MediaInfo | 6.1.6.2.5 | The media information related to the notification event. |
| DcMediaSpec | 6.1.6.2.6 | The data channel media specification information which includes media description attributes. |
| QosHintInfo | 6.1.6.2.7 | The QoS requirement of the media. |
| ChargingInfo | 6.1.6.2.8 | The charging information derived from the SIP headers. |
| EventType | 6.1.6.3.3 | The type of the event for which the notification is generated. |
| MediaType | 6.1.6.3.4 | The type of the media for which the notification is generated. |
| SessionCase | 6.1.6.3.5 | The type of the session case. |
| EventInitiator | 6.1.6.3.6 | The initiator of the event for which the notification is generated. |
| PsDataOffStatus | 6.1.6.3.7 | The active/inactive status of PS Data Off for a served UE. |

Table 6.1.6.1-2 specifies data types re-used by the Nimsas_SessionEventControl service based interface protocol from other specifications, including a reference to their respective specifications and when needed, a short description of their use within the Nimsas_SessionEventControl service based interface.

**Table 6.1.6.1-2: Nimsas_SessionEventControl re-used Data Types**

| Data type | Reference | Comments |
|---|---|---|
| ImsPublicId | TS 29.562 [15] | IMS Public Identity. |
| DcStream | TS 29.571 [16] | Data Channel mapping and configuration information. |
| MaxMessageSize | TS 29.571 [16] | Maximum message size of to be expected. |
| DcEndpoint | TS 29.571 [16] | local or remote endpoint for the Data Channel. |
| SessionId | TS 29.571 [16] | IMS Session Identity. |
| MediaId | TS 29.571 [16] | IMS media flow identity. |
| BdcUsedBy | TS 29.571 [16] | The party using the bootstrap data channel. |
| AppBindingInfo | TS 29.571 [16] | The application binding information of the application data channel. |
| MediaInfoExternal | 6.3.6.2.4 | The media related information. |
| RatType | TS 29.571 [16] | RAT type. |
| Ncgi | TS 29.571 [16] | NR Cell Global Identity. |
| Ecgi | TS 29.571 [16] | E-UTRA Cell Global Identity |
| PlmnId | TS 29.571 [16] | PLMN ID. |

## 6.1.6.2 Structured data types

### 6.1.6.2.1 Introduction

This clause defines the structures to be used in resource representations.

6.1.6.2.2 Type: SessionEventNotification

**Table 6.1.6.2.2-1: Definition of type SessionEventNotification**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| notificationEvent | NotificationEvent | M | 1 | The NotificationEvent corresponds to a report on one subscribed event. |
| sessionId | SessionId | M | 1 | The session ID is the identity of the IMS session for which the event relates to.<br>This IE contains the information in the Call-ID header which is the typical header of SIP message. |
| sessionInfo | SessionInfo | C | 0..1 | The sessionInfo is the IMS session related information, e.g. the calling and called identity. It shall be contained when the attribute eventType of the notificationEvent is one of the "SESSION_ESTABLISHMENT_REQUEST", "MEDIA_CHANGE_REQUEST", "EXTERNAL_SESSION_CREATE", "SESSION_ESTABLISHMENT_SUCCESS" (NOTE). |
| mediaInfoList | map(MediaInfo) | C | 1..N | Media info list includes a list of media related information, e.g. media identity, media type and media specifications. The mediaId attribute within the MediaInfo data type is the key of the map.<br>It shall be contained when the attribute eventType of the notificationEvent is one of the "SESSION_ESTABLISHMENT_REQUEST", "SESSION_ESTABLISHMENT_PROGRESS", "SESSION_ESTABLISHMENT_ALERTING", "SESSION_ESTABLISHMENT_SUCCESS", "MEDIA_CHANGE_REQUEST", "MEDIA_CHANGE_SUCCESS", "MEDIA_CHANGE_FAILURE", or "MEDIA_CHANGE_CANCEL". |
| psDataOff | PsDataOffStatus | C | 0..1 | It indicates the active/inactive status of PS Data Off for the served UE.<br>It shall be present if the attribute eventType of the notificationEvent is set to one of the following:<br>- "SESSION_ESTABLISHMENT_REQUEST", when IMS session establishment is invoked;<br>- "PS_DATA_OFF_STATUS_CHANGE", when the status change of PS Data Off happens without any other changes to the IMS session;<br>It may be present in other cases if the change of PS Data Off status happens together with other events (e.g. media changes). |
| mediaInfoSet | array(MediaInfoExternal) | C | 0..1 | Media information set includes a set of media information.<br>It shall be contained when the attribute eventType of the notificationEvent is one of the "EXTERNAL_SESSION_UPDATE" "EXTERNAL_SESSION_CREATE" |
| chargingInfo | ChargingInfo | O | 0..1 | Charging information derived from SIP headers.<br>It may be contained when the eventType attribute of the notificationEvent is one of the following:<br>- "SESSION_ESTABLISHMENT_REQUEST"<br>- "MEDIA_CHANGE_REQUEST"<br>- "EXTERNAL_SESSION_UPDATE" "EXTERNAL_SESSION_CREATE" |
| NOTE: The sessionInfo shall be present when the attribute eventType of the notificationEvent is set to "SESSION_ESTABLISHMENT_SUCCESS" and the originating IMS AS received the redirected-to party identity via the History-Info header field in the 200 OK responses to the initial INVITE request. | | | | |

#### 6.1.6.2.3 Type: NotificationEvent

**Table 6.1.6.2.3-1: Definition of type NotificationEvent**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| eventType | EventType | M | 1 | The event type of the event for which the notification is generated. |
| eventInitiator | EventInitiator | C | 0..1 | Event initiator indicates initiator of the event, e.g. "served IMS subscriber", "remote IMS subscriber" or "external party".<br>It shall be contained when the eventType is set to one of the following:<br>- "SESSION_ESTABLISHMENT_REQUEST",<br>- "MEDIA_CHANGE_REQUEST",<br>- "EXTERNAL_SESSION_CREATE". |

#### 6.1.6.2.4 Type: SessionInfo

**Table 6.1.6.2.4-1: Definition of type SessionInfo**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| callingIdentity | ImsPublicId | O | 0..1 | The IMS public identity of the calling IMS subscriber, or a specific Public Service Identity.<br>(NOTE 1) |
| calledIdentity | ImsPublicId | O | 0..1 | The IMS public identity of the called IMS Subscriber, or a specific Public Service Identity of DCSF.<br>(NOTE 2) |
| sessionCase | SessionCase | O | 0..1 | Indicator to indicate that whether this is an originating IMS session, a terminating IMS session or external initiated IMS session. |
| NOTE 1: The value of the callingIdentity shall be set to the value of the P-Asserted-Identity header field of the received initial INVITE request, see TS 24.186 [19] clause 9.3.2.2 and clause 9.3.3.2.<br>NOTE 2: The value of the calledIdentity shall be set to the Request URI of the received initial INVITE request, see TS 24.186 [19] clause 9.3.2.2 and clause 9.3.3.2. If communication diversion occurs and the originating IMS AS received the History-Info header field in after receiving the 200 OK responses to the initial INVITE request, the originating IMS AS shall set the calledIdentity attribute based on the received History-Info header field, see TS 24.604 [21] clause 4.5.2.7. | | | | |

#### 6.1.6.2.5 Type: MediaInfo

**Table 6.1.6.2.5-1: Definition of type MediaInfo**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| mediaId | MediaId | M | 1 | The media ID uniquely identifies this media item within the list. The identity is allocated by IMS AS. |
| mediaType | MediaType | M | 1 | The media type of the media for which the notification is generated. |
| dcMediaSpec | DcMediaSpec | C | 0..1 | The data channel media specification includes the relevant media attributes of interest to the consumer.<br>It shall be contained if the mediaType is set to "DC". |
| mediaSuspended | boolean | C | 0..1 | This IE indicates whether the media is suspended or not.<br><br>It shall be contained if the media is suspended or resumed as specified in TS 24.610 [18] and TS 24.186 [19].<br>- true: the media is suspended.<br><br>- false: the media is resumed. |
| mediaRemovalInd | boolean | C | 0..1 | This IE indicates whether the media is to be removed or not.<br><br>It shall be only present with true value if the media is to be removed, e.g. removal of bootstrap data channel media during the interaction with supplementary service, or removal of application data channel in other cases.<br><br>Presence with false value shall be prohibited. |

#### 6.1.6.2.6 Type: DcMediaSpec

**Table 6.1.6.2.6-1: Definition of type DcMediaSpec**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| streams | map(DcStream) | M | 1..N | Represents the data channel mapping and configuration information when originating/terminating data channel media flows on the Mb interface.<br>The streamId attribute within the DcStream data type is the key of the map.<br>(NOTE) |
| maxMessageSize | MaxMessageSize | O | 0..1 | Represents the maximum message size of to be expected.<br>(NOTE) |
| receivedDcEndpoint | DcEndpoint | M | 0..1 | Represents the SCTP and DTLS endpoint information for the Data Channel received from the SDP offer/answer.<br>(NOTE) |
| bdcUsedby | BdcUsedBy | C | 0..1 | Represents the party using the bootstrap data channel. It shall be present when the mediaId in the data type MediaInfo represents bootstrap data channel and the streamId set to 100 and 110. |
| appBindingInfo | AppBindingInfo | C | 0..1 | Represents the application binding information of the application data channel.<br><br>It shall be present when the mediaId in the data type MediaInfo represents application data channel, and the media is not multiplexed. |
| multiAppBindingInfo | array(AppBindingInfo) | C | 1..N | Represents the multiple application binding informations.<br><br>It shall be present when the mediaId in the data type MediaInfo represents application data channel and the media is multiplexed.<br><br>When present, the appDcInfoList within each of the AppBindingInfo shall be included. |
| qosHintInfo | QosHintInfo | C | 0..1 | Represents the QoS information that is provided within the SDP media level attribute "a=3gpp-qos-hint".<br>It shall be present when the received SDP offer and SDP answer contains the "a=3gpp-qos-hint" attribute line. |
| NOTE: The attribute and the value are derived from the SDP received by the IMS AS in a SIP INVITE or a re-INVITE request and the corresponding response related to an IMS Data Channel. | | | | |

#### 6.1.6.2.7 Type: QosHintInfo

**Table 6.1.6.2.7-1: Definition of type QosHintInfo**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| loss | number | O | 0..1 | Represents the maximum desirable end-to-end transport level packet loss rate in percent (without "%" sign) as a zero-based-integer or as a non-zero-real value. |
| localLoss | number | O | 0..1 | Represents the maximum desirable end-to-end transport level packet loss rate applied across the local link.<br>It may be included if the loss is included. |
| latency | number | O | 0..1 | Represents the maximum desirable end-to-end transport level packet latency in milliseconds as a zero-based-integer or as a non-zero-real value. |
| localLatency | number | O | 0..1 | Represents the maximum desirable end-to-end transport level packet latency applied across the local link.<br>It may be included if the loss is included. |

#### 6.1.6.2.8 Type: ChargingInfo

**Table 6.1.6.2.8-1: Definition of type ChargingInfo**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| ratTypes | array(RatType) | O | 1..N | When present, it indicates a list of RAT types. The IMS AS derives it from the access-type field of the P-Access-Network-Info header as specified in TS 24.229 [23]. |
| ncgi | Ncgi | O | 0..1 | When present, it indicates the NR Cell Global Identity. The IMS AS derives it from P-header Cellular-Network-Info header as specified in TS 24.229 [23]. |
| ecgi | Ecgi | O | 0..1 | When present, it indicates the E-UTRA Cell Global Identity. The IMS AS derives from P-header Cellular-Network-Info header as specified in TS 24.229 [23]. |
| imsServiceType | string | O | 0..1 | When present, it indicates the IMS service type identified by IMS Communication Service Identifier. The IMS AS derives it from P-Asserted-service header as specified in TS 24.229 [23]. |
| servingPlmnId | PlmnId | O | 0..1 | When present, it indicates the serving PLMN ID. The IMS AS derives it from P-Visited-Network-ID header as specified in TS 24.229 [23]. |

### 6.1.6.3 Simple data types and enumerations

#### 6.1.6.3.1 Introduction

This clause defines simple data types and enumerations that can be referenced from data structures defined in the previous clauses.

#### 6.1.6.3.2 Simple data types

The simple data types defined in table 6.1.6.3.2-1 shall be supported.

**Table 6.1.6.3.2-1: Simple data types**

| Type Name | Type Definition | Description | Applicability |
|---|---|---|---|
| | | | |

#### 6.1.6.3.3 Enumeration: EventType

The enumeration EventType represents the type of the event for which the notification is generated. It shall comply with the provisions defined in table 6.1.6.3.3-1.

**Table 6.1.6.3.3-1: Enumeration EventType**

| Enumeration value | Description |
|---|---|
| "SESSION_ESTABLISHMENT_REQUEST" | Session establishment request event. |
| "SESSION_ESTABLISHMENT_PROGRESS" | Session establishment progress event. |
| "SESSION_ESTABLISHMENT_ALERTING" | Session establishment alerting event. |
| "SESSION_ESTABLISHMENT_SUCCESS" | Session establishment success event. |
| "SESSION_ESTABLISHMENT_FAILURE" | Session establishment failure event. |
| "SESSION_ESTABLISHMENT_CANCEL" | Session establishment cancel event. |
| "MEDIA_CHANGE_REQUEST" | Media change request event. |
| "MEDIA_CHANGE_SUCCESS" | Media change success event. |
| "MEDIA_CHANGE_FAILURE" | Media change failure event. |
| "MEDIA_CHANGE_CANCEL" | Media change cancel event. |
| "SESSION_TERMINATION" | Session termination event. |
| "EXTERNAL_SESSION_UPDATE" | External session update event. |
| "EXTERNAL_SESSION_CREATE" | External session create event. |
| "PS_DATA_OFF_STATUS_CHANGE" | PS Data Off status change event. |

#### 6.1.6.3.4 Enumeration: MediaType

The enumeration MediaType represents the type of the media for which the notification is generated. It shall comply with the provisions defined in table 6.1.6.3.4-1.

**Table 6.1.6.3.4-1: Enumeration MediaType**

| Enumeration value | Description |
|---|---|
| "DC" | Data channel media type. |
| "AUDIO" | Audio media type. |
| "VIDEO" | Video media type. |

#### 6.1.6.3.5 Enumeration: SessionCase

The enumeration SessionCase represents the type of the session to for which the notification is generated. It shall comply with the provisions defined in table 6.1.6.3.5-1.

**Table 6.1.6.3.5-1: Enumeration SessionCase**

| Enumeration value | Description |
|---|---|
| "ORIGINATING_IMS_SESSION" | Originating IMS session. |
| "TERMINATING_IMS_SESSION" | Terminating IMS session. |
| "EXTERNAL_INITIATED_IMS_SESSION" | External party initiated IMS session. |

#### 6.1.6.3.6 Enumeration: EventInitiator

The enumeration EventInitiator represents the initiator of the event for which the notification is generated. It shall comply with the provisions defined in table 6.1.6.3.6-1.

**Table 6.1.6.3.6-1: Enumeration EventInitiator**

| Enumeration value | Description |
|---|---|
| "SERVED_IMS_SUBSCRIBER" | Served IMS subscriber. |
| "REMOTE_IMS_SUBSCRIBER" | Remote IMS subscriber. |
| "EXTERNAL_PARTY" | External party, i.e. 3rd party. |

#### 6.1.6.3.7 Enumeration: PsDataOffStatus

The enumeration PsDataOffStatus represents the active/inactive status of PS Data Off for a served UE. It shall comply with the provisions defined in table 6.1.6.3.7-1.

**Table 6.1.6.3.7-1: Enumeration PsDataOffStatus**

| Enumeration value | Description |
|---|---|
| "ACTIVE" | PS Data Off status is set to active. |
| "INACTIVE" | PS Data Off status is set to inactive. |

### 6.1.6.4 Data types describing alternative data types or combinations of data types

None in this release of the specification.

### 6.1.6.5 Binary data

None in this release of the specification.

## 6.1.7 Error Handling

### 6.1.7.1 General

For the Nimsas_SessionEventControl API, HTTP error responses shall be supported as specified in clause 4.8 of TS 29.501 [5]. Protocol errors and application errors specified in table 5.2.7.2-1 of TS 29.500 [4] shall be supported for an HTTP method if the corresponding HTTP status codes are specified as mandatory for that HTTP method in table 5.2.7.1-1 of TS 29.500 [4].

In addition, the requirements in the following clauses are applicable for the Nimsas_SessionEventControl API.

### 6.1.7.2 Protocol Errors

No specific procedures for the Nimsas_SessionEventControl service are specified.

### 6.1.7.3 Application Errors

The application errors defined for the Nimsas_SessionEventControl service are listed in Table 6.1.7.3-1.

**Table 6.1.7.3-1: Application errors**

| Application Error | HTTP status code | Description |
|---|---|---|
| USER_NOT_FOUND | 404 Not Found | The service user is not served by this DCSF, e.g. no data channel configurations for this service user is found. |
| NOTIFICATION_URI_NOT_FOUND | 404 Not Found | The session event notification URI is not recognized by the NF service consumer (i.e. DCSF). |

## 6.1.8 Feature negotiation

The optional features in table 6.1.8-1 are defined for the Nimsas_SessionEventControl API. They shall be negotiated using the extensibility mechanism defined in clause 6.6 of TS 29.500 [4].

**Table 6.1.8-1: Supported Features**

| Feature number | Feature Name | Description |
|---|---|---|
| | | |

## 6.1.9 Security

As indicated in TS 33.501 [8] and TS 29.500 [4], the access to the Nimsas_SessionEventControl API may be authorized by means of the OAuth2 protocol (see IETF RFC 6749 [9]), based on local configuration, using the "Client Credentials" authorization grant, where the NRF (see TS 29.510 [10]) plays the role of the authorization server.

If OAuth2 is used, an NF Service Consumer, prior to consuming services offered by the Nimsas_SessionEventControl API, shall obtain a "token" from the authorization server, by invoking the Access Token Request service, as described in TS 29.510 [10], clause 5.4.2.2.

NOTE: When multiple NRFs are deployed in a network, the NRF used as authorization server is the same NRF that the NF Service Consumer used for discovering the Nimsas_SessionEventControl service.

The Nimsas_SessionEventControl API defines a single scope "nimsas-sessioneventcontrol" for the entire service, and it does not define any additional scopes at resource or operation level.

## 6.1.10 HTTP redirection

An HTTP request may be redirected to a different IMS AS service instance when using direct or indirect communications (see TS 29.500 [4]).

An SCP that reselects a different NF consumer (e.g. DCSF) instance will return the NF Instance ID of the new NF consumer instance in the 3gpp-Sbi-Producer-Id header, as specified in clause 6.10.3.4 of TS 29.500 [4].

If an NF consumer (e.g. DCSF) redirects a service request to a different NF consumer using an 307 Temporary Redirect or 308 Permanent Redirect status code, the identity of the new NF consumer towards which the service request is redirected shall be indicated in the 3gpp-Sbi-Target-Nf-Id header of the 307 Temporary Redirect or 308 Permanent Redirect response as specified in clause 6.10.9.1 of TS 29.500 [4].

# 6.2 Nimsas_MediaControl Service API

## 6.2.1 API URI

The Nimsas_MediaControl service shall use the Nimsas_MediaControl API.

The API URI of the Nimsas_MediaControl API shall be:

**{apiRoot}/<apiName>/<apiVersion>**

The request URIs used in HTTP requests from the NF service consumer towards the NF service producer shall have the Resource URI structure defined in clause 4.4.1 of TS 29.501 [5], i.e.:

**{apiRoot}/<apiName>/<apiVersion>/<apiSpecificResourceUriPart>**

with the following components:

- The {apiRoot} shall be set as described in TS 29.501 [5].
- The <apiName> shall be "nimsas-mc".
- The <apiVersion> shall be "v1".
- The <apiSpecificResourceUriPart> shall be set as described in clause 6.2.3.

## 6.2.2 Usage of HTTP

### 6.2.2.1 General

HTTP/2, IETF RFC 7540 [11], shall be used as specified in clause 5 of TS 29.500 [4].

HTTP/2 shall be transported as specified in clause 5.3 of TS 29.500 [4].

The OpenAPI [6] specification of HTTP messages and content bodies for the Nimsas_MediaControl API is contained in Annex A.

### 6.2.2.2 HTTP standard headers

#### 6.2.2.2.1 General

See clause 5.2.2 of TS 29.500 [4] for the usage of HTTP standard headers.

#### 6.2.2.2.2 Content type

JSON, IETF RFC 8259 [12], shall be used as content type of the HTTP bodies specified in the present specification as specified in clause 5.4 of TS 29.500 [4]. The use of the JSON format shall be signalled by the content type "application/json".

"Problem Details" JSON object shall be used to indicate additional details of the error in a HTTP response body and shall be signalled by the content type "application/problem+json", as defined in IETF RFC 7807 [13].

#### 6.2.2.3 HTTP custom headers

The mandatory HTTP custom header fields specified in clause 5.2.3.2 of TS 29.500 [4] shall be supported, and the optional HTTP custom header fields specified in clause 5.2.3.3 of TS 29.500 [4] may be supported.

### 6.2.3 Resources

#### 6.2.3.1 Overview

This clause describes the structure for the Resource URIs and the resources and methods used for the service.

Figure 6.2.3.1-1 depicts the resource URIs structure for the Nimsas_MediaControl service API.

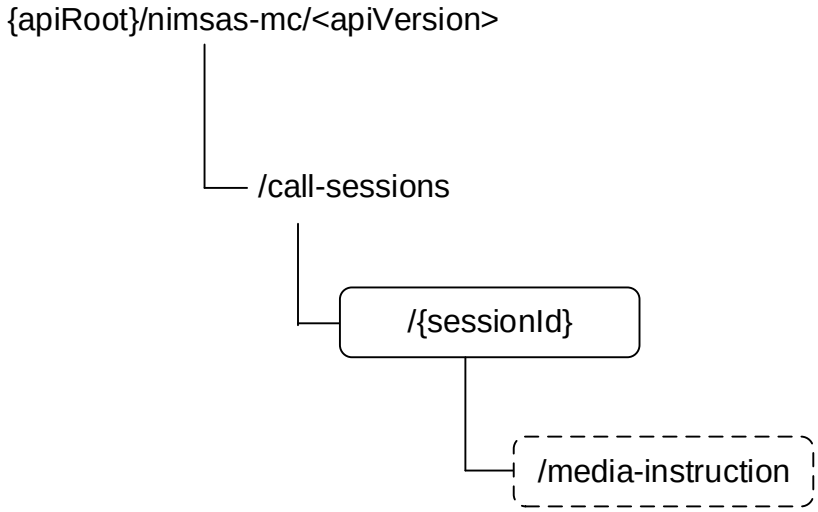



**Figure 6.2.3.1-1: Resource URI structure of the Nimsas_MediaControl API**

Table 6.2.3.1-1 provides an overview of the resources and applicable HTTP methods.

**Table 6.2.3.1-1: Resources and methods overview**

| Resource purpose/name | Resource URI (relative path after API URI) | HTTP method or custom operation | Description (service operation) |
|---|---|---|---|
| Individual call session (Document) | /call-sessions/{sessionId}/media-instruction | media-instruction (POST) | Instructs the IMS AS on how to control the media resources. |

#### 6.2.3.2 Resource: Individual call session

##### 6.2.3.2.1 Description

This resource represents an individual call session created in the IMS AS.

This resource is modelled with the Document resource archetype (see clause C.1 of TS 29.501 [5]).

##### 6.2.3.2.2 Resource Definition

Resource URI: **{apiRoot}/nimsas-mc/<apiVersion>/call-sessions/{sessionId}**

This resource shall support the resource URI variables defined in table 6.2.3.2.2-1.

**Table 6.2.3.2.2-1: Resource URI variables for this resource**

| Name | Data type | Definition |
|---|---|---|
| apiRoot | string | See clause 6.2.1 |
| sessionId | SessionId | Session ID assigned by the IMS AS during the IMS session setup and will be notified to consumer via Nimsas_SessionEventControl service. The consumer shall reuse the session ID it received from the IMS AS for referencing the same session. |

### 6.2.3.2.3 Resource Standard Methods

None.

### 6.2.3.2.4 Resource Custom Operations

#### 6.2.3.2.4.1 Overview

**Table 6.2.3.2.4.1-1: Custom operations**

| Operation Name | Custom operation URI | Mapped HTTP method | Description |
|---|---|---|---|
| media-instruction | {resourceUri}/media-instruction | POST | Update call session service operation. |

#### 6.2.3.2.4.2 Operation: media-instruction

##### 6.2.3.2.4.2.1 Description

##### 6.2.3.2.4.2.2 Operation Definition

This custom operation updates an individual call session resource and provide a set of media instruction information for a given IMS session, towards IMS AS.

This operation shall support the request data structures specified in table 6.2.3.2.4.2.2-1 and the response data structure and response codes specified in table 6.2.3.2.4.2.2-2.

**Table 6.2.3.2.4.2.2-1: Data structures supported by the POST Request Body on this resource**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| MediaInstructionData | M | 1 | Representation of the updates to apply to the call session. |

**Table 6.2.3.2.4.2.2-2: Data structures supported by the POST Response Body on this resource**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| MediaInstructionData | C | 0..1 | 200 OK | Successful update of the call session, when the IMS AS needs to return information in the response. |
| n/a | | | 204 No Content | Successful update of the call session, when the IMS AS does not need to return information in the response. |
| RedirectResponse | O | 0..1 | 307 Temporary Redirect | Temporary redirection.<br>(NOTE 2) |
| RedirectResponse | O | 0..1 | 308 Permanent Redirect | Permanent redirection.<br>(NOTE 2) |
| ProblemDetails | O | 0..1 | 400 Bad Request | Indicates the media instruction has failed due to application error.<br><br>The "cause" attribute may be used to indicate one of the following application errors:<br>- MEDIA_ID_NOT_FOUND |
| NOTE 1: The mandatory HTTP error status codes for the POST method listed in table 5.2.7.1-1 of TS 29.500 [4] also apply.<br>NOTE 2: RedirectResponse may be inserted by an SCP, see clause 6.10.9.1 of TS 29.500 [4]. | | | | |

**Table 6.2.3.3.4.2.2-3: Headers supported by the 307 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | A URI pointing to the endpoint of the NF service consumer instance to which the request should be sent. For the case, when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF instance ID towards which the request is redirected. |

**Table 6.2.3.3.4.2.2-4: Headers supported by the 308 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | A URI pointing to the endpoint of the NF service consumer instance to which the request should be sent. For the case, when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target SMF (service) instance ID towards which the request is redirected. |

## 6.2.4 Custom Operations without associated resources

None in this release of the specification.

## 6.2.5 Notifications

None in this release of the specification.

## 6.2.6 Data Model

### 6.2.6.1 General

This clause specifies the application data model supported by the API.

Table 6.2.6.1-1 specifies the data types defined for the Nimsas_MediaControl service based interface protocol.

**Table 6.2.6.1-1: Nimsas_MediaControl specific Data Types**

| Data type | Clause defined | Description |
|---|---|---|
| ArMediaSpecification | 6.2.6.2.5 | The AR media specification information needed for AR communication services from application layer. |
| AudioVideoReNegotiationInd | 6.2.6.2.7 | The audio/video media re-negotiation indication information to connect the UE's audio/video media stream to MF. |
| DcMediaSpecification | 6.2.6.2.4 | The data channel media specification information needed for data channel media stream from application layer. |
| InterworkingInstruction | 6.2.6.2.8 | The interworking instruction of data channel transcoding. |
| Mdc2EndpointInfo | 6.2.6.2.6 | The MDC2 media endpoint information. |
| Mdc2AudioVideoEndpoint | 6.2.6.2.9 | The endpoint information of DC AS for the audio or video media in MDC2 interface. |
| MediaInstruction | 6.2.6.3.3 | The media instruction for handling a specific media flow. |
| MediaInstructionData | 6.2.6.2.2 | The media instruction data for a specific session. |
| MediaInstructions | 6.2.6.2.3 | The media instructions for a specific media flow. |
| MediaConnSide | 6.2.6.3.4 | The side that the audio or video media need to be connected to MF. E.g., both the originating and terminating side. |
| TranscodeMode | 6.2.6.3.5 | The mode of data channel transcoding. |
| SendRecvMode | 6.2.6.3.6 | The send and recv mode of the media stream. |

Table 6.2.6.1-2 specifies data types re-used by the Nimsas_MediaControl service based interface protocol from other specifications, including a reference to their respective specifications and when needed, a short description of their use within the Nimsas_MediaControl service based interface.

**Table 6.2.6.1-2: Nimsas_MediaControl re-used Data Types**

| Data type | Reference | Comments |
|---|---|---|
| DcStream | TS 29.571 [16] | Data Channel mapping and configuration information. |
| MdcEndpoint | TS 29.571 [16] | The MDC1 and MDC2 Endpoint information. |
| MaxMessageSize | TS 29.571 [16] | Maximum SCTP user message size. |
| Mdc2Protocol | TS 29.176 [17] | Transport layer protocols for MDC2 interface. |
| MediaId | TS 29.571 [16] | IMS media flow identity. |
| MediaIdRm | TS 29.571 [16] | IMS media flow identity. |
| MediaProxy | TS 29.571 [16] | Media proxy configuration applicable to the media flow. |
| MediaResourceType | TS 29.571 [16] | IMS media resource type. |
| ReplaceHttpUrl | TS 29.571 [16] | A list of replacement HTTP URL per stream ID. |
| SessionId | TS 29.571 [16] | IMS session identity. |
| Uri | TS 29.571 [16] | URI |
| BdcUsedBy | TS 29.571 [16] | The party using the bootstrap data channel. |
| AppBindingInfo | TS 29.571 [16] | The application binding information of the application data channel. |
| QosHintInfo | 6.1.6.2.7 | The QoS hint information of the media. |
| AvatarMedia | TS 29.176 [17] | Avatar media specification. |
| Endpoint | TS 29.571 [16] | Endpoint information. |
| AudioVideoMedia | TS 29.571 [16] | The description of audio and video media. |

### 6.2.6.2 Structured data types

#### 6.2.6.2.1 Introduction

This clause defines the structures to be used in resource representations.

#### 6.2.6.2.2 Type: MediaInstructionData

**Table 6.2.6.2.2-1: Definition of type MediaInstructionData**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| sessionId | SessionId | M | 1 | The session ID is the identity of the IMS session for which the MediaInstructions applies. |
| mediaInstructionSet | map(MediaInstructions) | M | 1..N | The mediaInstructionSet includes a set of instructions for each media flow to control. The key of the map shall be a (unique) valid JSON string per clause 7 of IETF RFC 8259 [20], with a maximum of 32 characters. |

#### 6.2.6.2.3 Type: MediaInstructions

**Table 6.2.6.2.3-1: Definition of type MediaInstructions**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| mediaId | MediaIdRm | M | 1 | The media ID uniquely identifies this media item within the list and used by the producer (IMS AS) and the consumer for referencing purposes. The consumer reuses the media ID it received from the IMS AS for referencing the same media. This field shall be null for instructions related to originating new media or for data channel media components de-multiplexed from a data channel multiplexing media which do not reuse the media ID of that data channel multiplexing media. |
| associatedMediaId | MediaId | C | 0..1 | The identifier of the original multiplexed media stream which is de-multiplexed.<br>It shall be included only when the mediaId is null, mediaInstruction is set to "ORIGINATE" and the media stream identified by the value of this attribute is de-multiplexed. |
| mediaResourceType | MediaResourceType | M | 1 | The mediaResourceType identify the Media Resource types the Media instruction is intended for. |
| mediaInstruction | MediaInstruction | C | 0..1 | It shall be included if the mediaResourceType is set to "DC". When present, mediaInstruction includes instructions to the producer (IMS AS) for handling the media. |
| dcMediaSpecification | DcMediaSpecification | C | 0..1 | Represents the additional media specifications needed for data channel media stream from application layer.<br>It shall be contained if the mediaResourceType is set to "DC". |
| arMediaSpecification | ArMediaSpecification | O | 0..1 | Represents the additional media specifications needed for AR communication services from application layer. |
| mediaProcessingUrl | Uri | C | 0..1 | The mediaProcessingURL indicates the address where MF receive service-related media instructions. This field shall be contained in the response when originating a new media. |
| audioVideoReNegotiationInd | AudioVideoReNegotiationInd | C | 0..1 | Represents the audio/video media re-negotiation indication information to connect the UE's audio/video media stream to MF which is required by the application data channel.<br><br>It shall be included if the mediaResourceType is set to "AUDIO" or "VIDEO". |
| mdc2AudioEndpoint | Mdc2AudioVideoEndpoint | C | 0..1 | Represents the audio media endpoint information in MDC2 interface.<br>It shall be included when the attribute renderingMode within the avatarMedia is set to "NET_CENTRIC_DCAS". It may be included in other cases. |
| mdc2VideoEndpoint | Mdc2AudioVideoEndpoint | C | 0..1 | Represents the video media endpoint information in MDC2 interface.<br>It shall be included when the attribute renderingMode within the avatarMedia is set to "NET_CENTRIC_DCAS". It may be included in other cases. |
| avatarMedia | AvatarMedia | O | 0..1 | Represents the additional media specifications needed for avatar communication services. |

#### 6.2.6.2.4 Type: DcMediaSpecification

**Table 6.2.6.2.4-1: Definition of type DcMediaSpecification**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| mediaProxyConfig | MediaProxy | M | 1 | Represents the media proxy configuration on the MF.<br>The value shall be set to "HTTP_PROXY" if the mediaId represents the bootstrap data channel.<br>If the mediaId represents the application data channel, the values "HTTP_PROXY", "UDP_PROXY" and "DC_APPLICATION_PROXY" are applicable. (NOTE 2) |
| replaceHttpUrls | map(ReplaceHttpUrl) | C | 1..N | Represents a list of replacement HTTP URL per stream ID allocated by the application layer for the specific IMS subscriber when requesting the application list (e.g. graphical user interface) via the MDC1 interface.<br>It shall be included if the mediaId represents a bootstrap data channel and streamId is 0 or 100.<br>The streamId attribute within the ReplaceHttpUrl data type is the key of the map. |
| mdc1EndpointDcsf | MdcEndpoint | C | 0..1 | Represents the MDC1 media endpoint information reserved on DCSF, e.g., the IP address and port number of DCSF.<br>It shall be included in the request of the media instruction if the mediaId represents a bootstrap data channel.<br>(NOTE 1) |
| mdc1EndpointMf | MdcEndpoint | C | 0..1 | Represents the negotiated MDC1 media endpoint information on MF, e.g., the IP address and port number of DC capable MF.<br>It shall be included in the response of the media instruction if the mediaId represents a bootstrap data channel.<br>(NOTE 2) |
| mdc2EndpointInfo | Mdc2EndpointInfo | C | 0..1 | Represents the negotiated MDC2 media endpoint information, e.g., the protocol stack, IP address and port number.<br>It shall be included if the mediaId represents an application data channel involving DC AS, e.g. P2A or P2A2P application data channel.<br>If the mediaId represents an application data channel and this attribute is not included, this media is assumed to represent a P2P application data channel. |
| streams | map(DcStream) | M | 1..N | Represents the data channel mapping and configuration information when originating/terminating data channel media flows on the Mb interface.<br>The streamId attribute within the DcStream data type is the key of the map. |
| bdcUsedby | BdcUsedBy | C | 0..1 | Represents the party using the bootstrap data channel. It shall be contained when the mediaId in the data type MediaInstructions represents bootstrap data channel and the streamId sets to 100 and 110. |
| appBindingInfo | AppBindingInfo | C | 0..1 | Represents the application binding information of the application data channel. It shall be contained when the mediaId in the data type MediaInstrutions represents application data channel. |
| qosHintInfo | QosHintInfo | O | 0..1 | Represents the QoS information that is provided within the SDP media level attribute "a=3gpp-qos-hint". |
| interworkingInfo | InterworkingInstruction | O | 0..1 | Represents how to transcode the data channel media via MF. |
| dcIwkInd | boolean | O | 0..1 | Indicates the DC media is for DC interworking via DC AS. This attribute is set to "true" if the DC media is for DC interworking via DC AS. If the attribute is not present it means it is "false". |
| NOTE 1: The MDC1 interface shall use "TCP/TLS/HTTP" protocol stack and the corresponding IP, TCP, TLS and HTTP related attributes in MdcEndpoint should be included.<br>NOTE 2: The media proxy configuration is specified in the clause AC.6 of TS 23.228 [14]. | | | | |

#### 6.2.6.2.5 Type: ArMediaSpecification

**Table 6.2.6.2.5-1: Definition of type ArMediaSpecification**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| mediaProcessingSpec | string | M | 1 | It specifies how the AR media stream should be processed. It indicates how the MF assists in the AR media rendering function. |

#### 6.2.6.2.6 Type: Mdc2EndpointInfo

**Table 6.2.6.2.6-1: Definition of type Mdc2EndpointInfo**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| mdc2Protocol | Mdc2Protocol | C | 0..1 | Represents the transport layer protocol or protocol stack for MDC2 interface when the "HTTP_PROXY" media proxy configuration is used in MF.<br><br>It shall be included if the value of the mediaProxyConfig within the data type DcMediaSpecification is set to "HTTP_PROXY". |
| mdc2EndpointDcAs | MdcEndpoint | C | 0..1 | Represents the negotiated MDC2 media endpoint information on DC AS.<br>It shall be included in the POST request message for the media control.<br>The attributes "tlsId", "sctpPort" and "fingerprint" within the MdcEndpoint shall be present when the value of the mediaProxyConfig within the DcMediaSpecification is set to "UDP_PROXY" or "DC_APPLICATION_PROXY". (NOTE) |
| mdc2EndpointMf | MdcEndpoint | C | 0..1 | Represents the negotiated MDC2 media endpoint information on MF.<br>It shall be included in the successful POST response message for the media control API.<br>The attributes "tlsId", "sctpPort" and "fingerprint" within the MdcEndpoint shall not be present when the value of the mediaProxyConfig within the data type DcMediaSpecification is set to "UDP_PROXY".<br>The attributes "tlsId", "sctpPort" and "fingerprint" within the MdcEndpoint shall be present when the value of the mediaProxyConfig within the data type DcMediaSpecification is set to "DC_APPLICATION_PROXY". (NOTE) |
| NOTE: When the mdc2Protocol is set to "UDP/DTLS/SCTP", "TCP/TLS" or "SCTP/DTLS", the attributes "tlsId" and "fingerprint" within the data type MdcEndpoint shall be included, and the "securitySetup" may be included. When the value of mdc2Protocol is set to " UDP/DTLS/SCTP", the attribute "sctpPort" within the data type MdcEndpoint shall be included. | | | | |

#### 6.2.6.2.7 Type: AudioVideoReNegotiationInd

**Table 6.2.6.2.7-1: Definition of type AudioVideoReNegotiationInd**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| reNegotiation | boolean | M | 1 | When present, the IE shall indicate whether the audio or video media re-negotiation to connect the UE's media stream to MF is required by the application data channel. The media type will be represented by the value of the mediaResourceType.<br><br>- true: media re-negotiation to connect the UE's audio or video media stream to MF is required.<br>- false: media re-negotiation to connect the UE's audio or video media stream to MF is not required. |
| mediaConnSide | MediaConnSide | C | 0..1 | Represent the audio or video media connection side that the application data channel required.<br><br>It shall be included if the reNegotiation present and the attribute value is set to true. |
| mediaDirection | SendRecvMode | O | 0..1 | Represents the media direction of the audio/video media. |

#### 6.2.6.2.8 Type: InterworkingInstruction

**Table 6.2.6.2.8-1: Definition of type InterworkingInstruction**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| transcodeMode | TranscodeMode | M | 1 | Indicates the mode of the media transcoding. |
| addTransInfo | string | O | 0..1 | Additional information on how to transcode the data channel media. |

#### 6.2.6.2.9 Type: Mdc2AudioVideoEndpoint

**Table 6.2.6.2.9-1: Definition of type Mdc2AudioVideoEndpoint**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| mdc2AVEndpointDcAs | Endpoint | C | 0..1 | Represents the MDC2 endpoint information on DC AS for audio or video media.<br>It shall be included in the POST request message for the media control. |
| mdc2AVEndpointMf | Endpoint | C | 0..1 | Represents the MDC2 endpoint information on MF for audio or video media.<br>It shall be included in the successful POST response message for the media control API. |
| audioVideoMedia | AudioVideoMedia | C | 0..1 | This attribute is deprecated. |
| audioMediaParams | array(AudioVideoMedia) | O | 1..N | Represents the description of audio media parameters. |
| videoMediaParams | array(AudioVideoMedia) | O | 1..N | Represents the description of video media parameters. |

### 6.2.6.3 Simple data types and enumerations

#### 6.2.6.3.1 Introduction

This clause defines simple data types and enumerations that can be referenced from data structures defined in the previous clauses.

#### 6.2.6.3.2 Simple data types

The simple data types defined in table 6.2.6.3.2-1 shall be supported.

**Table 6.2.6.3.2-1: Simple data types**

| Type Name | Type Definition | Description |
|---|---|---|
| | | |

#### 6.2.6.3.3 Enumeration: MediaInstruction

The enumeration MediaInstruction represents the instructions to the producer (IMS AS) for handling the media. It shall comply with the provisions defined in table 6.2.6.3.3-1.

**Table 6.2.6.3.3-1: Enumeration MediaInstruction**

| Enumeration value | Description |
|---|---|
| "TERMINATE_MEDIA" | Terminate the offered media descriptor of the UE in the media resource. |
| "ORIGINATE_MEDIA" | Originate and offer a media descriptor from the mediaResource to the UE. |
| "TERMINATE_AND_ORIGINATE_MEDIA" | Terminate the offered media flow in the mediaResource from the UE and originate the same media flow from the mediaResource to the other UE. |
| "UPDATE_MEDIA" | Update a media flow of the mediaResource previously allocated by the instructions "TerminateMedia", "OriginateMedia" and "TerminateAndOriginateMedia". |
| "DELETE_MEDIA" | Delete a media flow of the mediaResource previously allocated by the instructions "TerminateMedia", "OriginateMedia" and "TerminateAndOriginateMedia". |
| "REJECT_MEDIA" | Remove an offered media flow. |

#### 6.2.6.3.4 Enumeration: MediaConnSide

The enumeration MediaConnSide represents the audio or video media connection side. It shall comply with the provisions defined in table 6.2.6.3.4-1.

**Table 6.2.6.3.4-1: Enumeration MediaConnSide**

| Enumeration value | Description |
|---|---|
| "BOTH" | Both the originating and terminating side of the audio or video media stream shall be connected to the MF. |
| "TERMINATING_SIDE" | Only the terminating side of the audio or video media stream are connected to the MF. |
| "ORIGINATING_SIDE" | Only the originating side of the audio or video media stream are connected to the MF. |

#### 6.2.6.3.5 Enumeration: TranscodeMode

The enumeration TranscodeMode represents the media transcode instruction. It shall comply with the provisions defined in table 6.2.6.3.5-1.

**Table 6.2.6.3.5-1: Enumeration TranscodeMode**

| Enumeration value | Description |
|---|---|
| "DC_TO_VIDEO" | Data channel media is to be transcoded to video media. |

#### 6.2.6.3.6 Enumeration: SendRecvMode

The enumeration SendRecvMode represents the audio or video media connection side. It shall comply with the provisions defined in table 6.2.6.3.6-1.

**Table 6.2.6.3.6-1: Enumeration SendRecvMode**

| Enumeration value | Description |
|---|---|
| "SENDRECV" | The media is send and recv. |
| "SENDONLY" | The media is send only. |
| "RECVONLY" | The media is receive only. |

### 6.2.6.4 Data types describing alternative data types or combinations of data types

None in this release of the specification.

### 6.2.6.5 Binary data

None in this release of the specification.

## 6.2.7 Error Handling

### 6.2.7.1 General

For the Nimsas_MediaControl API, HTTP error responses shall be supported as specified in clause 4.8 of TS 29.501 [5]. Protocol errors and application errors specified in table 5.2.7.2-1 of TS 29.500 [4] shall be supported for an HTTP method if the corresponding HTTP status codes are specified as mandatory for that HTTP method in table 5.2.7.1-1 of TS 29.500 [4].

In addition, the requirements in the following clauses are applicable for the Nimsas_MediaControl API.

### 6.2.7.2 Protocol Errors

No specific procedures for the Nimsas_MediaControl service are specified.

### 6.2.7.3 Application Errors

The application errors defined for the Nimsas_MediaControl service are listed in table 6.2.7.3-1.

**Table 6.2.7.3-1: Application errors**

| Application Error | HTTP status code | Description |
|---|---|---|
| MEDIA_ID_NOT_FOUND | 400 Bad Request | Indicates that the requested mediaId is not found in the IMS AS. |

## 6.2.8 Feature negotiation

The optional features in table 6.2.8-1 are defined for the Nimsas_MediaControl API. They shall be negotiated using the extensibility mechanism defined in clause 6.6 of TS 29.500 [4].

**Table 6.2.8-1: Supported Features**

| Feature number | Feature Name | Description |
|---|---|---|
| | | |

### 6.2.9 Security

As indicated in TS 33.501 [8] and TS 29.500 [4], the access to the Nimsas_MediaControl API may be authorized by means of the OAuth2 protocol (see IETF RFC 6749 [9]), based on local configuration, using the "Client Credentials" authorization grant, where the NRF (see TS 29.510 [10]) plays the role of the authorization server.

If OAuth2 is used, an NF Service Consumer, prior to consuming services offered by the Nimsas_MediaControl API, shall obtain a "token" from the authorization server, by invoking the Access Token Request service, as described in TS 29.510 [10], clause 5.4.2.2.

NOTE: When multiple NRFs are deployed in a network, the NRF used as authorization server is the same NRF that the NF Service Consumer used for discovering the Nimsas_MediaControl service.

The Nimsas_MediaControl API defines a single scope " nimsas-mc " for the entire service, and it does not define any additional scopes at resource or operation level.

### 6.2.10 HTTP redirection

An HTTP request may be redirected to a different IMS AS service instance when using direct or indirect communications (see TS 29.500 [4]).

An SCP that reselects a different IMS AS producer instance will return the NF Instance ID of the new IMS AS producer instance in the 3gpp-Sbi-Producer-Id header, as specified in clause 6.10.3.4 of TS 29.500 [4].

If an IMS AS redirects a service request to a different IMS AS using an 307 Temporary Redirect or 308 Permanent Redirect status code, the identity of the new IMS AS towards which the service request is redirected shall be indicated in the 3gpp-Sbi-Target-Nf-Id header of the 307 Temporary Redirect or 308 Permanent Redirect response as specified in clause 6.10.9.1 of TS 29.500 [4].

## 6.3 Nimsas_ImsSessionManagement Service API

### 6.3.1 API URI

The Nimsas_ImsSessionManagement service shall use the Nimsas_ImsSessionManagement API.

The API URI of the Nimsas_ImsSessionManagement API shall be:

**{apiRoot}/<apiName>/<apiVersion>**

The request URIs used in HTTP requests from the NF service consumer towards the NF service producer shall have the Resource URI structure defined in clause 4.4.1 of TS 29.501 [5], i.e.:

**{apiRoot}/<apiName>/<apiVersion>/<apiSpecificResourceUriPart>**

with the following components:

- The {apiRoot} shall be set as described in TS 29.501 [5].
- The <apiName> shall be "nimsas-ism".
- The <apiVersion> shall be "v1".
- The <apiSpecificResourceUriPart> shall be set as described in clause 6.3.3.

### 6.3.2 Usage of HTTP

#### 6.3.2.1 General

HTTP/2, IETF RFC 7540 [11], shall be used as specified in clause 5 of TS 29.500 [4].

HTTP/2 shall be transported as specified in clause 5.3 of TS 29.500 [4].

The OpenAPI [6] specification of HTTP messages and content bodies for the Nimsas_SessionEventControl API is contained in Annex A.

#### 6.3.2.2 HTTP standard headers

##### 6.3.2.2.1 General

See clause 5.2.2 of TS 29.500 [4] for the usage of HTTP standard headers.

##### 6.3.2.2.2 Content type

JSON, IETF RFC 8259 [12], shall be used as content type of the HTTP bodies specified in the present specification as specified in clause 5.4 of TS 29.500 [4]. The use of the JSON format shall be signalled by the content type "application/json".

"Problem Details" JSON object shall be used to indicate additional details of the error in a HTTP response body and shall be signalled by the content type "application/problem+json", as defined in IETF RFC 7807 [13].

#### 6.3.2.3 HTTP custom headers

The mandatory HTTP custom header fields specified in clause 5.2.3.2 of TS 29.500 [4] shall be supported, and the optional HTTP custom header fields specified in clause 5.2.3.3 of TS 29.500 [4] may be supported.

### 6.3.3 Resources

#### 6.3.3.1 Overview

This clause describes the structure for the Resource URIs and the resources and methods used for the service.

Figure 6.3.3.1-1 depicts the resource URIs structure for the Nimsas_ImsSessionManagement service API.

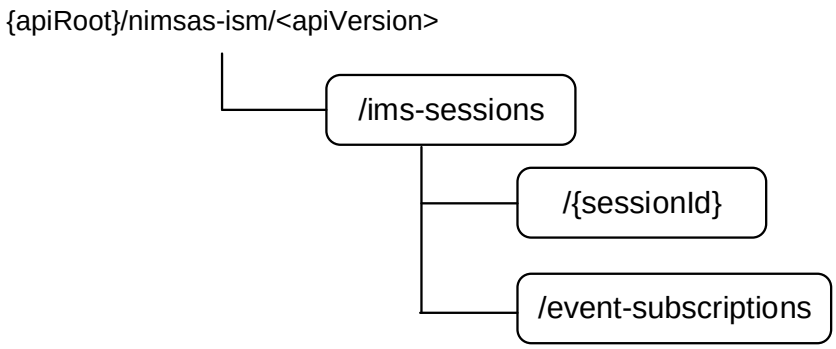



**Figure 6.3.3.1-1: Resource URI structure of the Nimsas_ImsSessionManagement API**

Table 6.3.3.1-1 provides an overview of the resources and applicable HTTP methods.

**Table 6.3.3.1-1: Resources and methods overview**

| Resource purpose/name | Resource URI (relative path after API URI) | HTTP method or custom operation | Description (service operation) |
|---|---|---|---|
| IMS Sessions (Collection) | /ims-sessions | POST | Creates a new Individual IMS Session resource. |
| Individual IMS Session (Document) | /ims-sessions/{sessionId} | PUT | Update an existing "Individual IMS Session" resource. |
| | | PATCH | Modify an existing Individual IMS Session resource. |
| | | DELETE | Delete an existing Individual IMS Session resource. |
| IMS Session Event Subscriptions (Collection) | /ims-sessions/event-subscriptions | POST | This is a pseudo resource. |

## 6.3.3.2 Resource: IMS Sessions

### 6.3.3.2.1 Description

The IMS Sessions Collection resource represents a collection of IMS Sessions created by NF service consumers of Nimsas_ImsSessionManagement service at a given IMS AS. The resource is modelled as Collection resource archetype (see clause C.2 of TS 29.501 [5]).

### 6.3.3.2.2 Resource Definition

Resource URI: **{apiRoot}/nimsas-ism/<apiVersion>/ims-sessions**

This resource shall support the resource URI variables defined in table 6.1.3.2.2-1.

**Table 6.3.3.2.2-1: Resource URI variables for this resource**

| Name | Data type | Definition |
|---|---|---|
| apiRoot | string | See clause 6.3.1 |

### 6.3.3.2.3 Resource Standard Methods

#### 6.3.3.2.3.1 POST

This method shall support the URI query parameters specified in table 6.3.3.2.3.1-1.

**Table 6.3.3.2.3.1-1: URI query parameters supported by the POST method on this resource**

| Name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| n/a | | | | | |

This method shall support the request data structures specified in table 6.3.3.2.3.1-2 and the response data structures and response codes specified in table 6.3.3.2.3.1-3.

**Table 6.3.3.2.3.1-2: Data structures supported by the POST Request Body on this resource**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| ImsSessionInfo | M | 1 | Creates a new Individual IMS Session resource |

**Table 6.3.3.2.3.1-3: Data structures supported by the POST Response Body on this resource**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| ImsSessionInfo | M | 1 | 201 Created | The creation of an Individual IMS Session resource is confirmed and a representation of that resource is returned. |
| RedirectResponse | O | 0..1 | 307 Temporary Redirect | Temporary redirection. (NOTE 2) |
| RedirectResponse | O | 0..1 | 308 Permanent Redirect | Permanent redirection. (NOTE 2) |
| NOTE 1: The manadatory HTTP error status code for the POST method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply.<br>NOTE 2: RedirectResponse may be inserted by an SCP, see clause 6.10.9.1 of TS 29.500 [4]. | | | | |

**Table 6.3.3.2.3.1-4: Headers supported by the 201 response code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | Contains the URI of the newly created resource, according to the structure: {apiRoot}/nimsas-ism/<apiVersion>/ims-sessions/{sessionId}. |

**Table 6.3.3.2.3.1-5: Headers supported by the 307 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative IMS AS (service) instance.<br>Or the same URI, if a request is redirected to the same target resource via a different SCP. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target IMS AS instance ID towards which the request is redirected. |

**Table 6.3.3.2.3.1-6: Headers supported by the 308 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative IMS AS (service) instance.<br>Or the same URI, if a request is redirected to the same target resource via a different SCP. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target IMS AS instance ID towards which the request is redirected. |

### 6.3.3.2.4 Resource Custom Operations

None.

## 6.3.3.3 Resource: Individual IMS Session

### 6.3.3.3.1 Description

The Individual IMS Session resource represents an individual IMS session created by the NF service consumers of Nimsas_ImsSessionManagement service at a given IMS AS. This resource is modelled as the Document resource archetype (see clause C.1 of TS 29.501 [5]).

### 6.3.3.3.2 Resource Definition

Resource URI: **{apiRoot}/nimsas_ism/<apiVersion>/ims-sessions/{sessionId}**

The <apiVersion> shall be set as described in clause 6.3.1.

This resource shall support the resource URI variables defined in table 6.3.3.3.2-1.

**Table 6.3.3.3.2-1: Resource URI variables for this resource**

| Name | Data type | Definition |
|---|---|---|
| apiRoot | string | See clause 6.3.1 |
| sessionId | string | Identifies an individual IMS session to the Nimsas_ImsSessionManagement service. |

### 6.3.3.3.3 Resource Standard Methods

#### 6.3.3.3.3.1 PATCH

This method shall support the URI query parameters specified in table 6.3.3.3.3.1-1.

**Table 6.3.3.3.3.1-1: URI query parameters supported by the PATCH method on this resource**

| Name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| supported-features | SupportedFeatures | O | 0..1 | see TS 29.500 [4] clause 6.6 | |

This method shall support the request data structures specified in table 6.3.3.3.3.1-2 and the response data structures and response codes specified in table 6.3.3.3.3.1-3.

**Table 6.3.3.3.3.1-2: Data structures supported by the PATCH Request Body on this resource**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| array(PatchItem) | M | 1..N | Document describes the modification(s) to an Individual IMS Session resource. |

**Table 6.3.3.3.3.1-3: Data structures supported by the PATCH Response Body on this resource**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| PatchResult | M | 1 | 200 OK | Upon success, the execution report is returned. (NOTE 3) |
| n/a | | | 204 No Content | Upon success, an empty response body is returned. (NOTE 3) |
| RedirectResponse | O | 0..1 | 307 Temporary Redirect | Temporary redirection. (NOTE 2) |
| RedirectResponse | O | 0..1 | 308 Permanent Redirect | Permanent redirection. (NOTE 2) |
| ProblemDetails | O | 0..1 | 409 Conflict | Indicates the creation of media context has failed due to application error.<br><br>The "cause" attribute may be used to indicate one of the following application errors:<br>- MEDIA_CORRELATION_ID_CONFLICT, if the media to be created has the identical mediaCorrelationId with one already created media. |
| ProblemDetails | O | 0..1 | 404 Not Found | Indicates the modification of IMS session has failed due to application error.<br><br>The "cause" attribute may be used to indicate one of the following application errors:<br>- SESSION_NOT_FOUND |
| ProblemDetails | O | 0..1 | 422 Unprocessable Entity | Indicates the modification of IMS session has failed due to application error.<br><br>The "cause" attribute may be used to indicate one of the following application errors:<br>- BDC_REMOVAL_NOT_ALLOWED |
| NOTE 1: The manadatory HTTP error status code for the PATCH method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply.<br>NOTE 2: RedirectResponse may be inserted by an SCP, see clause 6.10.9.1 of TS 29.500 [4].<br>NOTE 3: If all the modification instructions in the PATCH request have been implemented, the IMS AS shall respond with 204 No Content response; if some of the modification instructions in the PATCH request have been discarded, and the NF service consumer has included in the supported-feature query parameter the "PatchReport" feature number, the IMS AS shall respond with PatchResult. | | | | |

**Table 6.3.3.3.3.1-4: Headers supported by the 307 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative IMS AS (service) instance.<br>Or the same URI, if a request is redirected to the same target resource via a different SCP. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target IMS AS instance ID towards which the request is redirected. |

**Table 6.3.3.3.3.1-5: Headers supported by the 308 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative IMS AS (service) instance.<br>Or the same URI, if a request is redirected to the same target resource via a different SCP. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target IMS AS instance ID towards which the request is redirected. |

### 6.3.3.3.3.2 DELETE

This method shall support the URI query parameters specified in table 6.3.3.3.3.2-1.

**Table 6.3.3.3.3.2-1: URI query parameters supported by the DELETE method on this resource**

| Name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| n/a | | | | | |

This method shall support the request data structures specified in table 6.3.3.3.3.2-2 and the response data structures and response codes specified in table 6.3.3.3.3.2-3.

**Table 6.3.3.3.3.2-2: Data structures supported by the DELETE Request Body on this resource**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| n/a | | | |

**Table 6.3.3.3.3.2-3: Data structures supported by the DELETE Response Body on this resource**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| n/a | | | 204 No content | |
| RedirectResponse | O | 0..1 | 307 Temporary Redirect | Temporary redirection. (NOTE 2) |
| RedirectResponse | O | 0..1 | 308 Permanent Redirect | Permanent redirection. (NOTE 2) |
| Problem Details | O | 0..1 | 404 Not Found | Indicates the deletion of the IMS session has failed due to application error.<br><br>The "cause" attribute may be used to indicate one of the following application errors:<br>- SESSION_NOT_FOUND |
| NOTE 1: The manadatory HTTP error status code for the DELETE method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply.<br>NOTE 2: RedirectResponse may be inserted by an SCP, see clause 6.10.9.1 of TS 29.500 [4]. | | | | |

**Table 6.3.3.3.3.2-4: Headers supported by the 307 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative IMS AS (service) instance.<br>Or the same URI, if a request is redirected to the same target resource via a different SCP. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target IMS AS instance ID towards which the request is redirected. |

**Table 6.3.3.3.3.2-5: Headers supported by the 308 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative IMS AS (service) instance.<br>Or the same URI, if a request is redirected to the same target resource via a different SCP. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target IMS AS instance ID towards which the request is redirected. |

### 6.3.3.3.3.3 PUT

This method shall support the URI query parameters specified in table 6.3.3.3.3.3-1.

**Table 6.3.3.3.3.3-1: URI query parameters supported by the PUT method on this resource**

| Name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| n/a | | | | | |

This method shall support the request data structures specified in table 6.3.3.3.3.3-2 and the response data structures and response codes specified in table 6.3.3.3.3.3-3.

**Table 6.3.3.3.3.3-2: Data structures supported by the PUT Request Body on this resource**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| ImsSessionInfo | M | 1 | Parameters to update the "Individual IMS Session" resource. |

**Table 6.3.3.3.3.3-3: Data structures supported by the PUT Response Body on this resource**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| ImsSessionInfo | M | 1 | 200 OK | Successful response. The "Individual IMS Session" resource is successfully updated and the representation of that resource is returned. |
| n/a | | | 204 No Content | Successful response. The "Individual IMS Session" resource is successfully updated and no content is returned in the response body. |
| RedirectResponse | O | 0..1 | 307 Temporary Redirect | Temporary redirection. (NOTE 2) |
| RedirectResponse | O | 0..1 | 308 Permanent Redirect | Permanent redirection. (NOTE 2) |
| ProblemDetails | O | 0..1 | 404 Not Found | Indicates the modification of Individual IMS Session has failed due to application error.<br><br>The “cause” attribute may be used to indicate one of the following application errors:<br>- SESSION_NOT_FOUND |
| ProblemDetails | O | 0..1 | 409 Conflict | Indicates the creation of media context has failed due to application error.<br><br>The "cause" attribute may be used to indicate one of the following application errors:<br>- MEDIA_CORRELATION_ID_CONFLICT |
| NOTE 1: The manadatory HTTP error status code for the PUT method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply.<br>NOTE 2: RedirectResponse may be inserted by an SCP, see clause 6.10.9.1 of TS 29.500 [4]. | | | | |

**Table 6.3.3.3.3.3-4: Headers supported by the 307 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative IMS AS (service) instance.<br>Or the same URI, if a request is redirected to the same target resource via a different SCP. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target IMS AS instance ID towards which the request is redirected. |

**Table 6.3.3.3.3.3-5: Headers supported by the 308 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative IMS AS (service) instance.<br>Or the same URI, if a request is redirected to the same target resource via a different SCP. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target IMS AS instance ID towards which the request is redirected. |

#### 6.3.3.3.4 Resource Custom Operations

None.

### 6.3.4 Custom Operations without associated resources

None in this release of specification.

### 6.3.5 Notifications

#### 6.3.5.1 General

Notifications shall comply to clause 6.2 of TS 29.500 [4] and clause 4.6.2.3 of TS 29.501 [5].

**Table 6.3.5.1-1: Notifications overview**

| Notification | Callback URI | HTTP method or custom operation | Description (service operation) |
|---|---|---|---|
| IMS Session Event Notification | {notifUri} | POST | Report the IMS session event observed. |

#### 6.3.5.2 IMS Session Event Notification

##### 6.3.5.2.1 Description

The IMS Session Event Notification is used by the IMS AS to report the observed IMS Session Event to notification endpoints.

##### 6.3.5.2.2 Target URI

The Callback URI "{notifUri}" shall be used with the callback URI variables defined in table 6.3.5.2.2-1.

**Table 6.3.5.2.2-1: Callback URI variables**

| Name | Definition |
|---|---|
| notifUri | The notification URI of the Service Consumer (i.e., NEF, trusted AF) to receive the IMS session events. |

##### 6.3.5.2.3 Standard Methods

###### 6.3.5.2.3.1 POST

This method shall support the request data structures specified in table 6.3.5.2.3.1-1 and the response data structures and response codes specified in table 6.3.5.2.3.1-2.

**Table 6.3.5.2.3.1-1: Data structures supported by the POST Request Body**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| ImsSessionEventNotification | M | 1 | The IMS session event notification to notification endpoints. |

**Table 6.3.5.2.3.1-2: Data structures supported by the POST Response Body**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| n/a | | | 204 No Content | This case represents a successful notification of the event. |
| ProblemDetails | O | 0..1 | 404 Not Found | Indicates the session event notification has failed due to application error.<br><br>The "cause" attribute may be used to indicate one of the following application errors:<br>- CONTEXT_NOT_FOUND |
| RedirectResponse | O | 0..1 | 307 Temporary Redirect | Temporary redirection.<br>(NOTE 2) |
| RedirectResponse | O | 0..1 | 308 Permanent Redirect | Permanent redirection.<br>(NOTE 2) |
| NOTE 1: The mandatory HTTP error status codes for the POST method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply.<br>NOTE 2: RedirectResponse may be inserted by an SCP, see clause 6.10.9.1 of TS 29.500 [4]. | | | | |

**Table 6.3.5.2.3-3: Headers supported by the 307 Response Code on this endpoint**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | A URI pointing to the endpoint of the NF service consumer instance to which the request should be sent. For the case, when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF instance ID towards which the notification is redirected |

**Table 6.3.5.2.3-4: Headers supported by the 308 Response Code on this endpoint**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | A URI pointing to the endpoint of the NF service consumer instance to which the request should be sent. For the case, when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF instance ID towards which the notification is redirected |

## 6.3.6 Data Model

### 6.3.6.1 General

This clause specifies the application data model supported by the API.

Table 6.3.6.1-1 specifies the data types defined for the Nimsas_ImsSessionManagement service based interface protocol.

**Table 6.3.6.1-1: Nimsas_ImsSessionManagement specific Data Types**

| Data type | Clause defined | Description |
|---|---|---|
| ImsSessionInfo | 6.3.6.2.2 | The IMS session information and the media list used in the IMS session. |
| DcMediaPara | 6.3.6.2.5 | The parameters of the IMS DC media. |
| ImsSessionEventNotification | 6.3.6.2.6 | The IMS Session event notification. |
| MediaInfoExternal | 6.3.6.2.4 | The media information used in the external created or updated IMS session. |
| SessionInfoSM | 6.3.6.2.3 | The session information to be used. |
| TargetIdInfo | 6.3.6.2.7 | The public identity(ies) of the IMS Subscriber the DC is targeted. |
| DcType | 6.3.6.3.3 | The type of data channel. |
| AdcType | 6.3.6.3.4 | The type of the application data channel. |
| ImsSessionEventType | 6.3.6.3.6 | The event type of IMS session event. |

Table 6.3.6.1-2 specifies data types re-used by the Nimsas_ImsSessionManagement service based interface protocol from other specifications, including a reference to their respective specifications and when needed, a short description of their use within the Nimsas_ImsSessionManagement service based interface.

**Table 6.3.6.1-2: Nimsas_ImsSessionManagement re-used Data Types**

| Data type | Reference | Comments |
|---|---|---|
| AppBindingInfo | TS 29.571 [16] | The application binding information of the application data channel. |
| ImsPublicId | TS 29.562 [15] | IMS Public Identity. |
| MediaType | 6.1.6.3.4 | The media type used in the IMS session. |
| SessionId | TS 29.571 [16] | IMS Session Identity. |
| Mdc2EndpointInfo | 6.2.6.2.6 | The MDC2 media endpoint information. |
| MediaInfo | 6.1.6.2.5 | The media information related to the notification event. |
| Uri | TS 29.571 [16] | URI. |

## 6.3.6.2 Structured data types

### 6.3.6.2.1 Introduction

This clause defines the structures to be used in resource representations.

### 6.3.6.2.2 Type: ImsSessionInfo

**Table 6.3.6.2.2-1: Definition of type ImsSessionInfo**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| sessionId | SessionId | C | 0..1 | The session ID is the identity of the IMS session.<br>This IE contains the service producer generated identity which uniquely identifies the IMS session.<br>This IE shall be present when updating or deleting an existing IMS session. Otherwise, it shall not be present. |
| sessionInfo | SessionInfoSM | M | 1 | The sessionInfo is the IMS session related information. |
| mediaInfoSet | array(MediaInfo External) | M | 1..N | Media info set includes a list of media related information. |
| notifUri | Uri | O | 0..1 | This IE may be present if the NF consumer intends to receive IMS session events for the served IMS subscriber.<br>When present, it shall contain the URI of the NF consumer to receive IMS session event notification. |
| notifCorrelationId | string | C | 0..1 | This IE shall be present if notifUri is available.<br>When present, it shall contain the correlation ID to associate the IMS session event notification to be reported. |

#### 6.3.6.2.3 Type: SessionInfoSM

**Table 6.3.6.2.3-1: Definition of type SessionInfoSM**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| callingId | ImsPublicId | M | 1 | The public identity of the calling IMS subscriber or the identity of the consumer NF, e.g. DC AS. |
| calledId | ImsPublicId | M | 1 | The public identity of the called IMS Subscriber. |

#### 6.3.6.2.4 Type: MediaInfoExternal

**Table 6.3.6.2.4-1: Definition of type MediaInfoExternal**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| mediaType | MediaType | M | 1 | The media type used in the session.<br>In this release, the mediaType can only set to "DC". |
| mediaCorrelationId | string | M | 1 | The media correlation ID is the identity of the media. |
| dcMediaParas | array(DcMediaPara) | C | 1..N | The information of the created data channel media.<br>It shall be contained when the attribute "mediaType" is set to "DC". |

#### 6.3.6.2.5 Type: DcMediaPara

**Table 6.3.6.2.5-1: Definition of type DcMediaPara**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| dcType | DcType | M | 1 | The data channel type of the data channel. |
| adcType | AdcType | C | 0..1 | The type of application data type. This IE shall be contained when the attribute "dcType" is set to "ADC". |
| appBindingInfo | AppBindingInfo | C | 0..1 | Represents the application binding information of the application data channel.<br>It shall be contained when the attribute "dcType" is set to "ADC". |
| targetIds | array(TargetIdInfo) | C | 1..N | Representes the public identity(ies) of the IMS Subscriber the DC is targeted.<br>It shall be contained when the attribute "dcType" is set to "ADC". |
| targetId | ImsPublicId | C | 0..1 | the public identity of the IMS Subscriber the BDC is targeted.<br>It shall be contained when the attribute "dcType" is set to "BDC". |
| mdc2EndpointInfo | Mdc2EndpointInfo | C | 0..1 | Represents the negotiated MDC2 media endpoint information, e.g., the protocol stack, IP address and port number.<br>It shall be contained when the attribute "dcType" is set to "ADC" and adcType set to "P2A" or "P2A2P". |

#### 6.3.6.2.6 Type: ImsSessionEventNotification

**Table 6.3.6.2.6-1: Definition of type ImsSessionEventNotification**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| sessionId | SessionId | M | 1 | The session ID is the identity of the IMS session.<br>This IE contains the value of the sessionId attribute provided during the IMS session creation within the ImsSessionInfo data type. |
| eventType | ImsSessionEventType | M | 1 | This IE indicates the type of IMS session event. |
| notifCorrelationId | string | M | 1 | This IE indicates the correlation ID used to associate the IMS session event notification to be reported. |
| mediaInfoSet | array(MediaInfoExternal) | O | 1..N | Media info set includes a list of media related information. |
| callingIdentity | ImsPublicId | O | 0..1 | The IMS public identity of the calling party. It may be included if the eventType is set to "SESSION_ESTABLISHMENT_FAILURE", "MEDIA_CHANGE_SUCCESS", "MEDIA_CHANGE_FAILURE", or "SESSION_TERMINATION". |
| calledIdentity | ImsPublicId | O | 0..1 | The IMS public identity of the called party. It may be included if the eventType is set to "SESSION_ESTABLISHMENT_FAILURE", "MEDIA_CHANGE_SUCCESS", "MEDIA_CHANGE_FAILURE", or "SESSION_TERMINATION". |
| failureReason | string | O | 0..1 | The reason of the failure which is human readable. It may be included if the eventType is set to "SESSION_ESTABLISHMENT_FAILURE", or "MEDIA_CHANGE_FAILURE". |

#### 6.3.6.2.7 Type: TargetIdInfo

**Table 6.3.6.2.7-1: Definition of type TargetIdInfo**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| callingIdentity | ImsPublicId | C | 0..1 | The IMS public identity of the calling IMS subscriber.<br>It shall be contained when the attribute adcType set to "P2P" or "P2A2P".<br>(NOTE) |
| calledIdentity | ImsPublicId | C | 0..1 | The IMS public identity of the called IMS Subscriber.<br>It shall be contained when the attribute adcType set to "P2P" or "P2A2P".<br>(NOTE) |
| NOTE: One of the callingIdentity or calledIdentity shall be contained when the attribute adcType set to "P2A". | | | | |

### 6.3.6.3 Simple data types and enumerations

#### 6.3.6.3.1 Introduction

This clause defines simple data types and enumerations that can be referenced from data structures defined in the previous clauses.

#### 6.3.6.3.2 Simple data types

The simple data types defined in table 6.3.6.3.2-1 shall be supported.

**Table 6.3.6.3.2-1: Simple data types**

| Type Name | Type Definition | Description | Applicability |
|---|---|---|---|
| | | | |

### 6.3.6.3.3 Enumeration: DcType

The enumeration DcType represents the type of data channel. It shall comply with the provisions defined in table 6.3.6.3.3-1.

**Table 6.3.6.3.3-1: Enumeration DcType**

| Enumeration value | Description |
|---|---|
| "BDC" | Bootstrap data channel. |
| "ADC" | Application data channel. |

### 6.3.6.3.4 Enumeration: AdcType

The enumeration AdcType represents the type of the application data channel. It shall comply with the provisions defined in table 6.3.6.3.4-1.

**Table 6.3.6.3.4-1: Enumeration AdcType**

| Enumeration value | Description |
|---|---|
| "P2P" | P2P application data channel |
| "P2A" | P2A application data channel |
| "P2A2P" | P2A2P application data channel |

### 6.3.6.3.5 Void

### 6.3.6.3.6 Enumeration: ImsSessionEventType

The enumeration ImsSessionEventType represents the IMS session event type for which the notification is generated. It shall comply with the provisions defined in table 6.3.6.3.6-1.

**Table 6.3.6.3.6-1: Enumeration ImsSessionEventType**

| Enumeration value | Description |
|---|---|
| "SESSION_ESTABLISHMENT_SUCCESS" | Session establishment success event. |
| "SESSION_ESTABLISHMENT_FAILURE" | Session establishment failure event. |
| "MEDIA_CHANGE_SUCCESS" | Media change success event. |
| "MEDIA_CHANGE_FAILURE" | Media change failure event. |
| "SESSION_TERMINATION" | Session termination event. |

## 6.3.6.4 Data types describing alternative data types or combinations of data types

None in this release of the specification.

## 6.3.6.5 Binary data

None in this release of the specification.

## 6.3.7 Error Handling

### 6.3.7.1 General

For the Nimsas_ImsSessionManagement API, HTTP error responses shall be supported as specified in clause 4.8 of TS 29.501 [5]. Protocol errors and application errors specified in table 5.2.7.2-1 of TS 29.500 [4] shall be supported for an HTTP method if the corresponding HTTP status codes are specified as mandatory for that HTTP method in table 5.2.7.1-1 of TS 29.500 [4].

In addition, the requirements in the following clauses are applicable for the Nimsas_ImsSessionManagement API.

### 6.3.7.2 Protocol Errors

No specific procedures for the Nimsas_ImsSessionManagement service are specified.

### 6.3.7.3 Application Errors

The application errors defined for the Nimsas_ImsSessionManagement service are listed in Table 6.3.7.3-1.

**Table 6.3.7.3-1: Application errors**

| Application Error | HTTP status code | Description |
|---|---|---|
| SESSION_NOT_FOUND | 404 Not Found | The service user is not served by this DCSF, e.g. no data channel configurations for this service user is found. |
| CONTEXT_NOT_FOUND | 404 Not Found | The NF Service Consumer (e.g. NEF, trusted AF) receives a notification from IMS AS for session event that is not found for the service consumer. |
| MEDIA_CORRELATION_ID_CONFLICT | 409 Conflict | Indicates that a new media to be created has the identical mediaCorrelationId already assigned to one created media. |
| BDC_REMOVAL_NOT_ALLOWED | 422 Unprocessable Entity | Indicates the BDC is not allowed to be removed in a IMS session containing ADC. |

## 6.3.8 Feature negotiation

The optional features in table 6.3.8-1 are defined for the Nimsas_ImsSessionManagement API. They shall be negotiated using the extensibility mechanism defined in clause 6.6 of TS 29.500 [4].

**Table 6.3.8-1: Supported Features**

| Feature number | Feature Name | Description |
|---|---|---|
| 1 | PatchReport | If some of the modifications included in the PATCH request are not successfully implemented, the IMS AS reports the result of PATCH request execution to the consumer. See clause 5.2.7.2 of TS 29.500 [4]. |

## 6.3.9 Security

As indicated in TS 33.501 [8] and TS 29.500 [4], the access to the Nimsas_ImsSessionManagement API may be authorized by means of the OAuth2 protocol (see IETF RFC 6749 [9]), based on local configuration, using the "Client Credentials" authorization grant, where the NRF (see TS 29.510 [10]) plays the role of the authorization server.

If OAuth2 is used, an NF Service Consumer, prior to consuming services offered by the Nimsas_ImsSessionManagement API, shall obtain a "token" from the authorization server, by invoking the Access Token Request service, as described in TS 29.510 [10], clause 5.4.2.2.

NOTE: When multiple NRFs are deployed in a network, the NRF used as authorization server is the same NRF that the NF Service Consumer used for discovering the Nimsas_ImsSessionManagement service.

The Nimsas_ImsSessionManagement API defines a single scope "nimsas-imssessionmanagement" for the entire service, and it does not define any additional scopes at resource or operation level.

### 6.3.10 HTTP redirection

An HTTP request may be redirected to a different IMS AS service instance when using direct or indirect communications (see TS 29.500 [4]).

An SCP that reselects a different NF provider (e.g. IMS AS) instance will return the NF Instance ID of the new NF provider instance in the 3gpp-Sbi-Producer-Id header, as specified in clause 6.10.3.4 of TS 29.500 [4].

If an NF provider (e.g. IMS AS) redirects a service request to a different NF provider using a 307 Temporary Redirect or 308 Permanent Redirect status code, the identity of the new NF provider towards which the service request is redirected shall be indicated in the 3gpp-Sbi-Target-Nf-Id header of the 307 Temporary Redirect or 308 Permanent Redirect response as specified in clause 6.10.9.1 of TS 29.500 [4].

## 6.4 Nimsas_ImsEE Service API

### 6.4.1 API URI

The Nimsas_ImsEE service shall use the Nimsas_ImsEE API.

The API URI of the Nimsas_ImsEE API shall be:

**{apiRoot}/<apiName>/<apiVersion>**

The request URIs used in HTTP requests from the NF service consumer towards the NF service producer shall have the Resource URI structure defined in clause 4.4.1 of TS 29.501 [5], i.e.:

**{apiRoot}/<apiName>/<apiVersion>/<apiSpecificResourceUriPart>**

with the following components:

- The {apiRoot} shall be set as described in TS 29.501 [5].
- The <apiName> shall be "nimsas-ee".
- The <apiVersion> shall be "v1".
- The <apiSpecificResourceUriPart> shall be set as described in clause 6.4.3.

### 6.4.2 Usage of HTTP

#### 6.4.2.1 General

HTTP/2, IETF RFC 7540 [11], shall be used as specified in clause 5 of TS 29.500 [4].

HTTP/2 shall be transported as specified in clause 5.3 of TS 29.500 [4].

The OpenAPI [6] specification of HTTP messages and content bodies for the Nimsas_ImsEE API is contained in Annex A.

#### 6.4.2.2 HTTP standard headers

##### 6.4.2.2.1 General

See clause 5.2.2 of TS 29.500 [4] for the usage of HTTP standard headers.

#### 6.4.2.2.2 Content type

JSON, IETF RFC 8259 [12], shall be used as content type of the HTTP bodies specified in the present specification as specified in clause 5.4 of TS 29.500 [4]. The use of the JSON format shall be signalled by the content type "application/json".

"Problem Details" JSON object shall be used to indicate additional details of the error in a HTTP response body and shall be signalled by the content type "application/problem+json", as defined in IETF RFC 7807 [13].

### 6.4.2.3 HTTP custom headers

The mandatory HTTP custom header fields specified in clause 5.2.3.2 of TS 29.500 [4] shall be supported, and the optional HTTP custom header fields specified in clause 5.2.3.3 of TS 29.500 [4] may be supported.

## 6.4.3 Resources

### 6.4.3.1 Overview

This clause describes the structure for the Resource URIs and the resources and methods used for the service.

Figure 6.4.3.1-1 depicts the resource URIs structure for the Nimsas_EE service API.

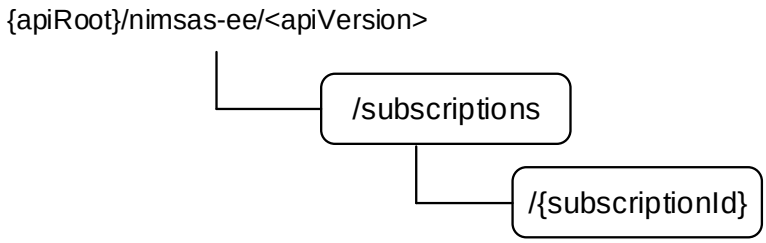



**Figure 6.4.3.1-1: Resource URI structure of the Nimsas_ImsEE API**

Table 6.4.3.1-1 provides an overview of the resources and applicable HTTP methods.

**Table 6.4.3.1-1: Resources and methods overview**

| Resource purpose/name | Resource URI (relative path after API URI) | HTTP method or custom operation | Description (service operation) |
|---|---|---|---|
| IMS Event Subscriptions (Collection) | /subscriptions | POST | Creates a new Individual IMS Event Subscription resource. |
| Individual IMS Event Subscription (Document) | /subscriptions/{subscriptionId} | GET | Get the information of a subscription identified by {subscriptionId}. |
| | | PUT | Updates an existing Individual IMS Event Subscription resource identified by {subscriptionId}. |
| | | PATCH | Updates an existing Individual IMS Event Subscription resource identified by {subscriptionId}. |
| | | DELETE | Delete an existing Individual IMS Subscription resource identified by {subscriptionId}. |

### 6.4.3.2 Resource: IMS Event Subscriptions

#### 6.4.3.2.1 Description

The IMS Event Subscriptions resource represents all subscriptions to the Nimsas_ImsEE service at a given IMS AS. The resource is modelled as Collection resource archetype (see clause C.2 of TS 29.501 [5]). The resource allows an NF service consumer to create a new Individual IMS Event Subscription resource.

#### 6.4.3.2.2 Resource Definition

Resource URI: **{apiRoot}/nimsas-ee/<apiVersion>/subscriptions**

This resource shall support the resource URI variables defined in table 6.4.3.2.2-1.

**Table 6.4.3.2.2-1: Resource URI variables for this resource**

| Name | Data type | Definition |
|---|---|---|
| apiRoot | string | See clause 6.4.1 |

#### 6.4.3.2.3 Resource Standard Methods

##### 6.4.3.2.3.1 POST

This method shall support the URI query parameters specified in table 6.4.3.2.3.1-1.

**Table 6.4.3.2.3.1-1: URI query parameters supported by the POST method on this resource**

| Name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| n/a | | | | | |

This method shall support the request data structures specified in table 6.4.3.2.3.1-2 and the response data structures and response codes specified in table 6.4.3.2.3.1-3.

**Table 6.4.3.2.3.1-2: Data structures supported by the POST Request Body on this resource**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| ImsEventsSubscription | M | 1 | Creates a new Individual IMS Event Subscription resource. |

**Table 6.4.3.2.3.1-3: Data structures supported by the POST Response Body on this resource**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| ImsEventsSubscriptionCreated | M | 1 | 201 Created | The creation of an Individual IMS Event Subscription resource is confirmed and a representation of that resource is returned. |
| RedirectResponse | O | 0..1 | 307 Temporary Redirect | Temporary redirection. (NOTE 2) |
| RedirectResponse | O | 0..1 | 308 Permanent Redirect | Permanent redirection. (NOTE 2) |
| ProblemDetails | O | 0..1 | 403 Forbidden | Indicates the creation of Individual IMS Event Subscription has failed due to application error.<br><br>The "cause" attribute may be used to indicate one of the following application errors:<br>- OPERATION_NOT_ALLOWED |
| NOTE 1: The manadatory HTTP error status code for the POST method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply.<br>NOTE 2: RedirectResponse may be inserted by an SCP, see clause 6.10.9.1 of TS 29.500 [4]. | | | | |

**Table 6.4.3.2.3.1-4: Headers supported by the 201 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | Contains the URI of the newly created resource, according to the structure: {apiRoot}/nimsas-ee/<apiVersion>/subscriptions/{subscriptionId}. |

**Table 6.4.3.2.3.1-5: Headers supported by the 307 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative IMS AS (service) instance.<br>Or the same URI, if a request is redirected to the same target resource via a different SCP. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target IMS AS instance ID towards which the request is redirected. |

**Table 6.4.3.2.3.1-6: Headers supported by the 308 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative IMS AS (service) instance.<br>Or the same URI, if a request is redirected to the same target resource via a different SCP. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target IMS AS instance ID towards which the request is redirected. |

#### 6.4.3.2.4 Resource Custom Operations

None.

### 6.4.3.3 Resource: Individual IMS Event Subscription

#### 6.4.3.3.1 Description

The Individual IMS Event Subscription resource represents an individual IMS event subscription created by the NF service consumers of Nimsas_ImsEE service at a given IMS AS. This resource is modelled as the Document resource archetype (see clause C.1 of TS 29.501 [5]).

#### 6.4.3.3.2 Resource Definition

Resource URI: **{apiRoot}/nimsas-ee/<apiVersion>/subscriptions/{subscriptionId}**

The <apiVersion> shall be set as described in clause 6.4.1.

This resource shall support the resource URI variables defined in table 6.4.3.3.2-1.

**Table 6.4.3.3.2-1: Resource URI variables for this resource**

| Name | Data type | Definition |
|---|---|---|
| apiRoot | string | See clause 6.4.1 |
| subscriptionId | string | Identifies an individual IMS event subscription to the Nimsas_ImsEE service. |

#### 6.4.3.3.3 Resource Standard Methods

##### 6.4.3.3.3.1 GET

This method shall support the URI query parameters specified in table 6.4.3.3.3.1-1.

**Table 6.4.3.3.3.1-1: URI query parameters supported by the GET method on this resource**

| Name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| n/a | | | | | |

This method shall support the request data structures specified in table 6.4.3.3.3.1-2 and the response data structures and response codes specified in table 6.4.3.3.3.1-3.

**Table 6.4.3.3.3.1-2: Data structures supported by the GET Request Body on this resource**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| n/a | | | |

**Table 6.4.3.3.3.1-3: Data structures supported by the GET Response Body on this resource**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| ImsEventsSubscription | M | 1 | 200 OK | A response body containing representation of the Individual IMS Event Subscription resource. |
| RedirectResponse | O | 0..1 | 307 Temporary Redirect | Temporary redirection. (NOTE 2) |
| RedirectResponse | O | 0..1 | 308 Permanent Redirect | Permanent redirection. (NOTE 2) |
| ProblemDetails | O | 0..1 | 404 Not Found | Indicates the information retrieval of Individual IMS Event Subscription has failed due to application error.<br><br>The "cause" attribute may be used to indicate one of the following application errors:<br>- SUBSCRIPTION_NOT_FOUND |
| ProblemDetails | O | 0..1 | 403 Forbidden | Indicates the information retrieval of Individual IMS Event Subscription has failed due to application error.<br><br>The "cause" attribute may be used to indicate one of the following application errors:<br>- OPERATION_NOT_ALLOWED |
| NOTE 1: The manadatory HTTP error status code for the GET method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply.<br>NOTE 2: RedirectResponse may be inserted by an SCP, see clause 6.10.9.1 of TS 29.500 [4]. | | | | |

**Table 6.4.3.3.3.1-4: Headers supported by the 307 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative IMS AS (service) instance.<br>Or the same URI, if a request is redirected to the same target resource via a different SCP. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target IMS AS instance ID towards which the request is redirected. |

**Table 6.4.3.3.3.1-5: Headers supported by the 308 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative IMS AS (service) instance.<br>Or the same URI, if a request is redirected to the same target resource via a different SCP. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target IMS AS instance ID towards which the request is redirected. |

6.4.3.3.3.2 PUT

This method shall support the URI query parameters specified in table 6.4.3.3.3.2-1.

**Table 6.4.3.3.3.2-1: URI query parameters supported by the PUT method on this resource**

| Name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| n/a | | | | | |

This method shall support the request data structures specified in table 6.4.3.3.3.2-2 and the response data structures and response codes specified in table 6.4.3.3.3.2-3.

**Table 6.4.3.3.3.2-2: Data structures supported by the PUT Request Body on this resource**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| ImsEventsSubscription | M | 1 | Parameters to replace a subscription to IMS Event Subscription resource. |

**Table 6.4.3.3.3.2-3: Data structures supported by the PUT Response Body on this resource**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| ImsEventsSubscriptionCreated | M | 1 | 200 OK | Represents a successful update on the subscription and the representation of that resource is returned. |
| n/a | | | 204 No Content | |
| RedirectResponse | O | 0..1 | 307 Temporary Redirect | Temporary redirection. (NOTE 2) |
| RedirectResponse | O | 0..1 | 308 Permanent Redirect | Permanent redirection. (NOTE 2) |
| ProblemDetails | O | 0..1 | 404 Not Found | Indicates the modification of Individual IMS Event Subscription has failed due to application error.<br><br>The "cause" attribute may be used to indicate one of the following application errors:<br>- SUBSCRIPTION_NOT_FOUND |
| ProblemDetails | O | 0..1 | 403 Forbidden | The "cause" attribute may be used to indicate one of the following application errors:<br>- OPERATION_NOT_ALLOWED |
| NOTE 1: The manadatory HTTP error status code for the PUT method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply.<br>NOTE 2: RedirectResponse may be inserted by an SCP, see clause 6.10.9.1 of TS 29.500 [4]. | | | | |

**Table 6.4.3.3.3.2-4: Headers supported by the 307 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative IMS AS (service) instance.<br>Or the same URI, if a request is redirected to the same target resource via a different SCP. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target IMS AS instance ID towards which the request is redirected. |

**Table 6.4.3.3.3.2-5: Headers supported by the 308 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative IMS AS (service) instance.<br>Or the same URI, if a request is redirected to the same target resource via a different SCP. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target IMS AS instance ID towards which the request is redirected. |

#### 6.4.3.3.3.3 PATCH

This method shall support the URI query parameters specified in table 6.4.3.3.3.3-1.

**Table 6.4.3.3.3.1-1: URI query parameters supported by the PATCH method on this resource**

| Name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| n/a | | | | | |

This method shall support the request data structures specified in table 6.4.3.3.3.3-2 and the response data structures and response codes specified in table 6.4.3.3.3.3-3.

**Table 6.4.3.3.3.3-2: Data structures supported by the PATCH Request Body on this resource**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| array(PatchItem) | M | 1..N | Parameters to replace a subscription to IMS Event Subscription resource. |

**Table 6.4.3.3.3.3-3: Data structures supported by the PATCH Response Body on this resource**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| ImsEventsSubscriptionCreated | M | 1 | 200 OK | Represents a successful update on the subscription and the representation of that resource is returned. |
| n/a | | | 204 No Content | |
| RedirectResponse | O | 0..1 | 307 Temporary Redirect | Temporary redirection. (NOTE 2) |
| RedirectResponse | O | 0..1 | 308 Permanent Redirect | Permanent redirection. (NOTE 2) |
| ProblemDetails | O | 0..1 | 404 Not Found | Indicates the modification of Individual IMS Event Subscription has failed due to application error.<br><br>The "cause" attribute may be used to indicate one of the following application errors:<br>- SUBSCRIPTION_NOT_FOUND |
| ProblemDetails | O | 0..1 | 403 Forbidden | The "cause" attribute may be used to indicate one of the following application errors:<br>- OPERATION_NOT_ALLOWED |
| NOTE 1: The manadatory HTTP error status code for the PATCH method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply.<br>NOTE 2: RedirectResponse may be inserted by an SCP, see clause 6.10.9.1 of TS 29.500 [4]. | | | | |

**Table 6.4.3.3.3.3-4: Headers supported by the 307 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative IMS AS (service) instance.<br>Or the same URI, if a request is redirected to the same target resource via a different SCP. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target IMS AS instance ID towards which the request is redirected. |

**Table 6.4.3.3.3.3-5: Headers supported by the 308 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative IMS AS (service) instance.<br>Or the same URI, if a request is redirected to the same target resource via a different SCP. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target IMS AS instance ID towards which the request is redirected. |

#### 6.4.3.3.3.4 DELETE

This method shall support the URI query parameters specified in table 6.4.3.3.3.4-1.

**Table 6.4.3.3.3.4-1: URI query parameters supported by the DELETE method on this resource**

| Name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| n/a | | | | | |

This method shall support the request data structures specified in table 6.4.3.3.3.4-2 and the response data structures and response codes specified in table 6.4.3.3.3.4-3.

**Table 6.4.3.3.3.4-2: Data structures supported by the DELETE Request Body on this resource**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| n/a | | | |

**Table 6.4.3.3.3.4-3: Data structures supported by the DELETE Response Body on this resource**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| n/a | | | 204 No content | |
| RedirectResponse | O | 0..1 | 307 Temporary Redirect | Temporary redirection. (NOTE 2) |
| RedirectResponse | O | 0..1 | 308 Permanent Redirect | Permanent redirection. (NOTE 2) |
| Problem Details | O | 0..1 | 404 Not Found | Indicates the deletion of the Invidudal IMS session has failed due to application error.<br><br>The “cause” attribute may be used to indicate one of the following application errors:<br>- SUBSCRIPTION_NOT_FOUND |
| NOTE 1: The manadatory HTTP error status code for the DELETE method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply.<br>NOTE 2: RedirectResponse may be inserted by an SCP, see clause 6.10.9.1 of TS 29.500 [4]. | | | | |

**Table 6.4.3.3.3.4-4: Headers supported by the 307 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative IMS AS (service) instance.<br>Or the same URI, if a request is redirected to the same target resource via a different SCP. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target IMS AS instance ID towards which the request is redirected. |

**Table 6.4.3.3.3.4-5: Headers supported by the 308 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative IMS AS (service) instance.<br>Or the same URI, if a request is redirected to the same target resource via a different SCP. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target IMS AS instance ID towards which the request is redirected. |

### 6.4.3.3.4 Resource Custom Operations

None.

## 6.4.4 Custom Operations without associated resources

None in this release of specification.

## 6.4.5 Notifications

### 6.4.5.1 General

Notifications shall comply with clause 6.2 of TS 29.500 [4] and clause 4.6.2.3 of TS 29.501 [5].

**Table 6.4.5.1-1: Notifications overview**

| Notification | Callback URI | HTTP method or custom operation | Description (service operation) |
|---|---|---|---|
| Event Notification | {notificationURI} | POST | Reports one or several observed Events. |

### 6.4.5.2 Event Notification

#### 6.4.5.2.1 Description

The Event Notification is used by the IMS AS to report one or several observed Events to an NF service consumer that has subscribed to such Notifications.

#### 6.4.5.2.2 Target URI

The Callback URI {notificationURI} shall be used with the callback URI variables defined in table 6.4.5.1-1.

**Table 6.4.5.2.2-1: Callback URI variables**

| Name | Data type | Definition |
|---|---|---|
| notificationURI | Uri | The Notification Uri as assigned within the Individual IMS Event Subscription and described within the ImsEventsSubscription type. |

#### 6.4.5.2.3 Standard Methods

##### 6.4.5.2.3.1 POST

This method shall support the URI query parameters specified in table 6.4.5.2.3-1.

**Table 6.4.5.2.3.1-1: URI query parameters supported by the POST method on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| n/a | | | | |

**Table 6.4.5.2.3-2: Data structures supported by the POST Request Body on this resource**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| array(ImsEventsNotification) | M | 1..N | Provides Information about observed IMS Events. |

**Table 6.4.5.2.3-3: Data structures supported by the POST Response Body on this resource**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| n/a | | | 204 No Content | The receipt of the Notification is acknowledged. |
| RedirectResponse | O | 0..1 | 307 Temporary Redirect | Temporary redirection, during the event notification. The response shall include a Location header field containing an alternative URI representing the end point of an alternative NF consumer (service) instance where the notification should be sent. |
| RedirectResponse | O | 0..1 | 308 Permanent Redirect | Permanent redirection, during the event notification. The response shall include a Location header field containing an alternative URI representing the end point of an alternative NF consumer (service) instance where the notification should be sent. |
| ProblemDetails | O | 0..1 | 404 Not Found | Indicates the notification of IMS event has failed due to application error.<br><br>The "cause" attribute may be used to indicate one of the following application errors:<br>- CONTEXT_NOT_FOUND |
| NOTE: The mandatory HTTP error status codes for the POST method listed in table 5.2.7.1-1 of TS 29.500 [4] also apply. | | | | |

**Table 6.4.5.2.3-4: Headers supported by the 307 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI representing the end point of an alternative NF consumer (service) instance towards which the notification should be redirected. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF consumer (service) instance towards which the notification request is redirected. |

**Table 6.4.5.2.3-5: Headers supported by the 308 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI representing the end point of an alternative NF consumer (service) instance towards which the notification should be redirected. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target NF consumer (service) instance towards which the notification request is redirected. |

## 6.4.6 Data Model

### 6.4.6.1 General

This clause specifies the application data model supported by the API.

Table 6.4.6.1-1 specifies the data types defined for the Nimsas_ImsEE service based interface protocol.

**Table 6.4.6.1-1: Nimsas_ImsEE specific Data Types**

| Data type | Clause defined | Description |
|---|---|---|
| ImsEventsSubscription | 6.4.6.2.2 | Represents an Individual IMS Event Subscription resource. |
| ImsEventsNotification | 6.4.6.2.4 | The events notification. |
| EventNotification | 6.4.6.2.5 | The notification information for a specific event. |
| ImsEventsSubscriptionCreated | 6.4.6.2.10 | Created Individual IMS Event Subscription |

Table 6.4.6.1-2 specifies data types re-used by the Nimsas_ImsEE service based interface protocol from other specifications, including a reference to their respective specifications and when needed, a short description of their use within the Nimsas_ImsEE service based interface.

**Table 6.4.6.1-2: Nimsas_ImsEE re-used Data Types**

| Data type | Reference | Comments |
|---|---|---|
| DateTime | TS 29.571 [16] | |
| ImsEvent | TS 29.571 [16] | The IMS events |
| ImsEventConfiguration | TS 29.571 [16] | The IMS event configuration info. |
| ImsPublicId | TS 29.562 [15] | IMS Public Identity. |
| ImsEventReport | TS 29.571 [16] | The IMS event report. |
| ImsEventReportInfo | TS 29.571 [16] | The IMS event report specific information. |
| ImsReportingOptions | TS 29.571 [16] | The IMS event reporting options. |
| SessionId | TS 29.571 [16] | The IMS session ID. |
| PatchItem | TS 29.571 [16] | The patch item. |
| Uri | TS 29.571 [16] | |
| ReferenceId | TS 29.503 [22] | Reference ID value of the event configuration and event report |

## 6.4.6.2 Structured data types

### 6.4.6.2.1 Introduction

This clause defines the structures to be used in resource representations.

#### 6.4.6.2.2 Type: ImsEventsSubscription

**Table 6.4.6.2.2-1: Definition of type ImsEventsSubscription**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| subscriberId | ImsPublicId | O | 0..1 | Represents the IMS public subscriber identifies that the subscription targeted for. (NOTE) |
| anyUeInd | boolean | O | 0..1 | Identifies whether the request applies to any UE. This attribute is set to "true" if applicable for any UE, otherwise, it is set to "false". (NOTE) |
| subscriptionId | string | C | 0..1 | Subscription ID.<br>This parameter shall be supplied by the IMS AS in HTTP responses that include an object of ImsEventsSubscription. |
| eventList | map(ImsEventConfiguration) | M | 1..N | A map (list of key-value pairs) where the key shall be a "Reference ID" value, which shall be a 64-bit unsigned integer (as defined by the "ReferenceId" data type specified in TS 29.503 [22], clause 6.4.6.3.2), converted to a string, and which shall be unique for each event configuration created by the NF consumer.<br><br>It contains the IMS events to be configured in the subscription request or the events successfully configured by the IMS AS for this subscription in a subscription response. |
| imsReportingOptions | ImsReportingOptions | O | 0..1 | This IE may be included if the NF service consumer wants to describe how the reports of the event to be generated,<br>If this attribute is absent, the default report mode shall be "ON_EVENT_DETECTION". |
| notificationURI | Uri | C | 0..1 | Identifies the recipient of Notifications sent by the NF consumer.<br>This parameter shall be supplied by the NF service consumer in the HTTP POST requests that create the subscriptions for event notifications and in the HTTP PUT requests that update the subscriptions for event notifications.<br>When an NF Service Consumer subscribes on behalf of another NF, the notificationURI identifies a resource under the authority of the other NF. |
| supportedFeatures | SupportedFeatures | O | 0..1 | These are the features supported by the sender of the message. |
| notificationCorrelationId | string | C | 0..1 | This IE shall be present if the notificationURI IE is present.<br>When present, this attribute identifies the notification correlation ID present by NF consumer in subscription.<br>The combination value of the notificationURI IE and this IE shall be unique per subscription for a given NF service consumer. |
| NOTE: If the event subscription applies for a specific UE, subscriberId shall be included; otherwise anyUeInd set to true shall be included. | | | | |

#### 6.4.6.2.3 Void

#### 6.4.6.2.4 Type: ImsEventsNotification

**Table 6.4.6.2.4-1: Definition of type ImsEventsNotification**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| eventNotifications | array(EventNotification) | M | 1..N | Notifications about Individual Events. |
| notificationCorrelationId | string | M | 1 | String identifying a notification correlation Id received in the subscribe . |

#### 6.4.6.2.5 Type: EventNotification

**Table 6.4.6.2.5-1: Definition of type EventNotification**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| event | ImsEvent | M | 1 | Event that is notified. |
| referenceId | ReferenceId | O | 0..1 | Reference ID value of the notified event |
| eventReport | ImsEventReportInfo | O | 0..1 | Report Information per event type |
| subscriberId | ImsPublicId | M | 1 | Represents the IMS public subscriber. |
| timestamp | DateTime | M | 1 | Time at which the event is observed. |
| sessionId | SessionId | O | 0..1 | Identifies the session ID. |

#### 6.4.6.2.6 Void

#### 6.4.6.2.7 Void

#### 6.4.6.2.8 Void

#### 6.4.6.2.9 Void

#### 6.4.6.2.10 Type: ImsEventsSubscriptionCreated

**Table 6.4.6.2.10-1: Definition of type ImsEventsSubscriptionCreated**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| imsEventsSubscription | ImsEventsSubscription | M | 1 | This IE shall contain the representation of the created IMS event subscription. |
| eventReports | array(ImsEventReport) | O | 1..N | If immediate report is requested, this IE shall contain the associated status to the event(s) being requested if the status associated to at least one of the requested event(s) is available at the time of subscription. |

#### 6.4.6.2.11 Void

### 6.4.6.3 Simple data types and enumerations

#### 6.4.6.3.1 Introduction

This clause defines simple data types and enumerations that can be referenced from data structures defined in the previous clauses.

#### 6.4.6.3.2 Simple data types

The simple data types defined in table 6.4.6.3.2-1 shall be supported.

**Table 6.4.6.3.2-1: Simple data types**

| Type Name | Type Definition | Description | Applicability |
|---|---|---|---|
| | | | |

### 6.4.6.4 Data types describing alternative data types or combinations of data types

None in this release of the specification.

#### 6.4.6.5 Binary data

None in this release of the specification.

### 6.4.7 Error Handling

#### 6.4.7.1 General

For the Nimsas_ImsEE API, HTTP error responses shall be supported as specified in clause 4.8 of TS 29.501 [5]. Protocol errors and application errors specified in table 5.2.7.2-1 of TS 29.500 [4] shall be supported for an HTTP method if the corresponding HTTP status codes are specified as mandatory for that HTTP method in table 5.2.7.1-1 of TS 29.500 [4].

In addition, the requirements in the following clauses are applicable for the Nimsas_ImsEE API.

#### 6.4.7.2 Protocol Errors

No specific procedures for the Nimsas_ImsEE service are specified.

#### 6.4.7.3 Application Errors

The application errors defined for the Nimsas_ImsEE service are listed in Table 6.4.7.3-1. The common application errors defined in the Table 5.2.7.2-1 in TS 29.500 [4] may also be used for the Nimsas_imsEE service. The following application errors listed in Table 6.4.7.3-1 are specific for the Nimsas_imsEE service.

**Table 6.4.7.3-1: Application errors**

| Application Error | HTTP status code | Description |
|---|---|---|
| SUBSCRIPTION_NOT_FOUND | 404 Not Found | The service user is not served by this IMS AS. |
| OPERATION_NOT_ALLOWED | 403 Forbidden | Indicates the IMS Event subscription is not allowed to be created/modified or the information of the IMS Event subscription is not allowed to be retrived. |
| CONTEXT_NOT_FOUND | 404 Not Found | The NF service consumer receives a notification from IMS AS for IMS event subscription that is not found for the service consumer. |

### 6.4.8 Feature negotiation

The optional features in table 6.4.8-1 are defined for the Nimsas_ImsEE API. They shall be negotiated using the extensibility mechanism defined in clause 6.6 of TS 29.500 [4].

**Table 6.4.8-1: Supported Features**

| Feature number | Feature Name | Description |
|---|---|---|
| | | |

### 6.4.9 Security

As indicated in TS 33.501 [8] and TS 29.500 [4], the access to the Nimsas_ImsEE API may be authorized by means of the OAuth2 protocol (see IETF RFC 6749 [9]), based on local configuration, using the "Client Credentials" authorization grant, where the NRF (see TS 29.510 [10]) plays the role of the authorization server.

If OAuth2 is used, an NF Service Consumer, prior to consuming services offered by the Nimsas_ImsEE API, shall obtain a "token" from the authorization server, by invoking the Access Token Request service, as described in TS 29.510 [10], clause 5.4.2.2.

NOTE: When multiple NRFs are deployed in a network, the NRF used as authorization server is the same NRF that the NF Service Consumer used for discovering the Nimsas_EE service.

The Nimsas_ImsEE API defines a single scope "nimsas-ee" for the entire service, and it does not define any additional scopes at resource or operation level.

### 6.4.10 HTTP redirection

An HTTP request may be redirected to a different IMS AS service instance when using direct or indirect communications (see TS 29.500 [4]).

An SCP that reselects a different NF provider (e.g. IMS AS) instance will return the NF Instance ID of the new NF provider instance in the 3gpp-Sbi-Producer-Id header, as specified in clause 6.10.3.4 of TS 29.500 [4].

If an NF provider (e.g. IMS AS) redirects a service request to a different NF provider using a 307 Temporary Redirect or 308 Permanent Redirect status code, the identity of the new NF provider towards which the service request is redirected shall be indicated in the 3gpp-Sbi-Target-Nf-Id header of the 307 Temporary Redirect or 308 Permanent Redirect response as specified in clause 6.10.9.1 of TS 29.500 [4].

## 6.5 Nimsas_ImsParameterProvision Service API

### 6.5.1 API URI

The Nimsas_ImsParameterProvision service shall use the Nimsas_ImsParameterProvision API.

The API URI of the Nimsas_ImsParameterProvision API shall be:

**{apiRoot}/<apiName>/<apiVersion>**

The request URIs used in HTTP requests from the NF service consumer towards the NF service producer shall have the Resource URI structure defined in clause 4.4.1 of TS 29.501 [5], i.e.:

**{apiRoot}/<apiName>/<apiVersion>/<apiSpecificResourceUriPart>**

with the following components:

- The {apiRoot} shall be set as described in TS 29.501 [5].
- The <apiName> shall be "nimsas-ipp".
- The <apiVersion> shall be "v1".
- The <apiSpecificResourceUriPart> shall be set as described in clause 6.5.3.

### 6.5.2 Usage of HTTP

#### 6.5.2.1 General

HTTP/2, IETF RFC 7540 [11], shall be used as specified in clause 5 of TS 29.500 [4].

HTTP/2 shall be transported as specified in clause 5.3 of TS 29.500 [4].

The OpenAPI [6] specification of HTTP messages and content bodies for the Nimsas_ImsParameterProvision API is contained in Annex A.

#### 6.5.2.2 HTTP standard headers

##### 6.5.2.2.1 General

See clause 5.2.2 of TS 29.500 [4] for the usage of HTTP standard headers.

#### 6.5.2.2.2 Content type

JSON, IETF RFC 8259 [12], shall be used as content type of the HTTP bodies specified in the present specification as specified in clause 5.4 of TS 29.500 [4]. The use of the JSON format shall be signalled by the content type "application/json".

"Problem Details" JSON object shall be used to indicate additional details of the error in a HTTP response body and shall be signalled by the content type "application/problem+json", as defined in IETF RFC 7807 [13].

### 6.5.2.3 HTTP custom headers

The mandatory HTTP custom header fields specified in clause 5.2.3.2 of TS 29.500 [4] shall be supported, and the optional HTTP custom header fields specified in clause 5.2.3.3 of TS 29.500 [4] may be supported.

## 6.5.3 Resources

### 6.5.3.1 Overview

This clause describes the structure for the Resource URIs and the resources and methods used for the service.

Figure 6.5.3.1-1 depicts the resource URIs structure for the Nimsas_ImsParameterProvision service API.

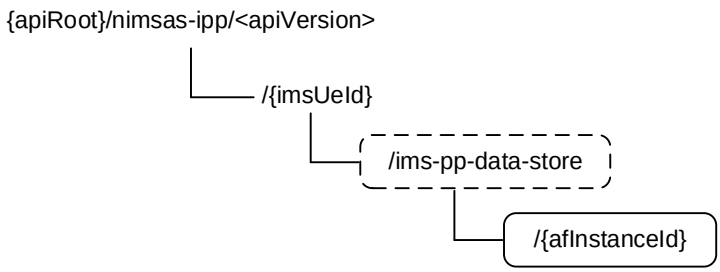



**Figure 6.5.3.1-1: Resource URI structure of the Nimsas_ImsParameterProvision API**

Table 6.5.3.1-1 provides an overview of the resources and applicable HTTP methods.

**Table 6.5.3.1-1: Resources and methods overview**

| Resource purpose/name | Resource URI (relative path after API URI) | HTTP method or custom operation | Description (service operation) |
|---|---|---|---|
| ImsParameterProvisioningDataEntry | /{imsUeId}/ims-pp-data-store/{afInstanceId} | PUT | Create or update a parameter provisioning data entry per AF for a specific IMS user. |
| | | DELETE | Delete a parameter provisioning data entry per AF for a specific IMS user. |

### 6.5.3.2 Resource: ImsParameterProvisioningDataEntry

#### 6.5.3.2.1 Description

The ImsParameterProvisioningDataEntry resource is used to allow the storage of a collection of parameter provisioning entries for different Application Functions.

In this release, this resource is used to store the IMS user related information created by NF service consumers of Nimsas_ImsParameterProvision service for a given IMS user.

The resource is modelled as Document resource archetype (see clause C.2 of TS 29.501 [5]).

#### 6.5.3.2.2 Resource Definition

Resource URI: **{apiRoot}/nimsas-ipp/<apiVersion>/{imsUeId}/ims-pp-data-store/{afInstanceId}**

This resource shall support the resource URI variables defined in table 6.5.3.2.2-1.

**Table 6.5.3.2.2-1: Resource URI variables for this resource**

| Name | Data type | Definition |
|---|---|---|
| apiRoot | string | See clause 6.5.1 |
| imsUeId | string | Represents the IMS Public Identity (i.e. IMPU).<br><br>Pattern: "^(impu-sip\:([a-zA-Z0-9_\-.!~*()&=+$,;?\/]+)\@([A-Za-z0-9]+([-A-Za-z0-9]+)\.)+[a-z]{2,}\|impu-tel\:\+[0-9]{5,15})$" |
| afInstanceId | string | The string identifying the AF instance. |

#### 6.5.3.2.3 Resource Standard Methods

##### 6.5.3.2.3.1 PUT

This method shall support the URI query parameters specified in table 6.5.3.2.3.1-1.

**Table 6.5.3.2.3.1-1: URI query parameters supported by the PUT method on this resource**

| Name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| n/a | | | | | |

This method shall support the request data structures specified in table 6.5.3.2.3.1-2 and the response data structures and response codes specified in table 6.5.3.2.3.1-3.

**Table 6.5.3.2.3.1-2: Data structures supported by the PUT Request Body on this resource**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| ImsPpDataEntry | M | 1 | Creates or updates an ImsParameterProvisioningDataEntry resource. |

**Table 6.5.3.2.3.1-3: Data structures supported by the PUT Response Body on this resource**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| ImsPpDataEntry | M | 1 | 200 OK | Successful resource update, a response body containing a representation of the modified ImsParameterProvisioningDataEntry resource is returned. |
| ImsPpDataEntry | M | 1 | 201 Created | Successful resource creation, a response body containing a representation of the created ImsParameterProvisioningDataEntry resource is returned. |
| n/a | | | 204 No Content | Successful resource update. |
| RedirectResponse | O | 0..1 | 307 Temporary Redirect | Temporary redirection. (NOTE 2) |
| RedirectResponse | O | 0..1 | 308 Permanent Redirect | Permanent redirection. (NOTE 2) |
| NOTE 1: The mandatory HTTP error status codes for the PUT method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply.<br>NOTE 2: RedirectResponse may be inserted by an SCP, see clause 6.10.9.1 of TS 29.500 [4]. | | | | |

**Table 6.5.3.2.3.1-4: Headers supported by the 307 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative IMS AS (service) instance.<br>Or the same URI, if a request is redirected to the same target resource via a different SCP. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target IMS AS instance ID towards which the request is redirected. |

**Table 6.5.3.2.3.1-5: Headers supported by the 308 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative IMS AS (service) instance.<br>Or the same URI, if a request is redirected to the same target resource via a different SCP. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target IMS AS instance ID towards which the request is redirected. |

**Table 6.5.3.2.3.1-6: Headers supported by the 201 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | Contains the URI of the newly created resource, according to the structure: {apiRoot}/nimsas-ipp/<apiVersion>/{imsUeId}/ims-pp-data-store/{afInstanceId} |

### 6.5.3.2.3.2 DELETE

This method shall support the URI query parameters specified in table 6.5.3.2.3.2-1.

**Table 6.5.3.2.3.2-1: URI query parameters supported by the DELETE method on this resource**

| Name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| n/a | | | | | |

This method shall support the request data structures specified in table 6.5.3.2.3.2-2 and the response data structures and response codes specified in table 6.5.3.2.3.2-3.

**Table 6.5.3.2.3.2-2: Data structures supported by the DELETE Request Body on this resource**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| n/a | | | |

**Table 6.5.3.2.3.2-3: Data structures supported by the DELETE Response Body on this resource**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| n/a | | | 204 No content | Successful resource deletion. |
| RedirectResponse | O | 0..1 | 307 Temporary Redirect | Temporary redirection. (NOTE 2) |
| RedirectResponse | O | 0..1 | 308 Permanent Redirect | Permanent redirection. (NOTE 2) |
| Problem Details | O | 0..1 | 404 Not Found | Indicates that the deletion of an ImsParameterProvisioningDataEntry resource failed due to application error.<br><br>The "cause" attribute may be used to indicate one of the following application errors:<br>- RESOURCE_URI_STRUCTURE_NOT_FOUND |
| NOTE 1: The mandatory HTTP error status codes for the DELETE method listed in table 5.2.7.1-1 of TS 29.500 [4] also apply.<br>NOTE 2: RedirectResponse may be inserted by an SCP, see clause 6.10.9.1 of TS 29.500 [4]. | | | | |

**Table 6.5.3.2.3.2-4: Headers supported by the 307 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative IMS AS (service) instance.<br>Or the same URI, if a request is redirected to the same target resource via a different SCP. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target IMS AS instance ID towards which the request is redirected. |

**Table 6.5.3.2.3.2-5: Headers supported by the 308 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location | string | M | 1 | An alternative URI of the resource located on an alternative IMS AS (service) instance.<br>Or the same URI, if a request is redirected to the same target resource via a different SCP. |
| 3gpp-Sbi-Target-Nf-Id | string | O | 0..1 | Identifier of the target IMS AS instance ID towards which the request is redirected. |

#### 6.5.3.2.4 Resource Custom Operations

None.

### 6.5.4 Custom Operations without associated resources

None in this release of specification.

### 6.5.5 Notifications

None in this release of specification.

### 6.5.6 Data Model

#### 6.5.6.1 General

This clause specifies the application data model supported by the API.

Table 6.5.6.1-1 specifies the data types defined for the Nimsas_ImsParameterProvision service based interface protocol.

**Table 6.5.6.1-1: Nimsas_ImsParameterProvision specific Data Types**

| Data type | Clause defined | Description |
|---|---|---|
| ImsPpDataEntry | 6.5.6.2.2 | The IMS parameter provision data entry. |

Table 6.5.6.1-2 specifies data types re-used by the Nimsas_ImsParameterProvision service based interface protocol from other specifications, including a reference to their respective specifications and when needed, a short description of their use within the Nimsas_ImsParameterProvision service based interface.

**Table 6.5.6.1-2: Nimsas_ImsParameterProvision re-used Data Types**

| Data type | Reference | Comments |
|---|---|---|
| RcdProperties | TS 29.571 [16] | Contains the RCD properties (i.e. RCD information, RCD server address or RCD URL). |

## 6.5.6.2 Structured data types

### 6.5.6.2.1 Introduction

This clause defines the structures to be used in resource representations.

### 6.5.6.2.2 Type: ImsPpDataEntry

**Table 6.5.6.2.2-1: Definition of type ImsPpDataEntry**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| rcdProp | RcdProperties | O | 0..1 | Contains the RCD properties (i.e. RCD information, RCD server address or RCD URL). |

## 6.5.6.3 Simple data types and enumerations

### 6.5.6.3.1 Introduction

This clause defines simple data types and enumerations that can be referenced from data structures defined in the previous clauses.

### 6.5.6.3.2 Simple data types

The simple data types defined in table 6.5.6.3.2-1 shall be supported.

**Table 6.5.6.3.2-1: Simple data types**

| Type Name | Type Definition | Description | Applicability |
|---|---|---|---|
| | | | |

## 6.5.6.4 Data types describing alternative data types or combinations of data types

None in this release of the specification.

## 6.5.6.5 Binary data

None in this release of the specification.

## 6.5.7 Error Handling

### 6.5.7.1 General

For the Nimsas_ImsParameterProvision API, HTTP error responses shall be supported as specified in clause 4.8 of TS 29.501 [5]. Protocol errors and application errors specified in table 5.2.7.2-1 of TS 29.500 [4] shall be supported for an HTTP method if the corresponding HTTP status codes are specified as mandatory for that HTTP method in table 5.2.7.1-1 of TS 29.500 [4].

In addition, the requirements in the following clauses are applicable for the Nimsas_ImsParameterProvision API.

### 6.5.7.2 Protocol Errors

No specific procedures for the Nimsas_ImsParameterProvision service are specified.

### 6.5.7.3 Application Errors

The application errors defined for the Nimsas_ImsParameterProvision service are listed in Table 6.5.7.3-1.

**Table 6.5.7.3-1: Application errors**

| Application Error | HTTP status code | Description |
|---|---|---|
| | | |

## 6.5.8 Feature negotiation

The optional features in table 6.5.8-1 are defined for the Nimsas_ImsParameterProvision API. They shall be negotiated using the extensibility mechanism defined in clause 6.6 of TS 29.500 [4].

**Table 6.5.8-1: Supported Features**

| Feature number | Feature Name | Description |
|---|---|---|
| | | |

## 6.5.9 Security

As indicated in TS 33.501 [8] and TS 29.500 [4], the access to the Nimsas_ImsParameterProvision API may be authorized by means of the OAuth2 protocol (see IETF RFC 6749 [9]), based on local configuration, using the "Client Credentials" authorization grant, where the NRF (see TS 29.510 [10]) plays the role of the authorization server.

If OAuth2 is used, an NF Service Consumer, prior to consuming services offered by the Nimsas_ImsParameterProvision API, shall obtain a "token" from the authorization server, by invoking the Access Token Request service, as described in TS 29.510 [10], clause 5.4.2.2.

NOTE: When multiple NRFs are deployed in a network, the NRF used as authorization server is the same NRF that the NF Service Consumer used for discovering the Nimsas_ImsParameterProvision service.

The Nimsas_ImsParameterProvision API defines a single scope "nimsas-ipp" for the entire service, and it does not define any additional scopes at resource or operation level.

## 6.5.10 HTTP redirection

An HTTP request may be redirected to a different IMS AS service instance when using direct or indirect communications (see TS 29.500 [4]).

An SCP that reselects a different NF provider (e.g. IMS AS) instance will return the NF Instance ID of the new NF provider instance in the 3gpp-Sbi-Producer-Id header, as specified in clause 6.10.3.4 of TS 29.500 [4].

If an NF provider (e.g. IMS AS) redirects a service request to a different NF provider using a 307 Temporary Redirect or 308 Permanent Redirect status code, the identity of the new NF provider towards which the service request is redirected shall be indicated in the 3gpp-Sbi-Target-Nf-Id header of the 307 Temporary Redirect or 308 Permanent Redirect response as specified in clause 6.10.9.1 of TS 29.500 [4].

# Annex A (normative): OpenAPI specification

## A.1 General

This Annex specifies the formal definition of the API(s) defined in the present specification. It consists of OpenAPI specifications in YAML format.

This Annex takes precedence when being discrepant to other parts of the specification with respect to the encoding of information elements and methods within the API(s).

NOTE 1: The semantics and procedures, as well as conditions, e.g. for the applicability and allowed combinations of attributes or values, not expressed in the OpenAPI definitions but defined in other parts of the specification also apply.

Informative copies of the OpenAPI specification files contained in this 3GPP Technical Specification are available on a Git-based repository that uses the GitLab software version control system (see clause 5.3.1 of TS 29.501 [5] and clause 5B of TR 21.900 [7]).

## A.2 Nimsas_SessionEventControl API

```
openapi: 3.0.0

info:

  title: 'IMS AS SessionEventControl Service'
  version: 1.1.1
  description: |
    Nimsas_SessionEventControl Service.
    © 2026, 3GPP Organizational Partners (ARIB, ATIS, CCSA, ETSI, TSDSI, TTA, TTC).
    All rights reserved.

externalDocs:
  description: >
    3GPP TS 29.175 V19.7.0; IP Multimedia Subsystem; IP Multimedia Subsystem (IMS) Application
    Server (AS) Services; Stage 3
  url: 'https://www.3gpp.org/ftp/Specs/archive/29_series/29.175/'

servers:
  - url: '{apiRoot}/nimsas-sec/v1'
    variables:
      apiRoot:
        default: https://example.com
        description: apiRoot as defined in clause 4.4 of 3GPP TS 29.501

security:
  - {}
  - oAuth2ClientCredentials:
      - nimsas-sec

paths:
  /session-event-subscriptions:
    post:
    # This is a pseudo operation, NF service consumers shall NOT invoke this method!
      summary: Nimsas_SessionEventControl Subscribe service operation, pseudo operation
      tags:
        - Session Event Subscriptions (Collection)
      operationId: Subscribe
      requestBody:
        required: true
        content:
          application/json:
            # Unspecified schema for the JSON body, since this is not used by consumer or producer.
            schema: {}
      responses:
        default:
          $ref: 'TS29571_CommonData.yaml#/components/responses/default'
```

```
      callbacks:
        eeNotification:
          '{SessionEventNotificationUri}':
          # The notification URI is provided by local configuration or via NRF.
            post:
              requestBody:
                required: true
                content:
                  application/json:
                    schema:
                      $ref: '#/components/schemas/SessionEventNotification'
              responses:
                '204':
                  description: No Content, notification was succesfull.
                '307':
                  description: Temporary Redirect
                  content:
                    application/json:
                      schema:
                        $ref: 'TS29571_CommonData.yaml#/components/schemas/RedirectResponse'
                  headers:
                    Location:
                      description: >
                        The URI pointing to the resource located on the redirect target NF
                        service consumer.
                      required: true
                      schema:
                        type: string
                '308':
                  description: Permanent Redirect
                  content:
                    application/json:
                      schema:
                        $ref: 'TS29571_CommonData.yaml#/components/schemas/RedirectResponse'
                  headers:
                    Location:
                      description: >
                        The URI pointing to the resource located on the redirect target NF
                        service consumer.
                      required: true
                      schema:
                        type: string
                '400':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/400'
                '401':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/401'
                '403':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/403'
                '404':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/404'
                '411':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/411'
                '413':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/413'
                '415':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/415'
                '429':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/429'
                '500':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/500'
                '502':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/502'
                '503':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/503'
                default:
                  $ref: 'TS29571_CommonData.yaml#/components/responses/default'

components:

  securitySchemes:
    oAuth2ClientCredentials:
      type: oauth2
      flows:
        clientCredentials:
          tokenUrl: '{nrfApiRoot}/oauth2/token'
          scopes:
            nimsas-sec: Access to the Nimsas_SessionEventControl API
```

```
  schemas:

# STRUCTURED DATA TYPES

    SessionEventNotification:
      description: The session event notification item.
      type: object
      required:
        - notificationEvent
        - sessionId
      properties:
        notificationEvent:
          $ref: '#/components/schemas/NotificationEvent'
        sessionId:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/SessionId'
        sessionInfo:
          $ref: '#/components/schemas/SessionInfo'
        mediaInfoList:
          description: >
            Contains a media related information. The mediaId attribute within the MediaInfo
            data type is the key of the map.
          type: object
          additionalProperties:
            $ref: '#/components/schemas/MediaInfo'
          minItems: 1
        psDataOff:
          $ref: '#/components/schemas/PsDataOffStatus'
        mediaInfoSet:
          type: array
          minItems: 1
          items:
            $ref: 'TS29175_Nimsas_ImsSessionManagement.yaml#/components/schemas/MediaInfoExternal'
        chargingInfo:
          $ref: '#/components/schemas/ChargingInfo'



    NotificationEvent:
      description: The notification event information.
      type: object
      required:
        - eventType
      properties:
        eventType:
          $ref: '#/components/schemas/EventType'
        eventInitiator:
          $ref: '#/components/schemas/EventInitiator'

    SessionInfo:
      description: The session information related to the notification event.
      type: object
      properties:
        callingIdentity:
          $ref: 'TS29562_Nhss_imsSDM.yaml#/components/schemas/ImsPublicId'
        calledIdentity:
          $ref: 'TS29562_Nhss_imsSDM.yaml#/components/schemas/ImsPublicId'
        sessionCase:
          $ref: '#/components/schemas/SessionCase'

    MediaInfo:
      description: The media information related to the notification event.
      type: object
      required:
        - mediaId
        - mediaType
      properties:
        mediaId:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/MediaId'
        mediaType:
          $ref: '#/components/schemas/MediaType'
        dcMediaSpec:
          $ref: '#/components/schemas/DcMediaSpec'
        mediaSuspended:
          type: boolean
        mediaRemovalInd:
          type: boolean
          enum:
```

```
            - true


    DcMediaSpec:
      description: >
        The data channel media specification information contains media description attributes.
      type: object
      required:
        - streams
        - receivedDcEndpoint
      properties:
        streams:
          description: >
            Contains a data channel mapping and configuration information. The streamId
            attribute within the DcStream data type is the key of the map.
          type: object
          additionalProperties:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/DcEndpoint'
          minItems: 1
        maxMessageSize:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/MaxMessageSize'
        receivedDcEndpoint:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/DcEndpoint'
        bdcUsedBy:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/BdcUsedBy'
        appBindingInfo:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/AppBindingInfo'
        multiAppBindingInfo:
          description: Includes a list of multiplexed application binding information.
          type: array
          minItems: 1
          items:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/AppBindingInfo'
        qosHintInfo:
          $ref: '#/components/schemas/QosHintInfo'

    QosHintInfo:
      description: The QoS hint information used for the media.
      type: object
      properties:
        loss:
          description: the maximum desirable end-to-end transport level packet loss rate
          type: number
          format: zero-based-integer or non-zero-real
        localLoss:
          description: the maximum desirable end-to-end transport level packet loss rate applied
across the local link
          type: number
          format: zero-based-integer or non-zero-real
        latency:
          description: the maximum desirable end-to-end transport level packet latency
          type: number
          format: zero-based-integer or non-zero-real
        localLatency:
          description: the maximum desirable end-to-end transport level packet latency applied
across the local link
          type: number
          format: zero-based-integer or non-zero-real

    ChargingInfo:
      description: The charging information derived from the SIP headers.
      type: object
      properties:
        ratTypes:
          description: List of RAT types derived from the SIP header
          type: array
          items:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/RatType'
          minItems: 1
        ncgi:
          description: NR Cell Global Identity derived from the SIP header
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Ncgi'
        ecgi:
          description: E-UTRA Cell Global Identity derived from the SIP header
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Ecgi'
        imsServiceType:
          description: IMS Communication Service Identifier derived from the SIP header
          type: string
```

```
        servingPlmnId:
          description: Serving PLMN ID derived from the SIP header
          $ref: 'TS29571_CommonData.yaml#/components/schemas/PlmnId'



# ENUMS

    EventType:
      description: The type of the event for which the notification is generated.
      anyOf:
        - type: string
          enum:
            - SESSION_ESTABLISHMENT_REQUEST
            - SESSION_ESTABLISHMENT_PROGRESS
            - SESSION_ESTABLISHMENT_ALERTING
            - SESSION_ESTABLISHMENT_SUCCESS
            - SESSION_ESTABLISHMENT_FAILURE
            - SESSION_ESTABLISHMENT_CANCEL
            - MEDIA_CHANGE_REQUEST
            - MEDIA_CHANGE_SUCCESS
            - MEDIA_CHANGE_FAILURE
            - MEDIA_CHANGE_CANCEL
            - SESSION_TERMINATION
            - EXTERNAL_SESSION_UPDATE
            - EXTERNAL_SESSION_CREATE
            - PS_DATA_OFF_STATUS_CHANGE
        - type: string
          description: >
            This string provides forward-compatibility with future extensions to the enumeration
            and is not used to encode content defined in the present version of this API.

    MediaType:
      description: The type of the media for which the notification is generated.
      anyOf:
        - type: string
          enum:
            - DC
            - AUDIO
            - VIDEO
        - type: string
          description: >
            This string provides forward-compatibility with future extensions to the enumeration
            and is not used to encode content defined in the present version of this API.

    SessionCase:
      description: The type of the session case.
      anyOf:
        - type: string
          enum:
            - ORIGINATING_IMS_SESSION
            - TERMINATING_IMS_SESSION
            - EXTERNAL_INITIATED_IMS_SESSION
        - type: string
          description: >
            This string provides forward-compatibility with future extensions to the enumeration
            and is not used to encode content defined in the present version of this API.

    EventInitiator:
      description: The initiator of the event for which the notification is generated.
      anyOf:
        - type: string
          enum:
            - SERVED_IMS_SUBSCRIBER
            - REMOTE_IMS_SUBSCRIBER
            - EXTERNAL_PARTY
        - type: string
          description: >
            This string provides forward-compatibility with future extensions to the enumeration
            and is not used to encode content defined in the present version of this API.

    PsDataOffStatus:
      description: The active/inactive status of PS Data Off.
      anyOf:
        - type: string
          enum:
            - ACTIVE
            - INACTIVE
```

```
        - type: string
          description: >
            This string provides forward-compatibility with future extensions to the enumeration
            and is not used to encode content defined in the present version of this API.
```

## A.3 Nimsas_MediaControl API

```
openapi: 3.0.0

info:
  title: 'IMS AS MediaControl Service'
  version: 1.1.3
  description: |
    Nimsas_MediaControl Service.
    © 2026, 3GPP Organizational Partners (ARIB, ATIS, CCSA, ETSI, TSDSI, TTA, TTC).
    All rights reserved.

externalDocs:
  description: >
    3GPP TS 29.175 V19.7.0; IP Multimedia Subsystem; IP Multimedia Subsystem (IMS) Application
    Server (AS) Services; Stage 3
  url: 'https://www.3gpp.org/ftp/Specs/archive/29_series/29.175'

servers:
  - url: '{apiRoot}/nimsas-mc/v1'
    variables:
      apiRoot:
        default: https://example.com
        description: apiRoot as defined in clause 4.4 of 3GPP TS 29.501

security:
  - {}
  - oAuth2ClientCredentials:
      - nimsas-mc

paths:

  /call-sessions/{sessionId}/media-instruction:
    post:
      summary: Update
      operationId: UpdateCallSession
      tags:
        - Individual call session (Document)
      parameters:
        - name: sessionId
          in: path
          description: Session ID assigned by the IMS AS during the IMS session setup.
          required: true
          schema:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/SessionId'
      requestBody:
        description: Representation of the updates to apply to the call session.
        required: true
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/MediaInstructionData'
      responses:
        '200':
          description: >
            Successful update of the call session, when the IMS AS needs to return information
            in the response.
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/MediaInstructionData'
        '204':
          description: >
            Successful update of the call session, when the IMS AS does not need to return
            information in the response.
        '307':
          description: Temporary Redirect
          content:
```

```
            application/json:
              schema:
                $ref: 'TS29571_CommonData.yaml#/components/schemas/RedirectResponse'
          headers:
            Location:
              description: The URI pointing to the resource located on the redirect target IMS AS.
              schema:
                type: string
        '308':
          description: Permanent Redirect
          content:
            application/json:
              schema:
                $ref: 'TS29571_CommonData.yaml#/components/schemas/RedirectResponse'
          headers:
            Location:
              description: The URI pointing to the resource located on the redirect target IMS AS.
              schema:
                type: string
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '406':
          $ref: 'TS29571_CommonData.yaml#/components/responses/406'
        '411':
          $ref: 'TS29571_CommonData.yaml#/components/responses/411'
        '413':
          $ref: 'TS29571_CommonData.yaml#/components/responses/413'
        '415':
          $ref: 'TS29571_CommonData.yaml#/components/responses/415'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '501':
          $ref: 'TS29571_CommonData.yaml#/components/responses/501'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          $ref: 'TS29571_CommonData.yaml#/components/responses/default'

components:

  securitySchemes:
    oAuth2ClientCredentials:
      type: oauth2
      flows:
        clientCredentials:
          tokenUrl: '{nrfApiRoot}/oauth2/token'
          scopes:
            nimsas-mc: Access to the Nimsas_MediaControl API.

  schemas:

# STRUCTURED DATA TYPES

    MediaInstructionData:
      description: The media instruction data for a specific session.
      type: object
      required:
        - sessionId
        - mediaInstructionSet
      properties:
        sessionId:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/SessionId'
        mediaInstructionSet:
          description: >
            Contains a set of instructions for each media flow to control. The mediaId
            attribute within the MediaInstructions data type is the key of the map.
          type: object
          additionalProperties:
            $ref: '#/components/schemas/MediaInstructions'
          minProperties: 1
```

```
    MediaInstructions:
      description: The media instructions for a specific media flow.
      type: object
      required:
        - mediaId
        - mediaResourceType
      properties:
        mediaId:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/MediaIdRm'
        associatedMediaId:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/MediaId'
        mediaResourceType:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/MediaResourceType'
        mediaInstruction:
          $ref: '#/components/schemas/MediaInstruction'
        dcMediaSpecification:
          $ref: '#/components/schemas/DcMediaSpecification'
        arMediaSpecification:
          $ref: '#/components/schemas/ArMediaSpecification'
        mediaProcessingUrl:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
        audioVideoReNegotiationInd:
          $ref: '#/components/schemas/AudioVideoReNegotiationInd'
        mdc2AudioEndpoint:
          $ref: '#/components/schemas/Mdc2AudioVideoEndpoint'
        mdc2VideoEndpoint:
          $ref: '#/components/schemas/Mdc2AudioVideoEndpoint'
        avatarMedia:
          $ref: 'TS29176_Nmf_MRM.yaml#/components/schemas/AvatarMedia'

    DcMediaSpecification:
      description: The data channel media specification information.
      type: object
      required:
        - streams
        - mediaProxyConfig
      properties:
        mediaProxyConfig:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/MediaProxy'
        replaceHttpUrls:
          description: >
            Contains a list of replacement HTTP URLs. The streamId
            attribute within the ReplaceHttpUrl data type is the key of the map.
          type: object
          additionalProperties:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/ReplaceHttpUrl'
          minProperties: 1
        mdc1EndpointDcsf:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/MdcEndpoint'
        mdc1EndpointMf:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/MdcEndpoint'
        mdc2EndpointInfo:
          $ref: '#/components/schemas/Mdc2EndpointInfo'
        streams:
          description: >
            Contains a data channel mapping and configuration information. The streamId
            attribute within the DcStream data type is the key of the map.
          type: object
          additionalProperties:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/DcStream'
          minProperties: 1
        bdcUsedBy:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/BdcUsedBy'
        appBindingInfo:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/AppBindingInfo'
        qosHintInfo:
          $ref: 'TS29175_Nimsas_SessionEventControl.yaml#/components/schemas/QosHintInfo'
        interworkingInfo:
          $ref: '#/components/schemas/InterworkingInstruction'
        dcIwkInd:
          type: boolean
          default: false



    ArMediaSpecification:
```

```
      description: The AR media specification information.
      type: object
      required:
        - mediaProcessingSpec
      properties:
        mediaProcessingSpec:
          type: string

    Mdc2EndpointInfo:
      description: Represents the negotiated MDC2 media endpoint information.
      type: object
      properties:
        mdc2Protocol:
          $ref: 'TS29176_Nmf_MRM.yaml#/components/schemas/Mdc2Protocol'
        mdc2EndpointDcAs:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/MdcEndpoint'
        mdc2EndpointMf:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/MdcEndpoint'

    AudioVideoReNegotiationInd:
      description: The audio/video media re-negotiation indication information.
      type: object
      required:
        - reNegotiation
      properties:
        reNegotiation:
          type: boolean
        mediaConnSide:
          $ref: '#/components/schemas/MediaConnSide'
        mediaDirection:
          $ref: '#/components/schemas/SendRecvMode'

    InterworkingInstruction:
      description: The interworking instruction of data channel transcoding.
      type: object
      required:
        - transcodeMode
      properties:
        transcodeMode:
          $ref: '#/components/schemas/TranscodeMode'
        addTransInfo:
          type: string

    Mdc2AudioVideoEndpoint:
      description: The audio or video media endpoint information in MDC2 interface.
      type: object
      properties:
        mdc2AVEndpointDcAs:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Endpoint'
        mdc2AVEndpointMf:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Endpoint'
        audioVideoMedia:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/AudioVideoMedia'
        audioMediaParams:
          description: the description of the audio media.
          type: array
          items:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/AudioVideoMedia'
        videoMediaParams:
          description: the description of the video media.
          type: array
          items:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/AudioVideoMedia'




# ENUMS

    MediaInstruction:
      description: The media instruction for handling a specific media flow.
      anyOf:
        - type: string
          enum:
            - TERMINATE_MEDIA
            - ORIGINATE_MEDIA
            - TERMINATE_AND_ORIGINATE_MEDIA
```

```
            - UPDATE_MEDIA
            - DELETE_MEDIA
            - REJECT_MEDIA
        - type: string
          description: >
            This string provides forward-compatibility with future extensions to the enumeration
            and is not used to encode content defined in the present version of this API.

    MediaConnSide:
      description: The audio or video media connection side.
      anyOf:
        - type: string
          enum:
            - BOTH
            - TERMINATING_SIDE
            - ORIGINATING_SIDE
        - type: string
          description: >
            This string provides forward-compatibility with future extensions to the enumeration
            and is not used to encode content defined in the present version of this API.

    SendRecvMode:
      description: The send and receive mode of media stream.
      anyOf:
        - type: string
          enum:
            - SENDRECV
            - SENDONLY
            - RECVONLY
        - type: string
          description: >
            This string provides forward-compatibility with future extensions to the enumeration
            and is not used to encode content defined in the present version of this API.

    TranscodeMode:
      description: The mode of the trancoding.
      anyOf:
        - type: string
          enum:
            - DC_TO_VIDEO
        - type: string
          description: >
            This string provides forward-compatibility with future extensions to the enumeration
            and is not used to encode content defined in the present version of this API.
```

# A.4 Nimsas_ImsSessionManagement API

```
openapi: 3.0.0

info:
  title: 'IMS AS ImsSessionManagement Service'
  version: 1.0.1
  description: |
    Nimsas_ImsSessionManagement Service.
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    All rights reserved.

externalDocs:
  description: >
    3GPP TS 29.175 V19.5.0; IP Multimedia Subsystem; IP Multimedia Subsystem (IMS) Application
    Server (AS) Services; Stage 3
  url: 'https://www.3gpp.org/ftp/Specs/archive/29_series/29.175'

servers:
  - url: '{apiRoot}/nimsas-ism/v1'
    variables:
      apiRoot:
        default: https://example.com
        description: apiRoot as defined in clause 4.4 of 3GPP TS 29.501

security:
  - {}
  - oAuth2ClientCredentials:
```

```
      - nimsas-ism

paths:
  /ims-sessions:
    post:
      summary: Create IMS Session
      operationId: ImsSessionCreate
      tags:
        - IMS Sessions (Collection)
      requestBody:
        description: Creates a new Individual IMS Session resource.
        required: true
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/ImsSessionInfo'
      responses:
        '201':
          description: >
            The creation of an Individual IMS Session resource is confirmed and
            a representation of that resource is returned.
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/ImsSessionInfo'
          headers:
            Location:
              description: >
                The URI of the newly created resource, according to the structure:
                {apiRoot}/nimsas-ism/<apiVersion>/ims-sessions/{sessionId}.
              schema:
                type: string
        '307':
          description: Temporary Redirect
          content:
            application/json:
              schema:
                $ref: 'TS29571_CommonData.yaml#/components/schemas/RedirectResponse'
          headers:
            Location:
              description: The URI pointing to the resource located on the redirect target IMS AS.
              schema:
                type: string
        '308':
          description: Permanent Redirect
          content:
            application/json:
              schema:
                $ref: 'TS29571_CommonData.yaml#/components/schemas/RedirectResponse'
          headers:
            Location:
              description: The URI pointing to the resource located on the redirect target IMS AS.
              schema:
                type: string
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '406':
          $ref: 'TS29571_CommonData.yaml#/components/responses/406'
        '411':
          $ref: 'TS29571_CommonData.yaml#/components/responses/411'
        '413':
          $ref: 'TS29571_CommonData.yaml#/components/responses/413'
        '415':
          $ref: 'TS29571_CommonData.yaml#/components/responses/415'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '501':
          $ref: 'TS29571_CommonData.yaml#/components/responses/501'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
```

```
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'

  /ims-sessions/{sessionId}:
    patch:
      summary: modify the IMS session
      operationId: ImsSessionModify
      tags:
        - IMS Session Modification
      parameters:
        - name: sessionId
          in: path
          description: Identifies an individual IMS session.
          required: true
          schema:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/SessionId'
        - name: supported-features
          in: query
          description: Features required to be supported by the target NF
          schema:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
      requestBody:
        content:
          application/json-patch+json:
            schema:
              type: array
              items:
                $ref: 'TS29571_CommonData.yaml#/components/schemas/PatchItem'
              minItems: 1
        required: true
      responses:
        '200':
          description: Successful update on the IMS session.
          content:
            application/json:
              schema:
                $ref: 'TS29571_CommonData.yaml#/components/schemas/PatchResult'
        '204':
          description: >
            No content, IMS AS accepts the request change which is to
            delete the existing media.
        '307':
          description: Temporary Redirect
          content:
            application/json:
              schema:
                $ref: 'TS29571_CommonData.yaml#/components/schemas/RedirectResponse'
          headers:
            Location:
              description: The URI pointing to the resource located on the redirect target IMS AS.
              schema:
                type: string
        '308':
          description: Permanent Redirect
          content:
            application/json:
              schema:
                $ref: 'TS29571_CommonData.yaml#/components/schemas/RedirectResponse'
          headers:
            Location:
              description: The URI pointing to the resource located on the redirect target IMS AS.
              schema:
                type: string
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '409':
          $ref: 'TS29571_CommonData.yaml#/components/responses/409'
        '411':
          $ref: 'TS29571_CommonData.yaml#/components/responses/411'
        '413':
          $ref: 'TS29571_CommonData.yaml#/components/responses/413'
        '415':
```

```
        $ref: 'TS29571_CommonData.yaml#/components/responses/415'
      '422':
        description: Unprocessable Entity
        content:
          application/problem+json:
            schema:
              $ref: 'TS29571_CommonData.yaml#/components/schemas/ProblemDetails'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '502':
        $ref: 'TS29571_CommonData.yaml#/components/responses/502'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'
  put:
    summary: Update an existing Individual IMS Session resource.
    operationId: ImsSessionUpdate
    tags:
      - Individual IMS Session (Document)
    requestBody:
      required: true
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/ImsSessionInfo'
    responses:
      '200':
        description: >
          OK. The Individual IMS Session resource is successfully updated and a representation
          of the updated resource is returned in the response body.
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/ImsSessionInfo'
      '204':
        description: >
          No Content. The Individual IMS Session Management resource is
          successfully updated and no content is returned in the response body.
      '307':
        $ref: 'TS29122_CommonData.yaml#/components/responses/307'
      '308':
        $ref: 'TS29122_CommonData.yaml#/components/responses/308'
      '400':
        $ref: 'TS29122_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29122_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29122_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29122_CommonData.yaml#/components/responses/404'
      '409':
        $ref: 'TS29122_CommonData.yaml#/components/responses/409'
      '411':
        $ref: 'TS29122_CommonData.yaml#/components/responses/411'
      '413':
        $ref: 'TS29122_CommonData.yaml#/components/responses/413'
      '415':
        $ref: 'TS29122_CommonData.yaml#/components/responses/415'
      '429':
        $ref: 'TS29122_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29122_CommonData.yaml#/components/responses/500'
      '503':
        $ref: 'TS29122_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29122_CommonData.yaml#/components/responses/default'

  delete:
    summary: delete the IMS session
    operationId: ImsSessionDelete
    tags:
      - IMS Session Deletion
    parameters:
      - name: sessionId
        in: path
```

```
        description: Identifies an individual IMS session.
        required: true
        schema:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/SessionId'
    responses:
      '204':
        description: No content, successful delete of the resource identified by sessionId.
      '307':
        description: Temporary Redirect
        content:
          application/json:
            schema:
              $ref: 'TS29571_CommonData.yaml#/components/schemas/RedirectResponse'
        headers:
          Location:
            description: The URI pointing to the resource located on the redirect target IMS AS.
            schema:
              type: string
      '308':
        description: Permanent Redirect
        content:
          application/json:
            schema:
              $ref: 'TS29571_CommonData.yaml#/components/schemas/RedirectResponse'
        headers:
          Location:
            description: The URI pointing to the resource located on the redirect target IMS AS.
            schema:
              type: string
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '502':
        $ref: 'TS29571_CommonData.yaml#/components/responses/502'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'

/ims-sessions/event-subscriptions:
  post:
  # This is a pseudo operation, NF service consumers shall NOT invoke this method!
    summary: Nimsas_ImsSessionManagement Subscribe service operation, pseudo operation
    tags:
      - IMS Session Event Subscriptions (Collection)
    operationId: Subscribe
    requestBody:
      required: true
      content:
        application/json:
          # Unspecified schema for the JSON body, since this is not used by consumer or producer.
          schema: {}
    responses:
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'

    callbacks:
      eventNotification:
        '{$request.body#/notifUri}':
        # The notification URI of the Service Consumer to receive the IMS session events.
          post:
            requestBody:
              required: true
              content:
                application/json:
                  schema:
                    $ref: '#/components/schemas/ImsSessionEventNotification'
            responses:
              '204':
```

```
                description: No Content, notification was succesfull.
              '307':
                description: Temporary Redirect
                content:
                  application/json:
                    schema:
                      $ref: 'TS29571_CommonData.yaml#/components/schemas/RedirectResponse'
                headers:
                  Location:
                    description: >
                      The URI pointing to the resource located on the redirect target NF
                      service consumer.
                    required: true
                    schema:
                      type: string
              '308':
                description: Permanent Redirect
                content:
                  application/json:
                    schema:
                      $ref: 'TS29571_CommonData.yaml#/components/schemas/RedirectResponse'
                headers:
                  Location:
                    description: >
                      The URI pointing to the resource located on the redirect target NF
                      service consumer.
                    required: true
                    schema:
                      type: string
              '400':
                $ref: 'TS29571_CommonData.yaml#/components/responses/400'
              '401':
                $ref: 'TS29571_CommonData.yaml#/components/responses/401'
              '403':
                $ref: 'TS29571_CommonData.yaml#/components/responses/403'
              '404':
                $ref: 'TS29571_CommonData.yaml#/components/responses/404'
              '411':
                $ref: 'TS29571_CommonData.yaml#/components/responses/411'
              '413':
                $ref: 'TS29571_CommonData.yaml#/components/responses/413'
              '415':
                $ref: 'TS29571_CommonData.yaml#/components/responses/415'
              '429':
                $ref: 'TS29571_CommonData.yaml#/components/responses/429'
              '500':
                $ref: 'TS29571_CommonData.yaml#/components/responses/500'
              '502':
                $ref: 'TS29571_CommonData.yaml#/components/responses/502'
              '503':
                $ref: 'TS29571_CommonData.yaml#/components/responses/503'
              default:
                $ref: 'TS29571_CommonData.yaml#/components/responses/default'

components:

  securitySchemes:
    oAuth2ClientCredentials:
      type: oauth2
      flows:
        clientCredentials:
          tokenUrl: '{nrfApiRoot}/oauth2/token'
          scopes:
            nimsas-ism: Access to the Nimsas_ImsSessionManagement API.

  schemas:

# STRUCTURED DATA TYPES

    ImsSessionInfo:
      description: The IMS session information and the media list used in the IMS session.
      type: object
      required:
        - sessionInfo
        - mediaInfoSet
      properties:
        sessionInfo:
          $ref: '#/components/schemas/SessionInfoSM'
```

```
        mediaInfoSet:
          description: Includes a list of media related information.
          type: array
          items:
            $ref: '#/components/schemas/MediaInfoExternal'
          minItems: 1
        sessionId:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/SessionId'
        notifUri:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
        notifCorrelationId:
          type: string

    SessionInfoSM:
      description: The session information to be used.
      type: object
      required:
        - callingId
        - calledId
      properties:
        callingId:
          $ref: 'TS29562_Nhss_imsSDM.yaml#/components/schemas/ImsPublicId'
        calledId:
          $ref: 'TS29562_Nhss_imsSDM.yaml#/components/schemas/ImsPublicId'

    MediaInfoExternal:
      description: The media information used in the external created or updated IMS session.
      type: object
      required:
        - mediaType
        - mediaCorrectionId
      properties:
        mediaType:
          $ref: 'TS29175_Nimsas_SessionEventControl.yaml#/components/schemas/MediaType'
        mediaCorrectionId:
          type: string
        dcMediaParas:
          description: The information of the created data channel media.
          type: array
          items:
            $ref: '#/components/schemas/DcMediaPara'
          minItems: 1

    DcMediaPara:
      description: The parameters of the IMS DC media.
      type: object
      required:
        - dcType
      properties:
        dcType:
          $ref: '#/components/schemas/DcType'
        adcType:
          $ref: '#/components/schemas/AdcType'
        appBindingInfo:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/AppBindingInfo'
        targetIds:
          description: Representes the public identity(ies) of the IMS Subscriber the DC is
targeted.
          type: array
          items:
            $ref: '#/components/schemas/TargetIdInfo'
          minItems: 1
        targetId:
          $ref: 'TS29562_Nhss_imsSDM.yaml#/components/schemas/ImsPublicId'
        mdc2EndpointInfo:
          $ref: 'TS29175_Nimsas_MediaControl.yaml#/components/schemas/Mdc2EndpointInfo'

    ImsSessionEventNotification:
      description: The IMS Session event notification.
      type: object
      required:
        - sessionId
        - eventType
        - notifCorrelationId
      properties:
        sessionId:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/SessionId'
        eventType:
```

```
        $ref: '#/components/schemas/ImsSessionEventType'
      notifCorrelationId:
        type: string
      mediaInfoSet:
        description: Includes a list of media related information.
        type: array
        items:
          $ref: '#/components/schemas/MediaInfoExternal'
        minItems: 1
      callingIdentity:
        $ref: 'TS29562_Nhss_imsSDM.yaml#/components/schemas/ImsPublicId'
      calledIdentity:
        $ref: 'TS29562_Nhss_imsSDM.yaml#/components/schemas/ImsPublicId'
      failureReason:
        type: string



  TargetIdInfo:
    description: The public identity(ies) of the IMS Subscriber the DC is targeted.
    type: object
    properties:
      callingIdentity:
        $ref: 'TS29562_Nhss_imsSDM.yaml#/components/schemas/ImsPublicId'
      calledIdentity:
        $ref: 'TS29562_Nhss_imsSDM.yaml#/components/schemas/ImsPublicId'



# ENUMS

  DcType:
    description: The type of data channel.
    anyOf:
      - type: string
        enum:
          - BDC
          - ADC
      - type: string
        description: >
          This string provides forward-compatibility with future extensions to the enumeration
          and is not used to encode content defined in the present version of this API.

  AdcType:
    description: The type of the application data channel.
    anyOf:
      - type: string
        enum:
          - P2P
          - P2A
          - P2A2P
      - type: string
        description: >
          This string provides forward-compatibility with future extensions to the enumeration
          and is not used to encode content defined in the present version of this API.

  ImsSessionEventType:
    description: The IMS session event type for which the notification is generated.
    anyOf:
      - type: string
        enum:
          - SESSION_ESTABLISHMENT_SUCCESS
          - SESSION_ESTABLISHMENT_FAILURE
          - MEDIA_CHANGE_SUCCESS
          - MEDIA_CHANGE_FAILURE
          - SESSION_TERMINATION
      - type: string
        description: >
          This string provides forward-compatibility with future extensions to the enumeration
          and is not used to encode content defined in the present version of this API.
```

# A.5 Nimsas_ImsEE API

```
openapi: 3.0.0

info:
  title: 'IMS AS ImsEventExposure Service'
  version: 1.0.2
  description: |
    Nimsas_ImsEE Service.
    © 2026, 3GPP Organizational Partners (ARIB, ATIS, CCSA, ETSI, TSDSI, TTA, TTC).
    All rights reserved.

externalDocs:
  description: >
    3GPP TS 29.175 V19.6.0; IP Multimedia Subsystem; IP Multimedia Subsystem (IMS) Application
    Server (AS) Services; Stage 3
  url: 'https://www.3gpp.org/ftp/Specs/archive/29_series/29.175'

servers:
  - url: '{apiRoot}/nimsas-ee/v1'
    variables:
      apiRoot:
        default: https://example.com
        description: apiRoot as defined in clause 4.4 of 3GPP TS 29.501

security:
  - {}
  - oAuth2ClientCredentials:
      - nimsas-ee

paths:
  /subscriptions:
    post:
      summary: Subscribe to IMS Event
      operationId: ImsEventSubscribe
      tags:
        - IMS Event Subscriptions (Collection)
      requestBody:
        description: Creates a new Individual IMS Event Subscription resource.
        required: true
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/ImsEventsSubscription'
      responses:
        '201':
          description: >
            Successful creation of the Individual IMS Event Subscription resource,
            when the IMS AS needs to return information in the response.
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/ImsEventsSubscriptionCreated'
          headers:
            Location:
              description: >
                Contains the URI of the newly created resource, according to the structure:
                {apiRoot}/nimsas-ee/<apiVersion>/subscriptions/{subscriptionId}.
              schema:
                type: string
        '307':
          description: Temporary Redirect
          content:
            application/json:
              schema:
                $ref: 'TS29571_CommonData.yaml#/components/schemas/RedirectResponse'
          headers:
            Location:
              description: The URI pointing to the resource located on the redirect target IMS AS.
              schema:
                type: string
        '308':
          description: Permanent Redirect
          content:
            application/json:
              schema:
                $ref: 'TS29571_CommonData.yaml#/components/schemas/RedirectResponse'
```

```
      headers:
        Location:
          description: The URI pointing to the resource located on the redirect target IMS AS.
          schema:
            type: string
    '400':
      $ref: 'TS29571_CommonData.yaml#/components/responses/400'
    '401':
      $ref: 'TS29571_CommonData.yaml#/components/responses/401'
    '403':
      $ref: 'TS29571_CommonData.yaml#/components/responses/403'
    '404':
      $ref: 'TS29571_CommonData.yaml#/components/responses/404'
    '406':
      $ref: 'TS29571_CommonData.yaml#/components/responses/406'
    '411':
      $ref: 'TS29571_CommonData.yaml#/components/responses/411'
    '413':
      $ref: 'TS29571_CommonData.yaml#/components/responses/413'
    '415':
      $ref: 'TS29571_CommonData.yaml#/components/responses/415'
    '429':
      $ref: 'TS29571_CommonData.yaml#/components/responses/429'
    '500':
      $ref: 'TS29571_CommonData.yaml#/components/responses/500'
    '501':
      $ref: 'TS29571_CommonData.yaml#/components/responses/501'
    '503':
      $ref: 'TS29571_CommonData.yaml#/components/responses/503'
  default:
    $ref: 'TS29571_CommonData.yaml#/components/responses/default'
callbacks:
  eventNotification:
    '{$request.body#/notificationURI}':
    # The notification URI is assigned within the Individual IMS Event Subscription.
      post:
        requestBody:
          required: true
          content:
            application/json:
              schema:
                type: array
                items:
                  $ref: '#/components/schemas/ImsEventsNotification'
                minItems: 1
        responses:
          '204':
            description: No Content, the receipt of the Notification is acknowledged.
          '307':
            description: Temporary Redirect
            content:
              application/json:
                schema:
                  $ref: 'TS29571_CommonData.yaml#/components/schemas/RedirectResponse'
            headers:
              Location:
                description: >
                  The URI pointing to the resource located on the redirect target NF
                  service consumer.
                required: true
                schema:
                  type: string
          '308':
            description: Permanent Redirect
            content:
              application/json:
                schema:
                  $ref: 'TS29571_CommonData.yaml#/components/schemas/RedirectResponse'
            headers:
              Location:
                description: >
                  The URI pointing to the resource located on the redirect target NF
                  service consumer.
                required: true
                schema:
                  type: string
          '400':
            $ref: 'TS29571_CommonData.yaml#/components/responses/400'
```

```
                '401':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/401'
                '403':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/403'
                '404':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/404'
                '411':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/411'
                '413':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/413'
                '415':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/415'
                '429':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/429'
                '500':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/500'
                '502':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/502'
                '503':
                  $ref: 'TS29571_CommonData.yaml#/components/responses/503'
                default:
                  $ref: 'TS29571_CommonData.yaml#/components/responses/default'

  /subscriptions/{subscriptionId}:
    get:
      summary: Retrieve the subscription information
      operationId: GetSubscriptionInfo
      tags:
        - Subscription information retrieval
      parameters:
        - name: subscriptionId
          in: path
          description: Individual IMS event subscription Identity
          required: true
          schema:
            type: string
      responses:
        '200':
          description: >
            A response body containing representation of the Individual IMS Event Subcription
resource.
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/ImsEventsSubscription'
        '307':
          description: Temporary Redirect
          content:
            application/json:
              schema:
                $ref: 'TS29571_CommonData.yaml#/components/schemas/RedirectResponse'
          headers:
            Location:
              description: The URI pointing to the resource located on the redirect target IMS AS.
              schema:
                type: string
        '308':
          description: Permanent Redirect
          content:
            application/json:
              schema:
                $ref: 'TS29571_CommonData.yaml#/components/schemas/RedirectResponse'
          headers:
            Location:
              description: The URI pointing to the resource located on the redirect target IMS AS.
              schema:
                type: string
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '405':
          $ref: 'TS29571_CommonData.yaml#/components/responses/405'
        '406':
```

```
        $ref: 'TS29571_CommonData.yaml#/components/responses/406'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '502':
        $ref: 'TS29571_CommonData.yaml#/components/responses/502'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'
  put:
    summary: Update IMS Event Subscription
    operationId: ImsEventSubsModify
    tags:
      - IMS Event Subscription Modification
    parameters:
      - name: subscriptionId
        in: path
        description: Individual IMS event subscription Identity
        required: true
        schema:
          type: string
    requestBody:
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/ImsEventsSubscription'
      required: true
    responses:
      '200':
        description: >
          Successful update of the subscription, when the IMS AS needs
          to return information in the response.
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/ImsEventsSubscriptionCreated'
      '204':
        description: No content
      '307':
        description: Temporary Redirect
        content:
          application/json:
            schema:
              $ref: 'TS29571_CommonData.yaml#/components/schemas/RedirectResponse'
        headers:
          Location:
            description: The URI pointing to the resource located on the redirect target IMS AS.
            schema:
              type: string
      '308':
        description: Permanent Redirect
        content:
          application/json:
            schema:
              $ref: 'TS29571_CommonData.yaml#/components/schemas/RedirectResponse'
        headers:
          Location:
            description: The URI pointing to the resource located on the redirect target IMS AS.
            schema:
              type: string
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '409':
        $ref: 'TS29571_CommonData.yaml#/components/responses/409'
      '411':
        $ref: 'TS29571_CommonData.yaml#/components/responses/411'
      '413':
        $ref: 'TS29571_CommonData.yaml#/components/responses/413'
      '415':
        $ref: 'TS29571_CommonData.yaml#/components/responses/415'
```

```
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '502':
        $ref: 'TS29571_CommonData.yaml#/components/responses/502'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        description: Unexpected error
  patch:
    summary: Partial update IMS Event Subscription
    operationId: ImsEventSubsPartialModify
    tags:
      - IMS Event Subscription Partial Modification
    parameters:
      - name: subscriptionId
        in: path
        description: Individual IMS event subscription Identity
        required: true
        schema:
          type: string
    requestBody:
      content:
        application/json-patch+json:
          schema:
            type: array
            items:
              $ref: 'TS29571_CommonData.yaml#/components/schemas/PatchItem'
            minItems: 1
      required: true
    responses:
      '200':
        description: >
          Successful update of the subscription, when the IMS AS needs
          to return information in the response.
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/ImsEventsSubscriptionCreated'
      '204':
        description: No content
      '307':
        description: Temporary Redirect
        content:
          application/json:
            schema:
              $ref: 'TS29571_CommonData.yaml#/components/schemas/RedirectResponse'
        headers:
          Location:
            description: The URI pointing to the resource located on the redirect target IMS AS.
            schema:
              type: string
      '308':
        description: Permanent Redirect
        content:
          application/json:
            schema:
              $ref: 'TS29571_CommonData.yaml#/components/schemas/RedirectResponse'
        headers:
          Location:
            description: The URI pointing to the resource located on the redirect target IMS AS.
            schema:
              type: string
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '411':
        $ref: 'TS29571_CommonData.yaml#/components/responses/411'
      '413':
        $ref: 'TS29571_CommonData.yaml#/components/responses/413'
      '415':
        $ref: 'TS29571_CommonData.yaml#/components/responses/415'
```

```
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '502':
          $ref: 'TS29571_CommonData.yaml#/components/responses/502'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          $ref: 'TS29571_CommonData.yaml#/components/responses/default'
    delete:
      summary: delete
      operationId: ImsEventSubsDelete
      tags:
        - IMS Event Subscription Deletion
      parameters:
        - name: subscriptionId
          in: path
          description: Individual IMS event subscription Identity
          required: true
          schema:
            type: string
      responses:
        '204':
          description: No content
        '307':
          description: Temporary Redirect
          content:
            application/json:
              schema:
                $ref: 'TS29571_CommonData.yaml#/components/schemas/RedirectResponse'
          headers:
            Location:
              description: The URI pointing to the resource located on the redirect target IMS AS.
              schema:
                type: string
        '308':
          description: Permanent Redirect
          content:
            application/json:
              schema:
                $ref: 'TS29571_CommonData.yaml#/components/schemas/RedirectResponse'
          headers:
            Location:
              description: The URI pointing to the resource located on the redirect target IMS AS.
              schema:
                type: string
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '502':
          $ref: 'TS29571_CommonData.yaml#/components/responses/502'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          $ref: 'TS29571_CommonData.yaml#/components/responses/default'

components:

  securitySchemes:
    oAuth2ClientCredentials:
      type: oauth2
      flows:
        clientCredentials:
          tokenUrl: '{nrfApiRoot}/oauth2/token'
          scopes:
            nimsas-ee: Access to the Nimsas_ImsEE API.

  schemas:
```

```
# STRUCTURED DATA TYPES

    ImsEventsSubscription:
      description: Represents an Individual IMS Event Subscription resource.
      type: object
      required:
        - eventList
      properties:
        eventList:
          description: >
            the IMS events to be subscribed or successfully subscribed in a
            subscription response within the referenceId converted from integer
            to string is the key of the map.
          type: object
          additionalProperties:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/ImsEventConfiguration'
          minProperties: 1
        subscriberId:
          $ref: 'TS29562_Nhss_imsSDM.yaml#/components/schemas/ImsPublicId'
        anyUeInd:
          type: boolean
        subscriptionId:
          type: string
        imsReportingOptions:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/ImsReportingOptions'
        notificationURI:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
        supportedFeatures:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
        notificationCorrelationId:
          type: string

    ImsEventsNotification:
      description: Notifications about Individual Events.
      type: object
      required:
        - eventNotifications
        - notificationCorrelationId
      properties:
        eventNotifications:
          description: Notifications about Individual Events.
          type: array
          items:
            $ref: '#/components/schemas/EventNotification'
          minItems: 1
        notificationCorrelationId:
          type: string

    EventNotification:
      description: The notification information for a specific event.
      type: object
      required:
        - event
        - subscriberId
        - timestamp
      properties:
        event:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/ImsEvent'
        referenceId:
          $ref: 'TS29503_Nudm_EE.yaml#/components/schemas/ReferenceId'
        subscriberId:
          $ref: 'TS29562_Nhss_imsSDM.yaml#/components/schemas/ImsPublicId'
        timestamp:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
        eventReport:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/ImsEventReportInfo'
        sessionId:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/SessionId'

    ImsEventsSubscriptionCreated:
      description: >
        Represents the subscription of the created/modified IMS event subscriptiona and
        the immediate reports
      type: object
      required:
        - imsEventsSubscription
      properties:
```

```
        imsEventsSubscription:
          $ref: '#/components/schemas/ImsEventsSubscription'
        eventReports:
          description: >
            Contains the associated status to the event(s) being requested if
            the status associated to at least one of the requested event(s)
            is available at the time of subscription.
          type: array
          items:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/ImsEventReport'
          minItems: 1


# ENUMS
```

# A.6 Nimsas_ImsParameterProvision API

```
openapi: 3.0.0

info:
  title: 'IMS AS ImsParameterProvision Service'
  version: 1.0.0
  description: |
    Nimsas_ImsParameterProvision Service.
    © 2025, 3GPP Organizational Partners (ARIB, ATIS, CCSA, ETSI, TSDSI, TTA, TTC).
    All rights reserved.

externalDocs:
  description: >
    3GPP TS 29.175 V19.4.0; IP Multimedia Subsystem; IP Multimedia Subsystem (IMS) Application
    Server (AS) Services; Stage 3
  url: 'https://www.3gpp.org/ftp/Specs/archive/29_series/29.175'

servers:
  - url: '{apiRoot}/nimsas-ipp/v1'
    variables:
      apiRoot:
        default: https://example.com
        description: apiRoot as defined in clause 4.4 of 3GPP TS 29.501

security:
  - {}
  - oAuth2ClientCredentials:
      - nimsas-ipp

paths:

  /{imsUeId}/ims-pp-data-store/{afInstanceId}:
    put:
      summary: Creates or updates the parameter provisioning data entry.
      operationId: CreateUpdatePpDataEntry
      tags:
        - ImsParameterProvisioningDataEntry
      parameters:
        - name: imsUeId
          in: path
          description: Contains the IMS public identity.
          required: true
          schema:
            type: string
            pattern: '^(impu-sip\:([a-zA-Z0-9_\-.!~*()&=+$,;?\/]+)\@([A-Za-z0-9]+([-A-Za-z0-
9]+)\.)+[a-z]{2,}|impu-tel\:\+[0-9]{5,15})$'
        - name: afInstanceId
          in: path
          description: Contains the AF instance identity.
          required: true
          schema:
            type: string
      requestBody:
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/ImsPpDataEntry'
        required: true
      responses:
```

```
        '201':
          description: >
            Created, a representation of the created ImsParameterProvisioningDataEntry resource
            is returned.
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/ImsPpDataEntry'
          headers:
            Location:
              description: >
                Contains the URI of the newly created resource, according to the structure:
                {apiRoot}/nhss-ims-uecm/v1/{imsUeId}/imsas-registration
              required: true
              schema:
                type: string
        '200':
          description: >
            OK, a representation of the updated ImsParameterProvisioningDataEntry resource
            is returned..
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/ImsPpDataEntry'
        '204':
          description: No content, the ImsParameterProvisioningDataEntry resource is updated.
        '307':
          $ref: 'TS29571_CommonData.yaml#/components/responses/307'
        '308':
          $ref: 'TS29571_CommonData.yaml#/components/responses/308'
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '411':
          $ref: 'TS29571_CommonData.yaml#/components/responses/411'
        '413':
          $ref: 'TS29571_CommonData.yaml#/components/responses/413'
        '415':
          $ref: 'TS29571_CommonData.yaml#/components/responses/415'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '502':
          $ref: 'TS29571_CommonData.yaml#/components/responses/502'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          $ref: 'TS29571_CommonData.yaml#/components/responses/default'

    delete:
      summary: Delete the parameter provisioning data entry.
      operationId: DeletePpDataEntry
      tags:
        - ImsParameterProvisioningDataEntry
      parameters:
        - name: imsUeId
          in: path
          description: Contains the IMS public identity.
          required: true
          schema:
            type: string
            pattern: '^(impu-sip\:([a-zA-Z0-9_\-.!~*()&=+$,;?\/]+)\@([A-Za-z0-9]+([-A-Za-z0-
9]+)\.)+[a-z]{2,}|impu-tel\:\+[0-9]{5,15})$'
        - name: afInstanceId
          in: path
          description: Contains the AF instance identity.
          required: true
          schema:
            type: string
      responses:
        '204':
          description: No content, the ImsParameterProvisioningDataEntry resource is deleted.
```

```
        '307':
          $ref: 'TS29571_CommonData.yaml#/components/responses/307'
        '308':
          $ref: 'TS29571_CommonData.yaml#/components/responses/308'
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '502':
          $ref: 'TS29571_CommonData.yaml#/components/responses/502'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          $ref: 'TS29571_CommonData.yaml#/components/responses/default'

components:

  securitySchemes:
    oAuth2ClientCredentials:
      type: oauth2
      flows:
        clientCredentials:
          tokenUrl: '{nrfApiRoot}/oauth2/token'
          scopes:
            nimsas-ipp: Access to the Nimsas_ImsParameterProvision API

  schemas:

# Structured data types

    ImsPpDataEntry:
      description: Contains the IMS parameter provision data i.e. the RCD properties.
      type: object
      properties:
        rcdProp:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/RcdProperties'
```

# Annex B (informative): Change history

| Change history | | | | | | | |
|---|---|---|---|---|---|---|---|
| **Date** | **Meeting** | **TDoc** | **CR** | **Rev** | **Cat** | **Subject/Comment** | **New version** |
| 2023-04 | CT4#115e | C4-231514 | | | | TS skeleton | 0.0.0 |
| 2023-04 | CT4#115e | C4-231551 | | | | Inclusion of pCRs agreed at CT4#115e, including C4-231214, C4-231217, C4-231224, C4-231225, C4-231514, C4-231515, C4-231516, C4-231517. | 0.1.0 |
| 2023-05 | CT4#116 | C4-232553 | | | | Inclusion of pCRs agreed at CT4#116, including C4-232142, C4-232145, C4-232504, C4-232505. | 0.2.0 |
| 2023-09 | CT4#117 | C4-233810 | | | | Inclusion of pCRs agreed at CT4#117, including C4-233250,C4-233254,C4-233255,C4-233256,C4-233700,C4-233701,C4-233704,C4-233856. | 0.3.0 |
| 2023-10 | CT4#118 | C4-234583 | | | | Inclusion of pCRs agreed at CT4#118, including C4-234172, C4-234174, C4-234175, C4-234177, C4-234545, C4-234547. | 0.4.0 |
| 2023-12 | CT4#119 | C4-235660 | | | | Inclusion of pCRs agreed at CT4#119, including C4-235221, C4-235464, C4-235465, C4-235466, C4-235467, C4-235468, C4-235471 | 0.5.0 |
| 2023-12 | CT#102 | CP-233022 | | | | TS presented for information | 1.0.0 |
| 2024-03 | CT4#121 | C4-240852 | | | | Inclusion of pCRs agreed at CT4#121, including C4-240405, C4-240406, C4-240407, C4-240409, C4-240410, C4-240470, C4-240704, C4-240705, C4-240708, C4-240716 | 1.1.0 |
| 2024-03 | CT#103 | CP-240025 | | | | TS presented for approval | 2.0.0 |
| 2024-03 | CT#103 | | | | | TS approved in CT#103 | 18.0.0 |
| 2024-06 | CT#104 | CP-241044 | 0001 | | F | Support of SDP attributes a=3gpp-bdc-used-by and a=3gpp-req-app | 18.1.0 |
| 2024-06 | CT#104 | CP-241044 | 0002 | 1 | F | Correction on the Nimsas_MediaControl service | 18.1.0 |
| 2024-06 | CT#104 | CP-241044 | 0003 | 1 | F | Correction on the Nimsas_SessionEventControl service | 18.1.0 |
| 2024-06 | CT#104 | CP-241044 | 0004 | | F | Correction on the Nimsas_MediaControl OpenAPI | 18.1.0 |
| 2024-06 | CT#104 | CP-241044 | 0006 | 1 | F | Update the presence condition and cardinality for mediaProxyConfig | 18.1.0 |
| 2024-06 | CT#104 | CP-241044 | 0007 | 1 | F | Update the DcMediaSpecification for Nimsas_SessionEventControl Service API | 18.1.0 |
| 2024-06 | CT#104 | CP-241044 | 0008 | 1 | F | Update the DcMediaSpecification Datatype for MdcEndpoint | 18.1.0 |
| 2024-06 | CT#104 | CP-241044 | 0009 | 1 | F | Update the SessionEventNotificationUri for session event notification | 18.1.0 |
| 2024-06 | CT#104 | CP-241044 | 0010 | 1 | F | Report the media HOLD to DCSF | 18.1.0 |
| 2024-06 | CT#104 | CP-241044 | 0011 | 2 | F | Add the Media re-negotiation indication to Nimsas_MediaControl API to support AR | 18.1.0 |
| 2024-06 | CT#104 | | 0012 | | F | 29.175 Rel-18 API version and External doc update | 18.1.0 |
| 2024-09 | CT#105 | CP-242048 | 0013 | | F | Corrections and Updates to Notify Service Operation | 18.2.0 |
| 2024-09 | CT#105 | CP-242048 | 0014 | | F | Remove the SecuritySetup in clause 6.1.6.1 | 18.2.0 |
| 2024-09 | CT#105 | CP-242048 | 0015 | 1 | F | Update the sessionInfo description for Nimsas_SessionEventControl API | 18.2.0 |
| 2024-12 | CT#106 | CP-243027 | 0017 | | F | DC resource release due to a CANCEL request | 19.0.0 |
| 2024-12 | CT#106 | CP-243027 | 0023 | 1 | F | Remove the MRF | 19.0.0 |
| 2024-12 | CT#106 | CP-243027 | 0020 | 1 | F | Report the QoS info to DCSF | 19.0.0 |
| 2024-12 | CT#106 | CP-243068 | 0032 | | F | API version and External doc update | 19.0.0 |
| 2024-12 | CT#106 | CP-243045 | 0019 | 1 | B | Supporting of network initiated IMS Data Channel | 19.0.0 |
| 2024-12 | CT#106 | CP-243045 | 0022 | 1 | B | Support of interworking with MTSI client | 19.0.0 |
| 2024-12 | CT#106 | CP-243045 | 0025 | | D | Update the description of EventInitiator data type | 19.0.0 |
| 2024-12 | CT#106 | CP-243045 | 0027 | | B | Support of the specific PSI within Called ID for standalone DC | 19.0.0 |
| 2024-12 | CT#106 | CP-243045 | 0021 | 2 | B | Definition of Nimsas_ImsSessionManagement Service | 19.0.0 |
| 2024-12 | CT#106 | CP-243045 | 0030 | 1 | B | Definition of Nimsas_ImsEE Service | 19.0.0 |
| 2024-12 | CT#106 | CP-243045 | 0024 | 2 | F | Event Notification for PS Data Off Status | 19.0.0 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2024-12 | CT#106 | CP-243045 | 0028 | 2 | B | Support of IMS AS Event Exposure Service | 19.0.0 |
| 2024-12 | CT#106 | CP-243069 | 0033 | | F | API version and External doc update | 19.0.0 |
| 2025-03 | CT#107 | CP-250047 | 0043 | 1 | B | Clarification on the MediaInfo of the IMS session management service | 19.1.0 |
| 2025-03 | CT#107 | CP-250047 | 0040 | 1 | B | Support of avatar communication | 19.1.0 |
| 2025-03 | CT#107 | CP-250047 | 0044 | 1 | B | Add Nimsas_ImsPP service to support 3rd party identity information | 19.1.0 |
| 2025-03 | CT#107 | CP-250047 | 0034 | 2 | B | Notify Service Operation of Nimsas_SessionManagement API | 19.1.0 |
| 2025-03 | CT#107 | CP-250047 | 0039 | 2 | B | Update Nimsas_ImsSessionManagement service | 19.1.0 |
| 2025-03 | CT#107 | CP-250047 | 0035 | 2 | B | Update the service description for terminating BDC establishment without DC SDP | 19.1.0 |
| 2025-03 | CT#107 | CP-250047 | 0036 | 2 | B | Update the Nimsas_SessionEventControl service to support Network Capability exposure | 19.1.0 |
| 2025-03 | CT#107 | CP-250047 | 0037 | 2 | B | Update the description of callingIdentity for Nimsas_SessionEventControl service | 19.1.0 |
| 2025-03 | CT#107 | CP-250047 | 0038 | 2 | B | Update the Nimsas_SessionEventControl service to support PS Data Off status report | 19.1.0 |
| 2025-03 | CT#107 | CP-250047 | 0045 | 2 | B | Corrections on service name and HTTP method | 19.1.0 |
| 2025-03 | CT#107 | CP-250047 | 0046 | 5 | B | Nimsas_EE subscribe operation | 19.1.0 |
| 2025-03 | CT#107 | CP-250137 | 0047 | | F | API version and External doc update | 19.1.0 |
| 2025-06 | CT#108 | CP-251049 | 0052 | 1 | A | Correction on the MediaInstructions data type for AR media | 19.2.0 |
| 2025-06 | CT#108 | CP-251049 | 0061 | 2 | A | Clarify the calling and called party identity | 19.2.0 |
| 2025-06 | CT#108 | CP-251049 | 0065 | 3 | A | Removal of Mediain Nimsas_SessionEventControl | 19.2.0 |
| 2025-06 | CT#108 | CP-251068 | 0050 | | D | Editorial modification on the description of AdcType | 19.2.0 |
| 2025-06 | CT#108 | CP-251068 | 0058 | | F | Definition of media correlation ID | 19.2.0 |
| 2025-06 | CT#108 | CP-251068 | 0059 | | B | Usage of common data types in IMS event exposure | 19.2.0 |
| 2025-06 | CT#108 | CP-251068 | 0048 | 1 | B | Update SessionEventControl Service to Notify Operation from External Party | 19.2.0 |
| 2025-06 | CT#108 | CP-251068 | 0053 | 1 | F | Update Nimsas_ImsSessionManagement service | 19.2.0 |
| 2025-06 | CT#108 | CP-251068 | 0054 | 1 | B | Updates the Nimsas_MediaControl service to support DC multiplexing | 19.2.0 |
| 2025-06 | CT#108 | CP-251068 | 0055 | 1 | B | Update the MediaProxy value | 19.2.0 |
| 2025-06 | CT#108 | CP-251068 | 0057 | 1 | F | Update the application errors for Nimsas_ImsEE service | 19.2.0 |
| 2025-06 | CT#108 | CP-251068 | 0063 | 1 | B | Remove the temporarily defined mediaInfoExternal | 19.2.0 |
| 2025-06 | CT#108 | CP-251068 | 0069 | | F | Correction on Nimsas_ImsEE service | 19.2.0 |
| 2025-06 | CT#108 | CP-251068 | 0066 | 1 | F | Corrections on PS Data Off | 19.2.0 |
| 2025-06 | CT#108 | CP-251068 | 0067 | 1 | B | Add Nimsas_ImsEE OpenAPI | 19.2.0 |
| 2025-06 | CT#108 | CP-251068 | 0068 | 1 | B | Add Nimsas_ImsSessionManagement OpenAPI | 19.2.0 |
| 2025-06 | CT#108 | CP-251068 | 0070 | 1 | F | Correction on Nimsas_ImsSessionManagement | 19.2.0 |
| 2025-06 | CT#108 | CP-251079 | 0072 | | F | API version and External doc update | 19.2.0 |
| 2025-09 | CT#109 | CP-252033 | 0085 | 2 | A | Missing condition for mediaInstruction | 19.3.0 |
| 2025-09 | CT#109 | CP-252050 | 0079 | | B | Nimsas_ImsPP_Delete service operation definition | 19.3.0 |
| 2025-09 | CT#109 | CP-252050 | 0080 | | B | Nimsas_ImsParameterProvision API: "201 Created" and "204 No Content" responses | 19.3.0 |
| 2025-09 | CT#109 | CP-252050 | 0086 | | B | Update the service operation description | 19.3.0 |
| 2025-09 | CT#109 | CP-252050 | 0089 | | B | Remove the editor note for Nimsas_ImsEE Service | 19.3.0 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2025-09 | CT#109 | CP-252050 | 0090 | | B | Remove the editor note in overview | 19.3.0 |
| 2025-09 | CT#109 | CP-252050 | 0073 | 1 | F | Correction on the data channel multiplexing | 19.3.0 |
| 2025-09 | CT#109 | CP-252050 | 0077 | 1 | B | Nimsas_ImsParameterProvision API: removal of EN on RCD | 19.3.0 |
| 2025-09 | CT#109 | CP-252050 | 0081 | 1 | F | Nimsas_ImsParameterProvision: correction on the Public User Identity | 19.3.0 |
| 2025-09 | CT#109 | CP-252050 | 0078 | 1 | D | Nimsas_ImsParameterProvision API: editorial corrections | 19.3.0 |
| 2025-09 | CT#109 | CP-252050 | 0082 | 1 | B | Nimsas_ImsParameterProvision API: specification of the OpenAPI file | 19.3.0 |
| 2025-09 | CT#109 | CP-252050 | 0083 | 2 | B | Instruction on transcoding for interworking | 19.3.0 |
| 2025-09 | CT#109 | CP-252050 | 0076 | 2 | F | Correction on the SessionId attribute description | 19.3.0 |
| 2025-09 | CT#109 | CP-252050 | 0092 | 2 | F | Add PUT operation in Nimsas_ImsSessionManagement_Update | 19.3.0 |
| 2025-09 | CT#109 | CP-252050 | 0091 | 2 | B | Add DC interworking indication in DcMediaSpecification | 19.3.0 |
| 2025-09 | CT#109 | CP-252176 | 0093 | | F | API version and External doc update | 19.3.0 |
| 2025-12 | CT#110 | CP-253142 | 0096 | 1 | A | Update the presence condition of arMediaSpecification | 19.4.0 |
| 2025-12 | CT#110 | CP-253142 | 0104 | 1 | A | Update the data type to MediaIdRm | 19.4.0 |
| 2025-12 | CT#110 | CP-253155 | 0100 | | F | Correction on Nimsas_ImsEE service operation | 19.4.0 |
| 2025-12 | CT#110 | CP-253155 | 0094 | 1 | F | API specification file name | 19.4.0 |
| 2025-12 | CT#110 | CP-253155 | 0098 | 1 | F | Add 404 Not Found application error in Nimsas_ImsEE notification | 19.4.0 |
| 2025-12 | CT#110 | CP-253155 | 0101 | 2 | F | Correction on the attribute notifUri and notifCorrectionId in ImsSessionInfo | 19.4.0 |
| 2025-12 | CT#110 | CP-253155 | 0109 | | F | Correction on Nimsas_ImsEE service operation | 19.4.0 |
| 2025-12 | CT#110 | CP-253155 | 0105 | 1 | F | Update on ImsSessionEventNotification for IMS session management API | 19.4.0 |
| 2025-12 | CT#110 | CP-253155 | 0106 | 1 | B | Audio/video support in the MDC2 interface | 19.4.0 |
| 2025-12 | CT#110 | CP-253155 | 0107 | 1 | B | Update error handling for IMS session management | 19.4.0 |
| 2025-12 | CT#110 | CP-253155 | 0108 | 1 | B | Update PUT method on Individual IMS Session Resource | 19.4.0 |
| 2025-12 | CT#110 | CP-253155 | 0110 | 1 | F | Correction on Nimsas_ImsSM service operation | 19.4.0 |
| 2025-12 | CT#110 | CP-253155 | 0099 | 2 | F | Add an application error for Nimsas_ImsSM_update operation | 19.4.0 |
| 2025-12 | CT#110 | CP-253155 | 0097 | 4 | B | Add the multiAppBindingInfo attribute to DcMediaSpec | 19.4.0 |
| 2025-12 | CT#110 | CP-253167 | 0112 | | F | API version and External doc update | 19.4.0 |
| 2026-03 | CT#111 | CP-260032 | 0114 | | F | Reference ID | 19.5.0 |
| 2026-03 | CT#111 | CP-260032 | 0113 | 1 | F | Presence Condition of Notification Correlation ID | 19.5.0 |
| 2026-03 | CT#111 | CP-260032 | 0116 | 2 | F | Update abbreviations and attribute description of ImsSessionInfo | 19.5.0 |
| 2026-03 | CT#111 | CP-260040 | 0117 | | F | API version and External doc update | 19.5.0 |
| 2026-06 | CT#112 | CP-261045 | 0122 | 1 | A | Correction on the DcMediaSpecification | 19.6.0 |
| 2026-06 | CT#112 | CP-261051 | 0118 | 1 | F | Correct Cardinality in Resource Operations | 19.6.0 |
| 2026-06 | CT#112 | CP-261051 | 0119 | 2 | F | Correction on the usage of "DC Application Proxy" in the description of mediaProxyConfig in DcMediaSpecification | 19.6.0 |
| 2026-06 | CT#112 | CP-261051 | 0123 | 1 | F | Update on the Mdc2 Info | 19.6.0 |
| 2026-06 | CT#112 | CP-261088 | 0128 | | F | API version and External doc update | 19.6.0 |
| 2026-09 | CT#113 | CP-262020 | 0133 | 1 | A | Update on the audio and video media on MDC2 interface | 19.7.0 |
| 2026-09 | CT#113 | CP-262185 | 0136 | 4 | F | Add charging information for IMS session event notification | 19.7.0 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2026-09 | CT#113 | CP-262043 | 0139 | | F | API version and External doc update | 19.7.0 |

# History

| Version | Date | Status |
|---|---|---|
| V19.3.0 | January 2026 | Publication |
| V19.4.0 | February 2026 | Publication |
| V19.5.0 | March 2026 | Publication |
| V19.6.0 | August 2026 | Publication |
| V19.7.0 | October 2026 | Publication |